

ORACLE
NetSuite

SuiteAnalytics Workbook

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SuiteAnalytics Workbook Overview

SuiteAnalytics Workbook is an analytics tool in NetSuite. With it, you can create custom workbooks that combine datasets, tables, pivot tables, and charts all in one place using the new analytics data source. The analytics data source ensures that fields show up the same way and give consistent results across all workbook visualizations. Workbook now supports multilevel joins, so you can use data from multiple record types, including custom records, when creating workbooks and datasets. Custom formula fields are also supported, letting you add fields with dynamic values to your workbooks. And you can link datasets in Workbook, so you can compare metrics from different datasets in a single workbook visualization.

The SuiteAnalytics Workbook user interface is designed so even users without much record schema or query language knowledge can create complex workbooks and datasets with simple actions like drag-and-drop. When you create a workbook or visualization, you can base it on any dataset you have access to in your account. You can also link multiple datasets to the same workbook, so you can analyze different metrics without switching tabs or workbooks. When you've connected multiple datasets to a workbook, you can link them to analyze their metrics in a pivot table or chart.

Note: You can't edit how datasets are linked using the Workbook interface right now, but you can use linked datasets to make new workbook visualizations. For more information, see [Dataset Linking in SuiteAnalytics Workbook](#).

The Workbook user interface is great for ad hoc analysis, with instant formula validation, quick data refresh, and drill-down through your query results. The interface also has a range of customization options for advanced formatting, filtering, and visualizing your data.

The latest version of SuiteAnalytics Workbook lets you create workbooks and datasets for lots of record types. You can also use analytical record types made specifically for Workbook, like the sales (invoiced) record type.

Important: SuiteAnalytics Workbook isn't supported by SuiteBundler. To avoid installation errors, don't include SuiteAnalytics Workbook objects in your bundles. For more information about SuiteBundler, see the help topic [SuiteBundler Overview](#).

To help familiarize yourself with SuiteAnalytics Workbook and better understand the content available in the Workbook Help Center, see [Getting Started with SuiteAnalytics Workbook](#).

Getting Started with SuiteAnalytics Workbook

To help familiarize yourself with SuiteAnalytics Workbook, see the following video and help topics:

 [Getting Started with SuiteAnalytics Workbook](#)

- To learn how to set up Workbook:
 - Workbook is enabled by default. However, if you do not see the Analytics tab in the NetSuite navigation menu, see [Enabling SuiteAnalytics Workbook in Your NetSuite Account](#).
 - There should be at least one user with the Analytics Administrator in your account. To learn what this user is enabled to do in Workbook, see [The Analytics Administrator Permission](#).
 - If you experience long load times for your datasets or workbooks using your datasets, you can enable Cached Data in Datasets. For more information, see [Optimized Data Refresh](#).

- To learn about custom workbooks, datasets, and preset templates:
 - [Custom Workbooks and Datasets](#) covers how to create your own workbooks and datasets. You'll find steps for defining, filtering, and pivoting your data, plus how to link datasets and create custom formula fields.
 - Workbook supports multiple predefined workbook and dataset templates. For more information, see [Workbook and Dataset Templates](#).
- To learn about the analytics data source:
 - Workbook uses the new analytics data source, so you might need different fields, record types, joins, or formulas to match your existing saved searches. Review the [Analytics Data Source Overview](#) for more information, including how to join record types and steps to create sample workbooks.
 - Some record types have been created specifically for Workbook and exist only in the analytics data source. For more information, see [Analytical Record Types](#).
- To learn about the Workbook user interface:
 - Workbook is built using a new user interface. To learn about the elements within the interface, see [Navigating SuiteAnalytics Workbook](#).
 - There is new terminology associated with Workbook. For more information, see the [SuiteAnalytics Workbook Glossary](#).
- To learn how to create and run SuiteAnalytics Workbook queries using SuiteScript:
 - [Scripting with the N/query Module](#) explains the different module objects and how to use them when defining your queries.
 - To learn about linking datasets as part of your query definitions, see [Linking Datasets](#).
- Additional resources:
 - Complete the [SuiteAnalytics Workbook Tutorial](#) to learn how to create a sample sales order workbook.
 - For information about how to add workbook-based portlets to your NetSuite dashboards, see [Workbook-based Portlets](#).
 - There are some known limitations with the current iteration of Workbook. For more information, see [Known Limitations in SuiteAnalytics Workbook](#).
 - You can also visit the [SuiteAnalytics Workbook Community](#) to find answers and other information about Workbook.
- Workbook videos:
 -  [Data Model Basics](#): Describes the different record types available in the analytics data source that is used by Workbook.
 -  [Joining Record Types in Workbook](#): Explains the implications of joining record types in Workbook. It also provides some information about the SQL that is used to accomplish these joins in the background, and some suggestions to improve performance.
 -  [Separation of Dataset and Workbook in 2020.1](#): Describes the change in NetSuite 2020.1 that separates datasets from workbooks.
 -  [Using Filters in SuiteAnalytics Workbook](#): Describes the different filtering options available in Workbook, including: data grid filters, value-based and measure-based filters for pivot tables, chart filtering options, and workbook criteria filters.
 -  [Transaction Data Model in 2020.1](#): Describes what happens when you join the transaction, transaction line, and transaction accounting line record types in NetSuite 2020.1. The video also

explains the main differences between a transaction saved search and a transaction workbook. You'll also get tips for using the transaction record type as the root record for a dataset.

Enabling SuiteAnalytics Workbook in Your NetSuite Account

By default, SuiteAnalytics Workbook is enabled in all NetSuite accounts. If you don't see the Analytics tab in the NetSuite menu, follow these steps to check if the feature is enabled. You need to have the Administrator role to enable SuiteAnalytics Workbook in your account.

To verify that SuiteAnalytics Workbook is enabled in your account:

1. Go to Setup > Company > Enable Features, and click the **Analytics** subtab.
2. Ensure that the **SuiteAnalytics Workbook** box is checked.
3. Click **Save**.

With the feature enabled, all standard roles have access to Workbook except for the following roles:

- Customer Center
- Vendor Center
- NetSuite Support Center
- NetSuite Support Center (Basic)
- Publisher Center

Since the Workbook feature is enabled for all NetSuite accounts, all users with standard roles have the SuiteAnalytics Workbook permission. But users with custom roles might not have access to Workbook. To give Workbook access to custom roles, add the SuiteAnalytics Workbook permission to the role and set the access to Edit under the Reports subtab of the Permissions subtab.

To enable users to edit and monitor workbooks created by others, add the Analytics Administrator permission to their role on the Setup subtab of the Permissions subtab. Alternatively, to enable Analytics Administrator for individual users, add the global permission to their employee record.

For more information about customizing roles, see the help topic [Customizing or Creating NetSuite Roles](#).

For more information about the Analytics Administrator permission, see [The Analytics Administrator Permission](#).

The Analytics Administrator Permission

The Analytics Administrator permission lets users delete, share, and edit workbooks and datasets in your account. Users with this permission can also track changes to workbooks and datasets through the audit trail and execution log record types. They can also set the audience for workbook and dataset templates.

The minimum access level for this permission is Full, and by default it's enabled for account administrators. You can assign this permission to a role under the Setup subtab of Permissions, or as a global permission on an employee record. Deleted workbooks and datasets can't be recovered, so this permission should only go to a small number of users. While Analytics Administrators can delete workbooks and datasets from other users, they can only view the fields and records allowed by their

NetSuite permissions. Analytics Administrators don't automatically get access to SuiteAnalytics Workbook, they must be assigned a role with the SuiteAnalytics Workbook permission.

Editing Employee Workbooks and Datasets

Users with the Analytics Administrator permission can access every workbook and dataset in your account from the Analytics Home page.

- To see workbooks or datasets created by others, pick Employee Workbooks or Employee Datasets in the dropdown of the right subtab.
- To edit workbook or dataset properties without opening them, click the Edit icon next to the name, description, or owner. You can also share or delete them using the icons in the Actions column. Right now, there are no notifications when workbooks are deleted, shared, or assigned to new owners.

Auditing Workbooks and Datasets



Important: As of 2020.1, the audit trail and execution log record types contain data only from release 2019.2 and earlier. You should not use these record types for auditing changes made to workbooks and datasets after 2020.1. If you want to query these record types using SuiteAnalytics Connect, use the NetSuite2.com data source and search for the usrauditlog, usrdsauditlog, and usrexecutionlong record type IDs. For more information, see the help topic [Connect Data Source](#).

To monitor Workbook use, Analytics Administrators have access to the audit trail and execution log record types. Each record type has a version for workbooks and datasets, which you can tell apart by the record ID. The dataset audit trail and execution log record types both have 'Ds' in the record ID. You can use these record types to make datasets that show what data is being accessed through Workbook and track changes to all saved workbooks and datasets in your account.

Record Type	Description	Fields
Audit trail	Displays changes made to each saved workbook or dataset in your account, including the date the change was saved and the user who saved it. Records for each saved workbook and dataset are stored indefinitely, or until the workbook or dataset is deleted. ■ To view the most recent saved changes, join fields from the change audit change record type to the audit trail dataset. ■ To view the details of each saved change in a workbook or dataset, including the fields that were changed and the new and original values, join the new value and old value fields from the detail audit detail record type to the audit trail dataset.	Date/Time, ID, User, Saved Workbook
Execution log	Displays all workbooks and datasets that were created in the past 30 days, including those that were not saved. For each workbook or dataset listed, the execution log displays the following data: ■ All fields used on the dataset ■ The base record type ■ The date the workbook or dataset was run ■ The name of the user who ran the workbook or dataset ■ If a workbook visualization or dataset was exported ■ Any formulas used	Base Record Type, Date/Time, Export, Expression, Fields, Formulas, ID, User Name, Saved Workbook

- | |
|--|
| ■ If applicable, the name of the associated saved workbook |
|--|

Defining the Audience for Workbook and Dataset Templates

Workbook offers many predefined workbook and dataset templates. For a detailed list including summaries of the content within each workbook or dataset, see [Workbook and Dataset Templates](#).

To share these templates, Analytics Administrators can share most templates with specific users or roles right from the Analytics Home page. If the Share icon isn't available, Analytics Administrators need to open the workbook or dataset, then save and share a copy. For more information about sharing workbooks and datasets, see [Saving and Sharing Workbooks and Datasets](#).

Data Refresh in SuiteAnalytics Workbook

By default in SuiteAnalytics Workbook, data in pivot tables and charts is cached after 60 minutes of inactivity, but datasets don't use cached data. However, you can turn on the Cached Data in Datasets feature so datasets use cached data too.

When Cached Data in Datasets is enabled, your datasets load faster and data refresh is optimized in all your datasets and workbooks. See the following options:

- **Optimized Data Refresh** - With the Cached Data in Datasets feature enabled, you can choose to use cached data or real-time data in your datasets and workbooks, and change the data refresh option at any time. For more information, see [Optimized Data Refresh](#).
- **Standard Data Refresh** - With the standard option, cached data is available for pivot tables and charts only. To better understand the default caching process and how to see real-time data in your pivot tables and charts, see [Standard Data Refresh](#).

For information about the differences between the optimized data refresh and the standard data refresh, see [Comparison of Standard Data Refresh and Optimized Data Refresh](#).

Comparison of Standard Data Refresh and Optimized Data Refresh

The following tables provides a comparison of the data refresh options available for datasets and workbook visualizations when using the standard data refresh and the optimized data refresh.

Datasets and Workbook Visualizations	Standard Data Refresh		Optimized Data Refresh	
	Cached Data	Real-time Data	Cached Data	Real-time Data
-	Cached Data in Datasets feature disabled (default option)		Cached Data in Datasets feature enabled	
-	Cached Data	Real-time Data	Cached Data	Real-time Data
Datasets and Table Views	Not available.	Always Available.	Available when set to Cached Response mode. Refresh frequency: adjusted to usage	Available when set to Real-time Response mode.

Datasets and Workbook Visualizations	Standard Data Refresh		Optimized Data Refresh	
Pivot Tables and Charts	Always available. Refresh frequency: after 60 minutes of inactivity. Note: Some changes in pivot tables and charts require loading real-time data. For more information, see Understanding Data Caching in Pivot Tables and Charts.	Available when clicking the refresh icon. In some scenarios, you need to clear the cached data first to see real-time data.	Available when set to Cached Response mode. Refresh frequency: after 60 minutes of inactivity. Note: Some changes in pivot tables and charts require loading real-time data. For more information, see Understanding Data Caching in Pivot Tables and Charts.	Available when set to Real-time Response mode.

For a better understanding of the two data refresh options, see the following topics:

- [Standard Data Refresh](#)
- [Optimized Data Refresh](#)

Standard Data Refresh

With standard data refresh, pivot tables and charts show cached data from storage for quicker loading. Datasets and table views don't use cached data and show only real-time data, which may take longer to load.

Note: You are using standard data refresh when the Cached Data in Datasets isn't enabled. For more information about this feature, see [Optimized Data Refresh](#).

This caching process speeds things up by returning results faster, but the data might not always be the most current. To see when your workbook data was last refreshed, check the **Last updated** field in the upper-right corner of the Viewer. To learn more about default caching and manual cache clearing, see the topics below:

- [Clearing the Cache for Pivot Tables and Charts](#)
- [Understanding Data Caching in Pivot Tables and Charts](#)

Understanding Data Caching in Pivot Tables and Charts

By default, the data is refreshed after 60 minutes of inactivity. To better understand when the data is refreshed, see the following scenarios:

- [Creating and modifying pivot tables and charts in an existing workbook](#)
- [Saving and re-opening a workbook before 60 minutes have elapsed](#)
- [Saving and re-opening a workbook before 60 minutes have elapsed](#)
- [Working with the Analytics Portlet](#)

Creating and modifying pivot tables and charts in an existing workbook

After you set the layout of a new pivot table or chart, click the **Refresh** icon  to apply the changes and update the results. If you make changes to a pivot table directly from the Pivot Table Viewer, the results are automatically displayed.

When you create or modify your pivot tables and charts, SuiteAnalytics Workbook creates a backend query. This query pulls in results for your pivot tables and charts, letting you see the data in the Viewer. If some pivot tables and charts share the same backend query, the data used during refresh can vary:

- If the backend query was refreshed less than 60 minutes ago from either the pivot or chart tab, the data from the cache is used for the refresh.
- If the backend query was refreshed more than 60 minutes ago from either the pivot or chart tab, the most current data is used for the refresh.

Note that some changes in pivot tables and charts require loading real-time data. For example, adding a field to **Rows**, **Columns**, or **Measures**.

You can verify the time of the last refresh in the upper-right corner of the pivot table or chart viewer.

Saving and re-opening a workbook before 60 minutes have elapsed

When you re-open a saved workbook before 60 minutes have elapsed and open a pivot table or chart, the Viewer shows cached data by default. The last refresh time is in the upper-right corner. To retrieve the most current data, click the **Refresh** icon.

Saving and re-opening a workbook after 60 minutes have elapsed

When you re-open a saved workbook after 60 minutes have elapsed, the cache has expired. When you open a pivot table or chart, the Viewer automatically shows the latest results.

Working with the Analytics Portlet

Pivot tables and charts in the Analytics portlet show the same data as their workbooks. To get the latest results, click the **Refresh** icon .

For more information about how to clear the cache manually, see [Clearing the Cache for Pivot Tables and Charts](#).

Clearing the Cache for Pivot Tables and Charts

The **Clear cache** option removes cached data from workbook pivot tables and charts so you can get the most current results. You can verify when the data was last updated by looking at the date and time in the upper-right corner of the Viewer. Here's how you can clear the cache for your open pivot tables and charts.

To clear the cache for pivot tables and charts:

1. On the Pivot or Chart tab, click the menu icon  in the upper right corner of the Viewer.
A popup message appears asking if you want to clear the cache.
2. Choose how you want to proceed:
 - **Clear cache** - Clears the cache for the entire workbook. The time and date in the upper-right corner of the Viewer is updated to reflect when the data was updated.
 - **Cancel** - The cache is not cleared. Data is updated following the default caching process.

3. If you clear the cache, a popup message appears asking you to refresh all pivot tables and charts and an icon is displayed next to each pivot table and chart in the workbook. There are three available icons:
 - **Warning icon**  - The displayed results are not up to date according to the date and time shown in the **Last updated** field. To retrieve the most current results, click the **Refresh** icon .
 - **Loading icon**  - The data is currently being updated according to the date and time shown in the **Last updated** field. After the refresh is completed, the icon changes to either the pivot or chart icon.
 - **Pivot and Chart icons**   - The data displayed is up to date according to the date and time shown in the **Last updated** field.
4. To retrieve the most current results in each open pivot table and chart, click the **Refresh** icon  in each tab.

For more information about the Pivot table and Chart tabs, see [Pivot Tab](#) and [Chart Tab](#).

To better understand the caching process of SuiteAnalytics Workbook, see [Understanding Data Caching in Pivot Tables and Charts](#).

Optimized Data Refresh

By default, only pivot tables and charts use cached data. If you want datasets and table views to load faster using cached data too, turn on the Cached Data in Datasets feature. After it's enabled, you can choose between two options:

- **Real-time Response** - Datasets, workbooks, and Analytics portlets always show the latest data. Results might take longer to load with this option.
- **Cached Response** - Datasets, workbooks, and Analytics portlets use cached data for quicker loading. The refresh rate adapts to your usage. Cached Response monitors when you view datasets, workbooks, or portlets and schedules data refreshes accordingly, refreshes can happen anywhere from every hour to every 24 hours, depending on your account usage.

For information about how to enable the Cached Data in Datasets feature, see [Enabling the Cached Data in Datasets Feature](#).

For information about how to work with cached data or real-time data in SuiteAnalytics Workbook, see the following topics:

- [Using Cached Data in SuiteAnalytics Workbook](#)
- [Using Real-Time Data in SuiteAnalytics Workbook](#)

Enabling the Cached Data in Datasets Feature

To begin using cached data in datasets and table views, you must first enable the feature.

To enable Cached Data in Datasets in your account:

1. Go to Setup > Company > Setup Tasks > Enable Features.
2. Click the **Analytics** subtab.
3. Check the **Cached Data in Datasets** box.
4. Click **Save**.

After you have enabled the Cached Data in Datasets feature, you can choose to use Real-time Response or Cached Response mode in your datasets and workbooks. For information about the two options, see the following topics:

- [Using Cached Data in SuiteAnalytics Workbook](#)
- [Using Real-Time Data in SuiteAnalytics Workbook](#)

Using Cached Data in SuiteAnalytics Workbook

You can use Cached Response mode to get faster results in your datasets and workbooks. To turn on Cached Response mode or learn how it works and how to monitor cached data, see the topics below:

- [Enabling Cached Response Mode](#)
- [Cached Data in Connected Datasets and Workbooks](#)
- [Working with Cached Response Mode](#)
- [Monitoring Cached Data for Datasets](#)
- [Known Limitations in Cached Response Mode](#)

Enabling Cached Response Mode

To use cached data, set the data refresh in your dataset or workbook to Cached Response mode. You can only do this for datasets and workbooks you can access. Make sure the Cached Data in Datasets feature is enabled in your account first. For more information, see [Enabling the Cached Data in Datasets Feature](#).

To enable Cached Response mode:

1. Open your dataset or workbook.

 **Note:** If you create a new workbook or dataset, you must save your dataset to change the data refresh to Cached Response mode.

2. On the upper-left corner, click the arrow next to **Real-time Response**.
3. In the Response Mode window, click **Cached**.
4. Click **Apply**.

The Cached Response mode is now enabled.

Cached Data in Connected Datasets and Workbooks

When you enable Cached Response mode in your datasets or workbooks, the mode also changes for the connected datasets and workbooks in some scenarios. For more information, see [Data Refresh Changes in Connected Datasets and Workbooks](#).

Working with Cached Response Mode

When you set data refresh to Cached Response, your datasets and workbooks load from stored data whenever you open them or make changes. To see when your data was last updated, check the time in the upper-left corner of the Viewer, next to **Cached Response**.

 **Note:** Some dataset changes require real-time data, even if Cached Response mode is on. Cached data isn't available if you add or remove fields from your dataset, modify its criteria filters, or change a formula used in your dataset. Save your dataset changes to start using cached data again.

When you work with cached data, you can get the most current data at any time. To refresh cached data and see the most up-to-date results, click the Refresh icon.

The frequency of cached data refresh can be different for datasets and workbook visualizations:

Frequency to refresh cached data in datasets

The refresh frequency for cached data in your datasets adapts to your usage. Cached Response tracks when you view datasets, workbooks, or portlets, then schedules refreshes so the data's updated when you need it. Refreshing can happen as often as every hour or as rarely as every 24 hours, depending on your usage.

Frequency to refresh cached data in pivot tables and charts

Cached data in your pivot tables and charts is refreshed after 60 minutes of inactivity, and after a standard data refresh. For detailed information about cached data in pivot tables and charts, see [Understanding Data Caching in Pivot Tables and Charts](#).

Monitoring Cached Data for Datasets

To see an overview of all datasets using Cached Response mode, admins can go to the Dataset Cache Management page. The Dataset Cache Management page lists all datasets using Cached Response mode, affected workbooks, workbook owners, status, and more. You can turn off Cached Response mode for any dataset on this page to switch it to Real-time Response mode. Only admins can access the Dataset Cache Management page.

To access the Dataset Cache Management Page

1. On the Analytics Home page, click the **Datasets** subtab.
2. In the Actions column, click the **More** icon.
3. Click **Dataset Cache Management**.

 **Note:** This option doesn't turn off Cached Data in Datasets, it only switches data refresh to Real-time Response mode. Workbook and dataset owners can still change the mode by opening the workbook or dataset and setting it back to Cached Response. For more information, see [Enabling Cached Response Mode](#).

Known Limitations in Cached Response Mode

Character large object (CLOB) values don't work in cached response mode. If you use CLOB fields with cached response, you'll get a message saying cached data can't be enabled. To keep working, check if you're using CLOB fields, then do one of the following:

- Remove CLOB fields – If you remove the fields from your dataset and save it, you can work with cached response mode.

 **Note:** The cached response mode is turned on considering the latest saved changes of the dataset. The response mode cannot be turned on if the dataset contained the CLOB field when was last saved. To work with cached response mode, ensure that you remove the CLOB fields from your dataset and then save it.

- Keep CLOB fields – If you keep the fields, you can work with real-time response mode only.

Identifying CLOB fields

There is no way to identify CLOB fields directly in your datasets or workbooks. However, you can use the Records Catalog to check field types.

Note: The Records Catalog is available for all users that have the Records Catalog permission assigned to their role. For more information, see the help topic [Records Catalog Overview](#).

To identify CLOB fields:

1. Go to Setup > Records Catalog.
2. Use the search field to find your record.
3. Click **SuiteScript and REST Query API**.
4. View the **Type** column on the right. All fields that appear with the **CLOBTEXT** type are CLOB fields.

For more information about the details available in the Records Catalog, see the help topic [Records Catalog Fields Available for SuiteScript and Rest Query API](#).

Using Real-Time Data in SuiteAnalytics Workbook

By default, all datasets, workbooks, and Analytics portlets use real-time data. This means that the most current data is displayed, but loading results may take longer. If you see cached data and want real-time data instead, set data refresh to Real-time Response mode.

Enabling Real-Time Response Mode

To enable Real-time Response mode:

1. Open your workbook or dataset.
2. On the upper-left corner, click the arrow next to **Cached Response**.
3. In the Response Mode window, click **Real-Time**.
4. Click **Apply**.

The Real-time Response mode is now enabled. When you enable Real-time Response mode in your datasets or workbooks, the mode also changes for the connected datasets and workbooks in some scenarios. For more information, see [Data Refresh Changes in Connected Datasets and Workbooks](#).

Users with the Administrator role can also change the Cached Response mode to Real-time Response mode for all datasets. For more information, see [Monitoring Cached Data for Datasets](#).

Working with Real-Time Response Mode

With Real-time Response mode, your datasets and workbooks always show the latest data. Each time you open or change them, you get the most current info, which might take longer to load. If you want to optimize resources, use cached data with your datasets and workbooks. For more information, see [Using Cached Data in SuiteAnalytics Workbook](#).

Data Refresh Changes in Connected Datasets and Workbooks

You can change the response mode to Cached Response or Real-time Response in several ways:

Changing the response mode from a dataset

When you change the response mode from the dataset, the mode changes for the workbook if your workbook uses one dataset only. If your workbook uses more than one dataset, you must change the

response mode in all linked datasets first. See the following example with the same workbooks and datasets:

- Example a: **Workbook 1a** uses **Dataset 1a**.
- Example b: **Workbook 1b** uses **Dataset 1b** and **Dataset 2b**.

In the first example, when you enable the mode from **Dataset 1a**, the mode for **Workbook 1a** is also enabled. However, when you enable the mode from **Dataset 1b**, the mode for **Workbook 1b** does not change. You must enable the mode in both datasets first, **Dataset 1b** and **Dataset 2b**. Then, the mode is also enabled for **Workbook 1b**.

For more information about how to work with Cached Response mode, see [Working with Cached Response Mode](#).

Changing the response mode from a workbook

When you change the response mode from your workbook, the mode is also enabled for all linked datasets that your workbook uses. See the following examples:

- Example a: **Workbook 1a** uses **Dataset 1a**.
- Example b: **Workbook 1b** uses **Dataset 1b** and **Dataset 2b**.

In the first example, when you enable the mode from **Workbook 1a**, the mode is also enabled for **Dataset 1a**. In the second example, when you enable the mode for **Workbook 1b**, the mode also changes for both datasets, **Dataset 1b** and **Dataset 2b**.

Accessing and Sharing Workbooks and Datasets



Important: Workbooks made before 2020.1 exist as separate workbook and dataset objects. If you shared a workbook before 2020.1, users have access to both. But since dataset changes update all linked workbooks, recipients can't edit a shared dataset unless they save their own copy. For more information, see [Custom Workbooks and Datasets](#).

When you first click the Analytics tab, you'll see all the workbooks and datasets you can access on the Analytics Home page. These include ones you own and those shared with you. To preview a workbook or dataset, click the Details link. If you have Analytics Administrator permission, you can also open workbooks and datasets other users created, as well as access or share built-in templates. For more information about the Analytics Administrator permission, see [The Analytics Administrator Permission](#).

The record types and fields you see in each workbook or dataset depend on which features are enabled and the permissions for your NetSuite role. For example, you need the Invoice permission to see or use the invoice record type. Some SuiteAnalytics Workbook actions depend on your NetSuite permissions. For example, to export a dataset to CSV, you need the Exports Lists permission at Create level or higher.

For more information about specific record type and field access in Workbook, download the following worksheet: [NetSuitePermissionsUsage.xls](#). If you do not see a specific record type or field that you need to access, contact your system administrator. Also, keep in mind that you can only access certain workbooks and datasets such as the **Sales (Invoiced)** workbook, if they are shared with you by other users in your account. For more information, see [Saving and Sharing Workbooks and Datasets](#).

Saving and Sharing Workbooks and Datasets

You can share workbooks and datasets with individual users or groups based on roles. To avoid data inconsistencies, shared workbook recipients can't edit the underlying datasets. For example, if you

share a workbook with users in the Accountant role, everyone in that role can access and save their own versions, but they can't edit the original datasets. If a workbook has multiple datasets, you need to save each before saving and sharing the workbook, including datasets you create during authoring but don't connect to a visualization. If you don't want to share datasets that aren't linked to a workbook visualization, delete them before saving and sharing.

Anyone with access to Workbook can share a workbook or dataset, however recipients can only view the record types and fields that they have access to based on their permissions. Record types and fields that a recipient does not have access to are simply not displayed. If you have the Analytics Administrator permission, you can also share workbooks and datasets with every user in your account using the **Share with all** option. Keep in mind however, if you use **Share with all** in a workbook, the original author won't be able to connect additional datasets to it. Make sure you contact the original owner of a workbook before you share it with everyone in your account.

Shared workbooks include all visualizations, including your Table View, Pivot, and Chart tab selections. If a visualization uses linked datasets, the link definition is shared too. You can't share a set of linked datasets without sharing the associated workbook. Shared datasets include all the joined record types, fields, and criteria filters that define them. If you have the export lists permission and only want to share a dataset, click Export in the dataset to download a CSV file of the Data Grid.

To save and share a workbook or dataset:

1. Click **Share** from anywhere within the workbook or dataset.
2. Select the roles or users you want to share the workbook or dataset with, then click the right arrow to add them to the distribution list. If you have the Analytics Administrator permission, check **Share with All** to share the object with everyone in your account.



Important: If you use the **Share with all** option in a workbook, the original author will not be able to connect additional datasets to it.

3. Click **Save and share**.

The object is shared and appears on the Analytics Home page for the selected recipients. Currently, notifications are not sent when a workbook or dataset is shared with a user.



Important: You cannot save and share a workbook if it has visualizations based on unsaved datasets. Before you begin, save or delete any datasets connected to the workbook. Remove these fields before you attempt to save and share.



Note: If you have the Analytics Administrator permission and you want to share workbook or dataset templates such as Open Sales Orders Lines or Number of Fulfillments, you must follow the steps in this procedure. These templates cannot be shared from the Analytics Home page.

Known Limitations in SuiteAnalytics Workbook

The following features and interactions in SuiteAnalytics Workbook have known limitations that are currently being addressed by our development teams:

- To mitigate the possibilities of a CSV injection, the following formatting is applied to all dataset and workbook results that are exported to CSV:
 - All text values are enclosed in double quotation marks
 - Values that begin with the characters -, +, =, @, tab, and EOL (end of line) have single apostrophes added in front of the value

This formatting also applies to workbook and dataset results that you export to CSV through SuiteScript and SuiteQL.
- The audit trail and execution log record types only contain data from 2019.2 and earlier. For more information, see [Auditing Workbooks and Datasets](#).
- On the Pivot tab, you cannot apply measure-based filters to fields that contain date or text values when the aggregation types for those fields are set to Min or Max.

For more information about measure-based filters, see [Workbook Visualization Filters](#).
- Some ad blocker plug-ins can prevent you from creating pivot tables or charts. To avoid errors with the Pivot or Chart tabs, deactivate these plug-ins and mark NetSuite as an exception. For more information about NetSuite browser support, see the help topic [Supported Browsers for NetSuite](#).
- If you define a criteria filter that is invalidated based on changes to the field used in the filter condition, you are currently unable to edit the filter.
- Users assigned to the employee center or partner center roles who have permission to access Workbook might not see the Analytics option in the NetSuite navigation menu.
- Filters that you create on the Dataset Builder based on custom duration values currently do not work as expected.
- When you query the analytics data source using SuiteScript, you currently cannot filter data using record types such as invoice.
- If you add multiple pivot tables or charts in a workbook, the user interface does not always display arrows so that you can switch between each table and chart.
- Certain ad blocking software can prevent data from loading in your workbook pivot tables.
- The search field on the Dataset Builder searches across record types that are a few joins away from the root record type. To find record types or fields that are further from the root record type, click **Show more results**.
- If you have multiple languages set up in your account, edits to workbook visualization texts are added as their own collections and not added to the existing workbook collection.
- Collection and string IDs on the Manage Translation page are not editable.
- You currently cannot manually disconnect a dataset from a workbook after it has been connected. For more information, see the [SuiteAnalytics Workbook FAQs](#).

- You currently cannot create Table Views based on linked datasets.
- If there are multiple fields in a dataset with the same ID but different join paths, using them to create a dataset link can cause issues. For more information, see [Link Datasets in a Workbook](#).
- Character large object (CLOB) data is not available in cached response mode. For more information, see [Known Limitations in Cached Response Mode](#).
- When you sort your results based on a field that contains CLOB values, Workbook currently only evaluates the first 250 characters for each result. Results with more than 250 characters may not be sorted properly.

Analytics Data Source Overview

Data Model Basics

The analytics data source is a collection of NetSuite data that is grouped according to record types and fields. The structure of the analytics data source enhances the capabilities of analyzing your NetSuite data which you can query using SuiteQL. This advanced query language lets you run complex SQL queries.

The analytics data source has been used in SuiteAnalytics Workbook since 2019.1. You can also retrieve data from the analytics data source using SuiteAnalytics Connect, SuiteScript, and SuiteTalk REST web services.

To help you understand the analytics data source, see the following topics:

- You can use the Records Catalog to see the fields and joins available for each record type. The Records Catalog displays also information such as the cardinality, field level help, and so on. The Records Catalog only shows record types you can access in NetSuite, based on your role. The Records Catalog is supported for the SuiteScript Analytic API and the REST Query API only. Therefore, the record types available in the Records Catalog and the dataset user interface may differ slightly. For more information, see the help topic [SuiteScript 2.x Analytic APIs](#).

 **Note:** The Records Catalog is available for all users that have the Records Catalog permission assigned to their role. For more information, see the help topic [Records Catalog Overview](#).

- You can use SuiteQL to run queries against the analytics data source through SuiteAnalytics Connect, SuiteScript, and SuiteTalk REST web services. For more information, see [SuiteQL](#).
- In Workbook, you can further customize the record types and fields available in the analytics data source for use in your workbooks. For more information, see [The Analytics Data Source and SuiteAnalytics Workbook](#).
- Some record types have been created specifically for Workbook and exist only in the analytics data source. For more information, see [Analytical Record Types](#).
- To locate fields in the analytics data source using the Workbook user interface, you can browse and search the Records and Fields list on the Dataset tab. For more information, see [Locating Fields in Workbook](#).

The Analytics Data Source and SuiteAnalytics Workbook

For every NetSuite account, all data is stored using a single database. The data sources used to expose this data through Saved Searches and Reports however, are slightly unique. For some record types, these unique data sources have resulted in inconsistent field naming and data exposure between the two tools.

In SuiteAnalytics Workbook, fields and record types are exposed using a new analytics data source that is designed to display consistent data across the Workbook application. Fields created in support of new NetSuite features are exposed to Workbook using the analytics data source, as of 2019.1.

The location, names, and IDs of some fields and record types might be different in the analytics data source. Also, some record types and fields have not been ported to the analytics data source. This includes fields that contain calculated values for certain record types. Consequently, you might need to use different record types and fields, or create new joins and custom formula fields to recreate your existing saved searches using Workbook.

To help you learn about the analytics data source, see the following topics:

- You can use the Records Catalog to see the fields and joins available for each record type. The Records Catalog displays also information such as the cardinality, field level help, and so on. Due to Records Catalog's reliance on the NetSuite role-based permissions, you can see only the record types for the data that you can access in the NetSuite user interface. The Records Catalog is supported for the SuiteScript Analytic API and the REST Query API only. Therefore, the record types available in the Records Catalog and the dataset user interface may differ slightly. For more information, see the help topic [SuiteScript 2.x Analytic APIs](#).



Note: The Records Catalog is available for all users that have the Records Catalog permission assigned to their role. For more information, see the help topic [Records Catalog Overview](#).

- The analytics data source supports hundreds of record types and fields. For more information about the record types and fields that you can find in Workbook, see [Available Record Types](#).
- The location and names of some fields in the analytics data source might be different from their Search and Report counterparts. Also, some fields such as fields with calculated values have not yet been ported to the analytics data source. For more information about differences in the new data source, see [Analytics Data Source Changes](#).
- After you select a root record type for a custom dataset or open a saved or shared dataset, all related record types that you have access to are displayed in the Records list. For more information about using the Dataset Builder to select and join related record types, see [Defining a Dataset](#).
- Workbook supports multilevel joins, however there are certain functions and interactions that you should be cautious of when adding specific record types or fields to a dataset. For more information, see [Guidelines for Joining Record Types in SuiteAnalytics Workbook](#).
- You can link datasets in a workbook even if they are based on record types that do have a predefined relationship in the analytics data source. For more information, see <Linking Datasets in a Workbook>.

Available Record Types

SuiteAnalytics Workbook supports hundreds of record types available through the analytics data source. The analytics data source includes most of the record types used throughout NetSuite.

You can use the Records Catalog to see the fields and joins available for each record type. The Records Catalog displays also information such as the cardinality, field level help, and so on. Due to Records Catalog's reliance on the NetSuite role-based permissions, you can see only the record types for the data that you can access in the NetSuite user interface. The Records Catalog is supported for the SuiteScript Analytic API and the REST Query API only. Therefore, the record types available in the Records Catalog and the dataset user interface may differ slightly. For more information, see the help topic [SuiteScript 2.x Analytic APIs](#).



Note: The Records Catalog is available for all users that have the Records Catalog permission assigned to their role. For more information, see the help topic [Records Catalog Overview](#).

To view each record type that you have access to in your account, you can also click New Dataset on the Analytics Home page. The record category and ID for each record type is displayed. Record types are divided into three categories:

- Standard – Record types that are included in each account of your NetSuite implementation and are available through all NetSuite data sources. For example, the transaction record type. Standard record types consist of concrete and generic record types. For more information, see [Analytics Data Source Changes](#).

- **Analytical** – Record types that have been created specifically for use with Workbook. These record types are based on standard record types but contain a unique set of fields. For more information, see [Analytical Record Types](#).
- **Custom** – Record types created by users in your company that are unique to your NetSuite account.

Access to record types and fields in Workbook depends on the same permissions used for NetSuite saved searches. For example, to access sales order data through the transaction record type using Workbook, you must have the Sales Order permission assigned to your role. Analytical record types such as the sales (invoiced) record type are only accessible to users with the Analytics Administrator permission. Also, some record types are only available in Workbook if you have specific features enabled in your account. For information about permissions—related access to record types and fields in the analytics data source, download the following worksheet: [NetSuitePermissionsUsage.xls](#)

After you select a root record type for a dataset or open an existing dataset, the Records list on the Dataset Builder shows all the related record types that you have access to. If you can't find a record type that you think you should have access to, contact your administrator.



Note: The search field in the Dataset Builder searches across record types that are a few joins away from the root record type. To search across record types that are further away, click **Show more results**.

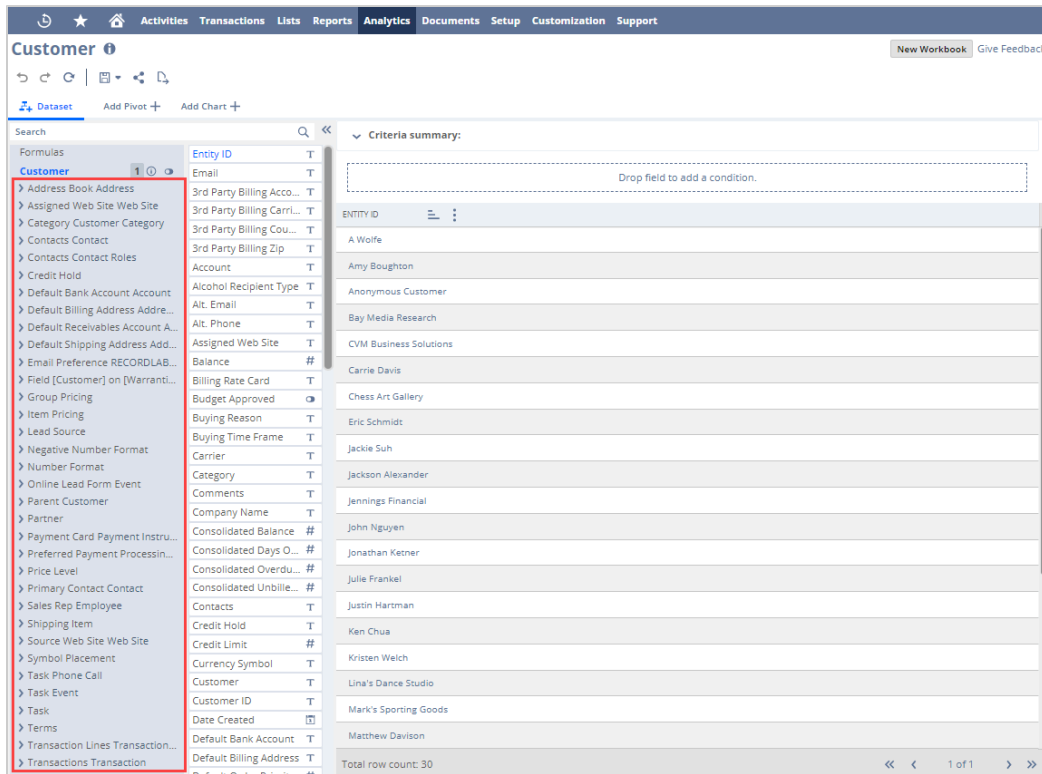
Analytics Data Source Changes

The analytics data source is different than the data sources used by the NetSuite Saved Search and Report applications. In the analytics data source, concrete record types that have a corresponding entry form in NetSuite remain mostly unchanged and use the same field labels. For example, the employee record type in SuiteAnalytics Workbook contains mostly all the of fields that are on the NetSuite employee form, with the same field labels that are used in saved searches and reports. Generic record types that don't have a single corresponding form in NetSuite, such as the transaction, entity, and item record types, use generic field labels. This is because generic record types are representative of multiple concrete record types. Also, some fields in the analytics data source are accessible through different record types than in saved searches and reports.

The layout of the analytics data source means that the names of some record types and fields have changed or appear differently than on NetSuite forms, saved searches, and reports. Some fields, such as those with calculated values, have also not been ported to the data source and other fields are only accessible by joining specific record types. Furthermore, the analytics data source includes some record types and fields that have been created specifically for SuiteAnalytics Workbook and aren't available through saved searches and reports.

To help you learn about the analytics data source, the following resources are available in the Workbook user interface and the NetSuite Help Center:

- To view the record types and fields that you have access to, use the Records list in the Dataset Builder. By default, datasets display all joinable record types and fields based on the root record type selected and the permissions assigned to the role you use to log in to NetSuite. For example, if you select the customer record type for your dataset, the following joinable record types are available:



If you click a record type, additional related record types are displayed.

- To recreate your existing saved searches using Workbook, the changes to the analytics data source mean that you might have to use different record types and fields, create custom formula fields, or join multiple record types. Click this link to access a Microsoft Excel worksheet that compares the transaction record type used by the saved search data source to the transaction record type used in Workbook: [TransactionMapping.xlsx](#)
- For annotated steps on how to recreate some of your saved searches using Workbook, see the following topics. Each sample includes the fields, record types, and formulas required to recreate the search:
 - [Open Sales Orders Lines](#)
 - [Number of Fulfillments](#)
 - [Journal Entry to Approve](#)

Open Sales Orders Lines

This saved search displays the items allocated to open sales orders in your account, so that you can better monitor your inventory.

To perform this search using the Saved Search application, all required criteria and results fields are available through the transaction record type. To recreate this search using SuiteAnalytics Workbook however, some required fields are only available through the transaction line and transaction accounting line record types. You must therefore join these record types in your dataset to recreate the search. Also, some required fields might have different labels in Workbook. The following table lists key differences in fields between the saved search and the workbook:

Key Field Differences in Workbook

Field Name in Saved Search	Record Type in Workbook	Required Join in Workbook	Field Name in Workbook
Main Line	Transaction Line	Transaction > Transaction Line	Main Line
Tax Line	Transaction Line	Transaction > Transaction Line	Tax Line
Closed	Transaction Line	Transaction > Transaction Line	Closed
Account Type	Account	Transaction > Transaction Line > Transaction Accounting Line > Account	Type
Shipping Line	Transaction Account Line	Transaction > Transaction Line > Transaction Accounting Line	Shipping Cost Item Type



Warning: In this workbook you must add fields from the transaction line and transaction accounting line record types. When you add fields from either the transaction line or transaction accounting line record type to a transaction dataset, data duplication can occur. For more information, see [Joining Transaction Line and Transaction Accounting Line in a Dataset](#).

To recreate this search using Workbook:

Create a new dataset with transaction as the root record type. Then, define the dataset as follows:



Important: Users assigned to the Sales role don't have access to the account record type in Workbook, which is required to properly recreate this saved search. If you create and share this workbook with users assigned to the Sales role, make sure you give them access to the account record type first.

Root Record Type	Joined Record Type(s)	Custom Formula Field(s)	Data Grid	Criteria
Transaction	Transaction Line > Transaction Accounting Line > Account	Create the following formula fields on the Dataset Builder:	Add the following fields to the grid from the	Set the following criteria in the Dataset Builder:

Important: Make sure you join the transaction accounting line record type from the transaction line record type. If you join both the transaction line and transaction accounting line record types directly from the transaction record, it can duplicate or otherwise skew your data. For more information, see Joining Transaction Line and Transaction Accounting Line in a Dataset.	- Name: Unfulfilled Items - Output Type: Float - Definition: <code>-(transactionlines.quantity)-NVL((transactionlines.quantitycommitted),0)-NVL((transactionlines.quantityshiprec),0)</code> - Use this formula to replace null values in the Transaction Line Quantity and Quantity committed fields with 0.00. - Name: Positive Item Quantity - Output Type: Float - Definition: <code>-{transactionlines.quantity}</code> - Use this formula to display your item quantities as positive values. - For more information about how to use formula fields in SuiteAnalytics Workbook, see Formula Fields.	transaction record type: - Transaction - Status - Posting Period - Entity Add the following fields to the grid from the transaction line record type: - Item - Location - Quantity - Committed - Quantity Shipped/Received/Picked Up Add the following custom formula fields to the grid: - Custom Formula Field 1 (Unfulfilled Items) - Custom Formula Field 2 (Positive Item Quantity)	[Custom Formula Field 2] greater than 0.00 - Type is Sales Order - Status any of Sales Order : Partially Fulfilled and Sales Order : Pending Fulfillment and Sales Order : Pending Billing/Partially Fulfilled - Transaction Line: Main Line is false - Transaction Line: Tax Line is false - Transaction Line: Item Type none of Shipping Cost Item - Account: Type is income Note: If the Multi-Book Accounting feature is enabled in your account, you can use the accounting book field in a criteria filter to view data for specific accounting books. To access the accounting book field, use the join path <code>transaction>transaction line>transaction accounting line</code>.
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Open Sales Orders Lines Pivot Tables

Create a new workbook using the open sales order lines dataset you created, and set the following fields to the appropriate dimensions for each pivot table:

- Open Sales Orders by Date
 - Rows:** Item
 - Columns:** Date (Quarter)
 - Values:** [Custom Formula Field 2] (Sum)
- Open Sales Orders by Location and Date
 - Rows:** Location, Item
 - Columns:** Date (Quarter)
 - Values:** [Custom Formula Field 2] (Sum)
- Open Sales Orders by Customer and Date
 - Rows:** Entity, Item
 - Columns:** Date (Quarter)
 - Values:** [Custom Formula Field 2] (Sum)

After you set each field to the appropriate table dimension, click the refresh  icon to generate the table.

You can also customize the table and add totals and grand totals, or rename each table. For more information, see [Pivot Table Customization](#).

Open Sales Orders Lines Charts

By recreating this saved search using Workbook, you can also chart your source data to visualize the information using different chart types.

Create the following charts by dragging the listed fields from the Dataset Panel to the corresponding section of the Layout panel, and select the chart type.

- Open Sales by Item
 - X-Axis:** Item
 - Series:** Not Applicable
 - Values:** Total Amount (Sum)
 - Chart type:** Column chart
- Open Sales by Date
 - X-Axis:** Date (Quarter)
 - Series:** Item
 - Values:** Total Amount (Sum)
 - Chart type:** Line chart

After you set each field and the chart type, click the Refresh  icon to generate the chart. You can also filter values, add a title and subtitle, and rename each axis. For more information, see [Workbook Charts](#).

Number of Fulfillments

 **Note:** Users with the Analytics Administrator permission can view a predefined version of this workbook from the Analytics Home Page. For more information, see [Workbook and Dataset Templates](#).

This saved search shows the number of items which have been fulfilled.

To perform this search using the Saved Search application, all required criteria and results fields are available through the transaction record type. To recreate this search using SuiteAnalytics Workbook however, some required fields are only available through the transaction line record type. You must therefore join this record type to your dataset to recreate the search. Also, some required fields might have different labels in Workbook. The following table lists key differences in fields between the saved search and the workbook:

Key Field Differences in Workbook

Field Name in Saved Search	Record Type in Workbook	Required Join in Workbook	Field Name in Workbook
Name	Transaction Line	Transaction > Transaction Line	Entity
Period	Transaction	—	Posting Period
Amount	Transaction Line	Transaction > Transaction Line	Amount (Net)

Field Name in Saved Search	Record Type in Workbook	Required Join in Workbook	Field Name in Workbook
Main Line	Transaction Line	Transaction > Transaction Line	Main Line

To recreate this search using Workbook:

Create a new dataset and select transaction as the root record type. Then, define the dataset as follows:

Root Record Type	Joined Record Type(s)	Custom Formula Field(s)	Data Grid	Criteria
Transaction	Transaction Line	—	Add the following fields to the grid from the transaction record type: - Transaction - Date - Due Date - Type - Total Amount (Transaction Currency) - Entity - Posting Period Add the following field to the grid from the transaction line record type: - Location	Set the following criteria on the Dataset Builder using fields from the transaction record type: Type is Item Fulfillment Set the following criteria using fields from the transaction line record type: Main Line — Yes

Number of Fulfillments Pivot Tables

Create a new workbook using the Number of Fulfillments dataset you created, and set the following fields to the appropriate dimensions for each pivot table:

■ Number of Fulfillments Per Customer

Rows: Entity

Columns: Date (Year)

Measures: Transaction (Count)

After you set each field to the appropriate table dimension, click the Refresh icon  to generate the table.

You can also customize the table and add totals and grand totals, or rename each table. For more information, see [Pivot Table Customization](#).

Number of Fulfillments Chart

By recreating this saved search using Workbook, you can chart your source data to visualize the information using different chart types.

Create the following chart by dragging the listed fields from the Dataset Panel to the Layout panel, and select the chart type.

■ Number of Fulfillments per Customer

X-Axis: Entity

Series: Date (Year)

Values: Transaction (Count)

Chart type: Column chart

After you set each field, click the Refresh  icon to generate the chart. You can also filter values, add a title and subtitle, and rename each axis. For more information, see [Workbook Charts](#).

Journal Entry to Approve



Important: This workbook is only useful if you've enabled the **Require Approvals on Journal Entries** permission in your NetSuite account. For more information, see the help topic [General Accounting Preferences](#).

This saved search shows the list of journal entries requiring approval. By recreating this saved search using SuiteAnalytics Workbook, you can use the drill down capability to open and approve each journal entry directly from the Data Grid.

To perform this search using the Saved Search application, all required criteria and results fields are available through the transaction record type. To recreate this search using Workbook however, some required fields are only available through the transaction line and transaction accounting line record types. You must therefore join these records in your dataset to recreate the search. Also, some required fields might have different labels in Workbook. The following table lists key differences in fields between the saved search and the workbook:

Key Field Differences in Workbook

Field Name in Saved Search	Record Type in Workbook	Required Join in Workbook	Field Name in Workbook
Period	Transaction	–	Posting Period
Tax Period	–	–	— This field has been removed from Workbook
Name	Transaction Line	Transaction > Transaction Line	Entity
Memo	Transaction Line	Transaction > Transaction Line	Memo
Created From	Transaction Line	Transaction > Transaction Line	Created From
Account	Transaction Accounting Line	Transaction > Transaction Line > Transaction Accounting Line	Account
Amount	Transaction Accounting Line	Transaction > Transaction Line > Transaction Accounting Line	Amount



Warning: In this workbook you must add fields from the transaction line and transaction accounting line record types. When you add fields from either the transaction line or transaction accounting line record type to a transaction dataset, data duplication can occur. For more information, see [Joining Transaction Line and Transaction Accounting Line in a Dataset](#).

To recreate this search using Workbook:

Create a new dataset and select transaction as the root record type. Then, define the dataset as follows:

Root Record Type	Joined Record Type(s)	Custom Formula Field(s)	Data Grid	Criteria
Transaction	Transaction Line > Transaction Accounting Line	—	Add the following fields to the grid from the transaction record type: - Transaction - Date - Posting Period - Created By Add the following fields from the Transaction Line record type: - Entity - Memo - Created From - Location Add the following fields from the Transaction Accounting Line record type: - Account - Amount	Set the following criteria on the Dataset Builder: Type is Journal Status is Journal: Pending Approval

Guidelines for Joining Record Types in SuiteAnalytics Workbook

The following video explains some basics for joining record types in SuiteAnalytics Workbook.

 [Joining Record Types in Workbook](#)

The following video describes how to join the transaction, transaction line, and transaction accounting line record types.

 [Transaction Data Model](#)

Workbook enables you to add fields from multiple record types to a single dataset. This includes record types that are more than one join away from the root record type of a dataset, enabling you to compile workbook source data from a more diverse set of record types.

By default, the Dataset Builder lists all joinable record types that you have access to. To complete a join, you add fields from the desired record type to the Data Grid or create a Criteria filter.

If joining record types is a new concept for you, there are several considerations you should make before you attempt to join different record types in a dataset. To best take advantage of this powerful new capability, see the following topics:

- The analytics data source contains a variety of record types with predefined relationships to one another. The type of relationship between record types can cause data duplication. For more information, see [Data Duplication Based on Record Joins](#).

- The transaction and transaction line record types share a one-to-many relationship which can result in duplication of your workbook source data. Joining the transaction accounting line record type to a transaction dataset can also cause data duplication. For more information about using the transaction record type in a dataset, see [Joining Transaction Line and Transaction Accounting Line in a Dataset](#).
- Based on the relationship between each record type that you add to a dataset, the order in which you join record types can also impact your source data. For more information, see [Join Order in SuiteAnalytics Workbook](#).

Data Duplication Based on Record Joins

The analytics data source contains different record types with predefined relationships to one another. In the SuiteAnalytics Workbook user interface, you can see which record types are related to one another in the Records list on the Dataset Builder.

The screenshot shows the SuiteAnalytics Workbook interface. The top navigation bar includes tabs for Activities, Transactions, Lists, Reports, Analytics, Documents, Setup, Customization, and Support. The main area displays a dataset named 'Transaction' with a criteria summary: 'Document Number/ID AND Entity'. The criteria are 'Document Number/ID is 1008' and 'Entity is Phillip Van Hook'. Below the criteria, a table of records is shown with columns: TRANSACTION, DATE, TOTAL AMOUNT (TRANSACTION CURRENCY), ENTITY, and DOCUMENT NUMBER/ID. The first record is 'Sales Order #1008' with a date of '2/13/2002', a total amount of '\$6,982.85', and an entity of 'Phillip Van Hook'. The bottom of the table shows summary statistics: 'Sum: CAD 6,982.85', 'Average: CAD 6,982.85', 'Count: 1', 'Min: CAD 6,982.85', and 'Max: CAD 6,982.85'.

Note: The record types listed on the Dataset Builder are based on the root record type of the dataset, the features enabled in your account, and the permissions assigned to the role you use to log in to NetSuite. If you don't see a record type that you think you should have access to, contact your NetSuite administrator.

There are two types of relationships that you should be cautious of when you join record types in a dataset: those with a one-to-many relationship with the source record type, and those with a many-to-many relationship with the source record type.

One-to-many relationships between record types are those in which the source record type can be associated to multiple iterations of the target record type. For example, consider the relationship between the transaction and transaction line record types. For each transaction record there can be multiple transaction line records. When records with a one-to-many relationship are joined in a dataset, the cardinality of the data from the source record type is duplicated for each instance of the target record type. This duplication can result in inaccurate data aggregations in your datasets and workbooks.

To better understand the implications of joining record types with a one-to-many relationship, consider the total amount field on the transaction record type.

In a transaction dataset, for each transaction record listed there's a single value for the total amount field.

Transaction

Criteria summary: Document Number/ID AND Entity Show details

Document Number/ID is 1008 AND Entity is Phillip Van Hook

TRANSACTION	DATE	TOTAL AMOUNT (TRANSACTION CURRENCY)	ENTITY	DOCUMENT NUMBER/ID
Sales Order #1008	2/13/2002	\$6,982.85	Phillip Van Hook	1008

Sum: CAD 6,982.85 Average: CAD 6,982.85 Count: 1 Min: CAD 6,982.85 Max: CAD 6,982.85

However, if you join the transaction line record type to the dataset by adding a transaction line field to the Data Grid, notice that the number of values listed for the total amount field is multiplied by the number of associated transaction line records.

Transaction

Criteria summary: Document Number/ID AND Entity Show details

Document Number/ID is 1008 AND Entity is Phillip Van Hook

TRANSACTION	DATE	TOTAL AMOUNT (TRANSACTION CURRENCY)	ENTITY	TRANSACTION LINE AMOUNT (NET) (TRANSACTION CURRENCY)	DOCUMENT NUMBER
Sales Order #1008	2/13/2002	\$6,982.85	Phillip Van Hook		1008
Sales Order #1008	2/13/2002	\$6,982.85	Phillip Van Hook		1008
Sales Order #1008	2/13/2002	\$6,982.85	Phillip Van Hook		1008
Sales Order #1008	2/13/2002	\$6,982.85	Phillip Van Hook		1008
Sales Order #1008	2/13/2002	\$6,982.85	Phillip Van Hook		1008
Sales Order #1008	2/13/2002	\$6,982.85	Phillip Van Hook		1008
Sales Order #1008	2/13/2002	\$6,982.85	Phillip Van Hook		1008
Sales Order #1008	2/13/2002	\$6,982.85	Phillip Van Hook		1008
Sales Order #1008	2/13/2002	\$6,982.85	Phillip Van Hook		1008
Sales Order #1008	2/13/2002	\$6,982.85	Phillip Van Hook		1008
Sales Order #1008	2/13/2002	\$6,982.85	Phillip Van Hook		1008

Sum: CAD 76,811.35 Average: CAD 6,982.85 Count: 11 Min: CAD 6,982.85 Max: CAD 6,982.85

The result is that aggregations based on the duplicated source record fields are inaccurate. This includes the summaries displayed at the bottom of the Data Grid, and the totals and grand totals that you set up in any pivot tables based on the dataset.

Transaction

Dataset: Pivot 1

Layout

Fields

Amount (Net) (Transacti... #

Date

Document Number/ID T

Entity

Item

Total Amount (Tran... #

Transaction T

Rows

Item

Columns

Total Amount (Transaction Currency) (Sum)

Measures

Total Amount (Transaction Cur...

Grand Total \$76,811.35

This data duplication can also occur when you create criteria filters using fields from record types that have a one-to-many relationship, even if you don't explicitly add fields from the joined record type to the Data Grid.

For example, assume you are working on a transaction dataset with only one transaction record that matches your selected criteria.

Transaction

Dataset: Add Pivot + Add Chart +

Criteria summary: Entity AND Document Number/ID Show details

Entity is Jackson Alexander

Document Number/ID is 1010

Drop field to add a condition.

TRANSACTION TOTAL AMOUNT (TRANSACTION CURRENCY) ENTITY DOCUMENT NUMBER/ID

Sales Order #1010 \$2,719.61 Jackson Alexander 1010

Total row count: 1

If you create an additional criteria filter based on the transaction line record type, the results are again multiplied based on the number of associated transaction line records.

The screenshot shows the SuiteAnalytics Workbook interface. On the left is a search sidebar with various record types. The main area displays a dataset with a criteria filter. The criteria filter is set to 'Entity AND Document Number/ID AND Transaction Line: Amount (Net) (Transaction Currency)'. The 'Transaction Line: Amount (Net) (Transaction Currency) (consolidated) not equal 0.00' condition is highlighted with a red box. The dataset table shows 12 rows of transaction data for 'Sales Order #1010'.

TRANSACTION	TOTAL AMOUNT (TRANSACTION CURRENCY)	ENTITY	DOCUMENT NUMBER/ID
Sales Order #1010	\$2,719.61	Jackson Alexander	1010
Sales Order #1010	\$2,719.61	Jackson Alexander	1010
Sales Order #1010	\$2,719.61	Jackson Alexander	1010
Sales Order #1010	\$2,719.61	Jackson Alexander	1010
Sales Order #1010	\$2,719.61	Jackson Alexander	1010
Sales Order #1010	\$2,719.61	Jackson Alexander	1010
Sales Order #1010	\$2,719.61	Jackson Alexander	1010
Sales Order #1010	\$2,719.61	Jackson Alexander	1010
Sales Order #1010	\$2,719.61	Jackson Alexander	1010
Sales Order #1010	\$2,719.61	Jackson Alexander	1010
Sales Order #1010	\$2,719.61	Jackson Alexander	1010
Sales Order #1010	\$2,719.61	Jackson Alexander	1010

In record types with a many-to-many relationship, the same duplication can occur. Many-to-many relationships between record types exist when multiple iterations of the source record type are associated to multiple iterations of the target record type. For example, in a multi-subsidiary account, each account record can be assigned to multiple subsidiaries and each subsidiary record can be associated with multiple accounts. Consequently, the account and subsidiary record types share a many-to-many relationship, and joining these record types in a dataset can cause data duplication.

You should always be cautious when you create joins between record types that have a one-to-many or many-to-many relationship. If you do perform these types of joins, be aware of the impact they'll have on your datasets and the aggregations throughout your workbooks.

Joining Transaction Line and Transaction Accounting Line in a Dataset

In the Saved Search application, by default a Transaction saved search could include both accounting and operational data. In SuiteAnalytics Workbook, to include accounting and operational data in a transaction workbook, you might need to join three record types in a single dataset: transaction, transaction line, and transaction accounting line. These record types exist in one-to-many and many-to-many relationships, which can cause data duplication. For more information, see [Data Duplication Based on Record Joins](#). There are additional implications you should be aware of when joining these record types in a dataset:

- If you need to join the transaction accounting line record type in a dataset, you must use the join path **Transaction > Transaction Line > Transaction Accounting Line**. This join path prevents increased data duplication which would occur if you used the join path **Transaction > Transaction Accounting Line**.
- If you have the Multi-Book Accounting feature enabled in your account and you join the transaction accounting line record type in a dataset, data duplication can be increased. This is because each transaction accounting line stores data for each accounting book in your NetSuite account. To limit this, consider creating a criteria filter using the accounting book field from the transaction accounting line record type, so that data from only one accounting book is included.
- The transaction accounting line record type stores monetary values in the base currency of the subsidiary assigned to the transaction. The amount fields in this record type can only be consolidated and are typically used for accounting purposes, such as calculating financial results for your entire company, including all subsidiaries.

The transaction and transaction line record types store monetary values in the currency of the transaction and can only be converted. These amount fields are typically used for operational purposes, such as analyzing total sales of a specific item. For more information about currency consolidation and conversion in Workbook, see [Currency in Datasets and Workbooks](#).

The following video demonstrates how data can be impacted by joining the transaction line and transaction accounting line record types in a dataset.



Join Order in SuiteAnalytics Workbook

As specified in the [Data Duplication Based on Record Joins](#) topic, data duplication can occur when you join record types with a one-to-many or many-to-many relationship in a dataset. We can also demonstrate this duplication by examining the order in which record types are added to a dataset.

Consider the relationship between the entity and transaction record types: for each transaction record, there can only be one associated entity or customer record. Each customer record however, can be associated with multiple transaction records. This means that summary information such as field counts can vary from dataset to dataset even if they contain the same fields and record types, depending on the order that the record types are joined in a dataset.

For example, assume you want to see a list of customers and their associated transactions. If you create a new dataset beginning with the entity record type, you can then join the transaction record type to view each associated transaction. If you select the entity field to view a count of the total number of distinct customers, it is displayed at the bottom of the Data Grid.

The screenshot shows the SuiteAnalytics Workbook interface. On the left, the 'Entity' record type is selected. The 'Criteria summary' section shows a join between 'Entity ID' and 'Transaction ID'. The 'Data Grid' displays a list of transactions with columns for 'Entity ID', 'Type', 'Name', and 'Transactions: Transaction'. The bottom of the grid shows a count of 229 and a distinct count of 54.

Entity ID	Type	Name	Transactions: Transaction
-Accountant-	Vendor	-Accountant-	
-No Company-	Company	-No Company-	
-System-	Internal	-System-	
A Wolfe	Customer, Employee	A Wolfe	
Alice Stein	Contact	Stein Investments - Alice	
Allister Sullivan	Contact	Sullivan Distributors, Inc.	
Amy Boughton	Customer	Amy Boughton	Item Fulfillment #1001
Amy Boughton	Customer	Amy Boughton	Sales Order #1001
Amy Boughton	Customer	Amy Boughton	Cash Sale #1001
Amy Nguyen	Employee	Amy Nguyen	
Angela A Hitchcock	Employee	Angela A Hitchcock	
Anita Jennings	Contact	Jennings Financial - Anita	
Anonymous Customer	Customer	Anonymous Customer	
BC Electric	Vendor	BC Electric	Bill Payment #To Print
BC Electric	Vendor	BC Electric	Bill 3/10/2002
BC Electric	Vendor	BC Electric	Bill 1/10/2002
BC Electric	Vendor	BC Electric	Bill 2/10/2002

Count: 229 Count (Distinct): 54

If you create your dataset beginning with the transaction record type and join the entity record type however, notice that the number of distinct customers decreases.

The screenshot shows the SuiteAnalytics Workbook interface. On the left, there's a sidebar with a search bar and a list of fields categorized under 'Transaction' and 'Entity'. The 'Entity' category is expanded, showing fields like 'Address Book Address', 'Contact', 'Customer', 'Employee', 'Entity Parent', 'Entity Top Level Parent', 'Group', 'Job', 'Other Name', 'Partner', 'Task Phone Call', 'Task Event', 'Task', 'Transaction Lines Transac...', 'Transactions Transaction', 'Vendor', 'Entity Last Modified By', 'Fulfillment Type', and 'Incoterm'. The main area displays a table of transactions. The columns are: TRANSACTION, TYPE, TOTAL AMOUNT (TRANSACTION CURRENCY), ENTITY, and ENTITY-CUSTOMER. The 'ENTITY-CUSTOMER' column is highlighted with a red box. A red arrow points from this column to the 'Criteria summary' section at the top, which has a placeholder text 'Drop field to add a condition.' Below the table, there's a summary row showing 'Count: 56' and 'Count (Distinct): 20'.

TRANSACTION	TYPE	TOTAL AMOUNT (TRANSACTION CURRENCY)	ENTITY	ENTITY-CUSTOMER
Cash Refund #1001	Cash Refund	-445.95	Sandra Burns	Sandra Burns
Cash Sale #1001	Cash Sale	\$799.25	Amy Boughton	Amy Boughton
Cash Sale #1002	Cash Sale	\$2,910.55	Travis Gilbert	Travis Gilbert
Cash Sale #1003	Cash Sale	\$96.43	Eric Schmidt	Eric Schmidt
Cash Sale #1004	Cash Sale	\$1,009.53	Tony Matsuda	Tony Matsuda
Cash Sale #1005	Cash Sale	\$1,930.73	Jackie Suh	Jackie Suh
Cash Sale #1006	Cash Sale	\$6,982.85	Phillip Van Hook	Phillip Van Hook
Cash Sale #1007	Cash Sale	\$302.33	Tim Griffin	Tim Griffin
Cash Sale #1008	Cash Sale	\$2,879.25	Jackson Alexander	Jackson Alexander
Cash Sale #1009	Cash Sale	\$2,064.25	Julie Frankel	Julie Frankel
Cash Sale #1010	Cash Sale	\$1,183.23	Monica Parker	Monica Parker
Cash Sale #1011	Cash Sale	\$1,344.30	John Nguyen	John Nguyen
Cash Sale #1012	Cash Sale	\$1,493.85	Ken Chua	Ken Chua
Cash Sale #1013	Cash Sale	\$45.95	Sandra Burns	Sandra Burns
Check #2001	Check	-\$3,450.00	McGeever Property Man	
Check #2002	Check	-\$228.85	Safety Net Security	
Check #2003	Check	-\$159.90	Koka Office Supplies	

This discrepancy is another implication of the one-to-many relationship between specific record types. To avoid confusion over these discrepancies, make sure you join related record types in the correct order according to the type of datasets and results you want to generate.

SuiteQL

SuiteQL is a query language based on the SQL-92 revision of the SQL database query language. It offers advanced query capabilities you can use to access your NetSuite records and data, and it supports querying the analytics data source. For more information, see [Analytics Data Source Overview](#).

SuiteQL is currently available using SuiteAnalytics Connect, the N/query module in SuiteScript, and SuiteTalk REST web services. For more information, see [Using SuiteQL](#).

Some benefits of using SuiteQL include the following:

- Support for the analytics data source:** SuiteQL lets you query the analytics data source. This data source enhances the capabilities of querying your NetSuite data. The exposed data is consistent with SuiteAnalytics Workbook, which resolves previous data exposure inconsistencies in saved searches and reports. For more information, see [Analytics Data Source Overview](#).
- Improved security:** SuiteQL enforces the same role-based access restrictions used in SuiteAnalytics Workbook. This approach means that SuiteQL lets you query the same data you can access in the SuiteAnalytics Workbook user interface, which helps to increase security. SuiteQL also includes a list of supported SQL functions and doesn't allow you to use unsupported SQL functions in your query, which prevents SQL injection.

SuiteQL supports the syntax for both SQL-92 and Oracle SQL. However, you can't use both syntaxes in the same query.

To help you understand how to use SuiteQL, see the following help topics:

- [Using SuiteQL](#)
- [SuiteQL Join Types](#)
- [SuiteQL Syntax and Examples](#)

- [SuiteQL Limitations and Exceptions](#)
- [SuiteQL Supported and Unsupported Functions](#)
- [SuiteQL Supported Built-in Functions](#)

Using SuiteQL

SuiteQL lets you query your NetSuite data using advanced query capabilities. SuiteQL includes a list of supported SQL functions and doesn't allow you to use unsupported SQL functions in your query, which prevents SQL injection and other unauthorized access to data.

SuiteQL is currently available using SuiteAnalytics Connect and the N/query module in SuiteScript. You can also learn how to find record type names and field names to use in SuiteQL queries. See the following help topics:

- [Using SuiteQL with the Connect Service](#)
- [Using SuiteQL with the N/query Module in SuiteScript](#)
- [Using SuiteQL with SuiteTalk REST Web Services](#)
- [Finding Record Type and Field Names](#)

Using SuiteQL with the Connect Service

You can use SuiteQL to query your NetSuite data through the Connect Service. To do so, you must consider the following:

- [SuiteQL Performance and Best Practices for Connect](#)
- [Querying the Analytics Data Source](#)
- [Unsupported Features in SQL-92](#)
- [Supported and Unsupported Functions](#)
- [Supported Built-in Functions](#)

SuiteQL Performance and Best Practices for Connect

You can improve query performance by following guidelines and best practices. These guidelines apply even if a SuiteQL is later transformed to SQL by a process hidden to the user.

You should consider the following tips for optimal query performance:

Testing Queries

Evaluate the performance of the query after it is written. If the performance of the query is not ideal, you should experiment with query simplification or by replacing with semantically equivalent notation with better performance.

Syntax and Performance Best Practices

As you create queries using SuiteQL and the Connect Service, you must follow specific SuiteQL syntax requirements and best practices. For more information, see [SuiteQL Performance and Best Practices](#).

Multiple Connections

Running multiple concurrent sessions can slow performance and the retrieval of results. For information about running multiple connections when working with SuiteAnalytics Connect, see the help topic [Multiple Concurrent Sessions](#).

Roles and permissions

Roles and restrictions can result in performance issues. Ensure that you apply the necessary access restrictions for human interaction and consider adapting the access restrictions for automated machine-to-machine transfers. For more information about roles and permissions, see the help topic [Connect Permissions](#).

Oracle SQL

Due to Oracle SQL syntax working internally with NetSuite Analytics, you should use Oracle SQL syntax when creating SuiteQL queries. Although SuiteQL supports ANSI SQL-92, there's a risk of critical performance issues, such as time outs that aren't operationally remediable, that may occur if a SuiteQL is converted to SQL. This approach helps avoid the worst performance scenario, regardless of other factors. The following example shows how you can change your queries to use Oracle SQL syntax.

- The following query doesn't follow the Oracle SQL syntax:

```
1 | select t.id, t.period, t.amount, a.name, p.name
2 | from transaction t
3 | join account a on a.id = t.account
4 | left outer join accountingperiod p on p.id = t.period
```

- The following query follows the Oracle SQL syntax:

```
1 | select t.id, t.period, t.amount, a.name, p.name
2 | from transaction t,
3 | account a,
4 | accountingperiod p
5 | where a.id = t.account
6 | and on p.id(+) = t.period
```

- Note that you can't use right outer joins in Oracle syntax. You can use ANSI syntax for outer joins. For example, the following Oracle SQL is not valid in SuiteQL:

```
1 | select a1.id from account a1, account a2 where a1.id (+) = a2.id
```

Querying the Analytics Data Source

To find all record types and fields that are currently available for the analytics data source schema, see the help topic [Record Types and Fields](#).

The analytics data source isn't accessible for some roles and permissions. For more information, see the help topic [Role and Permission Considerations for NetSuite2.com](#).

Unsupported Features in SQL-92

When using SQL-92 syntax, there are several features that aren't supported, including embedded null values in CHAR fields, DEFAULT clauses for column values, and subqueries in SELECT lists.

Supported and Unsupported Functions

There are several functions that you can use when you run queries using SuiteQL. For a full list of supported and unsupported functions, see [SuiteQL Supported and Unsupported Functions](#).

Supported Built-in Functions

You can use built-in functions to perform certain operations in SuiteQL queries. For a full list of supported built-in functions, see [SuiteQL Supported Built-in Functions](#).

Using SuiteQL with the N/query Module in SuiteScript

You can use SuiteQL to query your NetSuite data using the N/query module in SuiteScript. The following object and methods are available in the N/query module to work with SuiteQL:

- The `query.SuiteQL` object represents a SuiteQL query. It includes properties that represent the query type, the SuiteQL string representation of the query, result columns to return from the query, and any additional parameters the query requires. You can use `SuiteQL.run()` to run the query as a non-paged query or `SuiteQL.runPaged(options)` to run the query as a paged query.
- The `Query.toSuiteQL()` method converts an existing query (as a `query.Query` object) to its SuiteQL representation (as a `query.SuiteQL` object).
- The `query.runSuiteQL(options)` and `query.runSuiteQLPaged(options)` methods let you run an arbitrary SuiteQL query. You can specify this query as a SuiteQL string, `query.SuiteQL` object, or generic JavaScript Object containing a required query property and an optional params property.

Note: If the SuiteAnalytics Connect feature is enabled in your NetSuite account, there's no limit to the number of results these methods can return. If the SuiteAnalytics Connect feature isn't enabled, these methods can return a maximum of 100,000 results across all pages in the result set. For more information about SuiteAnalytics Connect, see the help topic [SuiteAnalytics Connect](#).

Here's an example of how to load an existing query, convert it to SuiteQL, and run it:

```
1 var myLoadedQuery = query.load({ id: 'custworkbook237'
2 }); var mySuiteQLQuery = myLoadedQuery.toSuiteQL();
3 var myResultSet = mySuiteQLQuery.run();
```

For more information, see the help topic [SuiteQL in the N/query Module](#).

For a list of supported functions, see [SuiteQL Supported and Unsupported Functions](#) and [SuiteQL Supported Built-in Functions](#).

Using SuiteQL with SuiteTalk REST Web Services

Important: You must specify a sorting order in the query definition when using the `SuiteQL.runPaged(options)` method to avoid duplicate or missing results. The query definition must provide a unique and unambiguous sorting order with a specified precedence.

To run SuiteQL queries through REST web services, send a POST request to the `suiteql` resource, and specify the query in the request body after the query parameter `q`. The following example shows a SuiteQL query executed through REST web services.

```
1 > POST https://demo123.suitetalk.api.netsuite.com/services/rest/query/v1/suiteql
2 > Prefer: transient
3 . {
4 . "q": "SELECT email, COUNT(*) as count FROM transaction GROUP BY email"
5 . }
```

For more information about executing SuiteQL queries through REST web services, see the help topic [Executing SuiteQL Queries Through REST Web Services](#).

Finding Record Type and Field Names

SuiteQL queries include references to record types and fields. For example, the following SuiteQL string queries for entityid field values that are included in employee records:

```
1 | SELECT employee.entityid AS entityidRAW FROM employee
```

To create a query using SuiteQL, you must know the names of the record types and fields that you want to use in your query. You can find these names in several ways:

- [Using the NetSuite UI](#)
- [Using the SuiteAnalytics Workbook UI](#)
- [Using the Export Option from SuiteAnalytics Workbook](#)
- [Using the Records Catalog](#)

Using the NetSuite UI

On the Customization tab, you can see the IDs of the available record types and fields. For example, to see the IDs for custom CRM fields, go to Customization > Lists, Records, & Fields > CRM Fields (Administrator). The ID column appears in the list of custom fields.

Using the SuiteAnalytics Workbook UI

You can use the SuiteAnalytics Workbook UI to find record type and field names. SuiteAnalytics Workbook uses the same data source that SuiteQL does (the analytics data source), so you can use the SuiteAnalytics Workbook UI to explore the available record types and fields before you create your query.

To find record type and field names using the SuiteAnalytics Workbook UI:

1. In your NetSuite account, click the **Analytics** tab in the NetSuite navigation menu.
2. On the **Workbooks** page, click **New Workbook**.
3. Search for the record type that you want to use as the root record of your query.

The name of each supported record type is listed in the Record ID column. You can use this page to find all supported record type names, but you can't use this page to find field names. If you need to find field names, continue to step 4.

New Workbook: Select a Record Type

[Give Feedback](#)

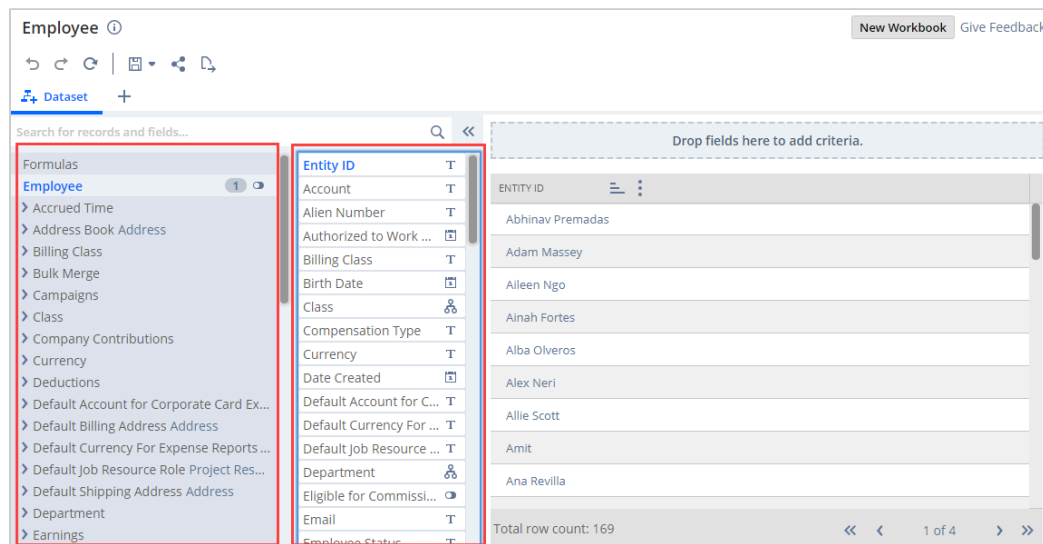
Search for a record type that includes fields that you would like to show in your workbook.

SEARCH RECORD TYPES

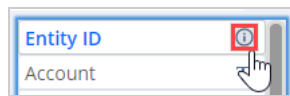
RECORD TYPE	RECORD CATEGORY	RECORD ID
Employee	Standard	employee
Employee Directory	Standard	employeeList
Employee Status	Standard	EmployeeStatus
Employee-Subsidiary Relationship	Standard	employeeSubsidiaryRelationship
Employee Type	Standard	EmployeeType
RelationshipSelectEmployeeRecord	Standard	RelationshipSelectEmployeeRecord

- Click a record type. A workbook is created with the selected record type as the root record.

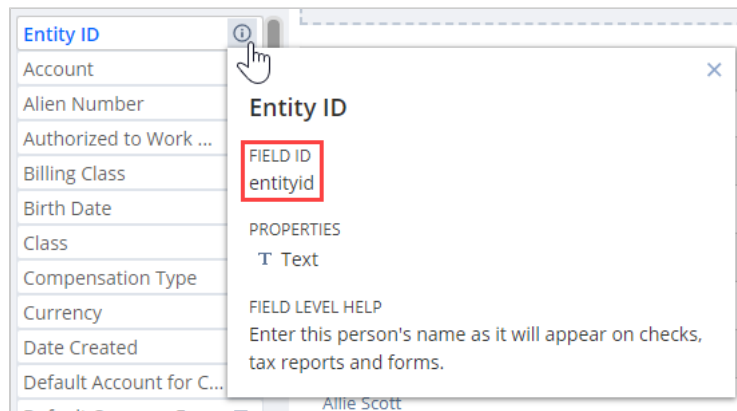
The left column lists the record types that you can join with the root record. The middle column lists the fields that are available on the selected record.



- In the middle column, find the field that you want to use in your query. Hover over the field, and click the information button.



Information about the field is displayed. The field name is listed under Field ID. You can also see the type of data that the field contains (text, date, boolean, and so on).



If you want to find field names for fields that are on a different root record, repeat this procedure from step 2 and choose a new root record. To learn more about creating workbooks, see [SuiteAnalytics Workbook Tutorial](#).

Using the Export Option from SuiteAnalytics Workbook

When you use SuiteAnalytics Workbook, you can export your saved dataset definitions to a SuiteQL TXT file. This allows you to understand how fields are joined in your dataset so that you can build your SuiteQL queries.

Note: When using the exported SuiteQL file, ensure that you use the same user and role used for the export with SuiteAnalytics Workbook. The export option is only available for users that have the AnalyticsAdmin permission assigned to their roles. For more information about Permissions, see the help topic [Setting Permissions](#).

For information about how to export your dataset definitions, see [Exporting Dataset Definitions to SuiteQL](#).

Using the Records Catalog

You can use the Records Catalog to see the fields and joins available for each record type when you use the following channels:

- If you are using the N/query module in SuiteScript and SuiteAnalytics Workbook.
- If you are using SuiteAnalytics Connect with the NetSuite2.com data source.

Note: The Records Catalog is supported for the SuiteScript Analytic API and the REST Query API only. Therefore, the record types available in the Records Catalog and the record types accessible through the Connect Service may differ slightly.

The Records Catalog displays information such as the cardinality, field level help, and the feature and permission requirements for each record type. The Records Catalog relies on NetSuite role-based permissions, so you can see only the record types for the data that you can access in the NetSuite user interface.

Note: The Records Catalog is available for all users that have the Records Catalog permission assigned to their role. For more information, see the help topic [Records Catalog Overview](#).

See the following topics:

- For information about the SuiteScript Analytic APIs, see the help topic [SuiteScript 2.x Analytic APIs](#).
- For information about SuiteAnalytic Connect, see the help topic [Record Types and Fields](#).

SuiteQL Join Types

SuiteQL supports several SQL join types. You can use these join types to customize the results you receive from your SuiteQL queries. By default, when you join record types in SuiteAnalytics Workbook, the join performed is a left outer join. This join type is appropriate for many use cases, but in some situations, you may want to use a different join type to obtain a more customized result set.

For more information about joining record types in SuiteAnalytics Workbook, see [Guidelines for Joining Record Types in SuiteAnalytics Workbook](#). For more information about SQL joins in general, see [Join \(SQL\)](#).

SuiteQL supports the following SQL join types:

- **Cross Joins** — A cross join is used to produce all row combinations between two tables. The result is known as the Cartesian product of the rows in both tables.
- **Inner Joins** — An inner join is used to produce row combinations between two tables based on a common value. The results include only those rows that share the common value.
- **Outer Joins** — An outer join is also used to produce row combinations between two tables based on a common value. The results include all rows from one or both tables, depending on the type of outer join.

To demonstrate each join type, the examples in this topic use the following simplified database tables. These tables include some of the fields that appear on customer and employee records in NetSuite. You can use the Records Catalog to determine the fields you can use in SuiteQL queries. For more information, see the help topic [Records Catalog Overview](#).

customer table		
entityid	email	salesrep
Big Computers	bigcomputers@example.com	5
Fantastic Laptops	fantasticlaptops@example.com	3
PennyPack Systems	pennypacksystems@example.com	NULL
Tall Manufacturing	tallmanufacturing@example.com	1
Vision Corporation	visioncorporation@example.com	1

employee table	
id	entityid
1	Allie Anderson
2	Billy Brown
3	Carol Connors
4	David Davis
5	Eugene Evans

Cross Joins

A cross join is the Cartesian product of the rows in two tables. It combines each row in the first table with each row in the second table. This type of join is computationally expensive, so you should use this type of join rarely. You can filter the results using the WHERE keyword, which produces an inner join.

An explicit cross join uses the CROSS JOIN keywords. Here's an example of an explicit cross join:

```
1 | SELECT *
2 | FROM customer CROSS JOIN employee
```

This join is equivalent to the following implicit cross join, which doesn't use the CROSS JOIN keywords:

```
1 | SELECT *
2 | FROM customer, employee
```

When using SuiteQL with SuiteAnalytics Connect, the CROSS JOIN keywords aren't supported. To perform a cross join, you can use the implicit notation. You can also use a full outer join to simulate a cross join, as follows:

```
1 | SELECT *
2 | FROM customer
3 | FULL OUTER JOIN employee ON 1=1
```

Using the example database tables, these joins produce 25 result rows:

customer.entityid	customer.email	customer.salesrep	employee.id	employee.entityid
Big Computers	bigcomputers@example.com	5	1	Allie Anderson
Big Computers	bigcomputers@example.com	5	2	Billy Brown
Big Computers	bigcomputers@example.com	5	3	Carol Connors
Big Computers	bigcomputers@example.com	5	4	David Davis
Big Computers	bigcomputers@example.com	5	5	Eugene Evans
Fantastic Laptops	fantasticlaptops@example.com	3	1	Allie Anderson
Fantastic Laptops	fantasticlaptops@example.com	3	2	Billy Brown
...	...	...	...	...

The first row in the first table is combined with each row in the second table, the second row in the first table is combined with each row in the second table, and so on. Additional result rows have been omitted from this table, but the same pattern continues for the remaining result rows.

Inner Joins

An inner join combines rows in the first table with rows in the second table based on a common value. The results of an inner join include only those rows that share the common column (or field) value.

An explicit inner join uses the `INNER JOIN` and `ON` keywords. Here's an example of an explicit inner join:

```

1 | SELECT customer.entityid, customer.email, employee.entityid
2 | FROM customer
3 | INNER JOIN employee ON customer.salesrep = employee.id

```

This explicit join is equivalent to the following implicit inner join, which uses the `WHERE` keyword instead of the `INNER JOIN` and `ON` keywords:

```

1 | SELECT customer.entityid, customer.email, employee.entityid
2 | FROM customer, employee
3 | WHERE customer.salesrep = employee.id

```

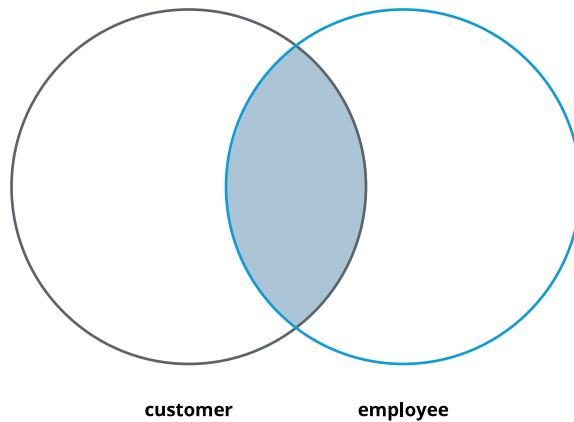
Using the example database tables, these joins produce four result rows:

customer.entityid	customer.email	employee.entityid
Big Computers	bigcomputers@example.com	Eugene Evans
Fantastic Laptops	fantasticlaptops@example.com	Carol Connors
Tall Manufacturing	tallmanufacturing@example.com	Allie Anderson
Vision Corporation	visioncorporation@example.com	Allie Anderson

In this example inner join, a result row is included only if the value of the `salesrep` field in the customer table matches the value of the `id` field in the employee table.

The following diagram illustrates an inner join:

Inner Join



Outer Joins

Similar to an inner join, an outer join combines rows in the first table with rows in the second table based on a common value. However, an outer join includes all of the rows in one or both tables in the results, depending on the type of outer join being performed.

SuiteQL supports the following outer join types:

- [Left Outer Joins](#)
- [Right Outer Joins](#)
- [Full Outer Joins](#)

Left Outer Joins

A left outer join combines all rows in the first (left) table with rows in the second (right) table based on a common value. The results of this join always contain every row in the first table, even if the join condition doesn't find any matching row in the second table.

This join is the default join type when you join record types in SuiteAnalytics Workbook. For more information, see [Guidelines for Joining Record Types in SuiteAnalytics Workbook](#).

An explicit left outer join uses the `LEFT OUTER JOIN` and `ON` keywords. Here's an example of an explicit left outer join:

```
1 | SELECT customer.entityid, customer.email, employee.entityid
2 | FROM customer
3 | LEFT OUTER JOIN employee ON customer.salesrep = employee.id
```

The `OUTER` keyword is optional, so this explicit join is equivalent to the following:

```
1 | SELECT customer.entityid, customer.email, employee.entityid
2 | FROM customer
3 | LEFT JOIN employee ON customer.salesrep = employee.id
```

This join is also equivalent to the following implicit left outer join, which uses the `(+)` syntax:

```
1 | SELECT customer.entityid, customer.email, employee.entityid
```

```

2 | FROM customer, employee
3 | WHERE customer.salesrep = employee.id(+)

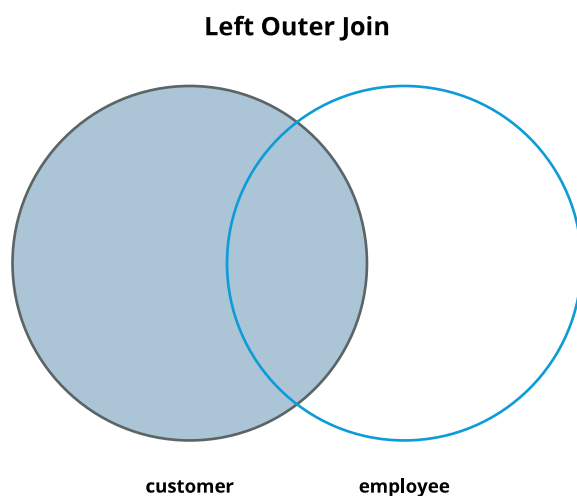
```

Using the example database tables, these joins produce five result rows:

customer.entityid	customer.email	employee.entityid
Big Computers	bigcomputers@example.com	Eugene Evans
Fantastic Laptops	fantasticlaptops@example.com	Carol Connors
PennyPack Systems	pennypacksystems@example.com	NULL
Tall Manufacturing	tallmanufacturing@example.com	Allie Anderson
Vision Corporation	visioncorporation@example.com	Allie Anderson

In this example left outer join, a row for PennyPack Systems is included in the results even though there's no matching value for that row in the employee table.

The following diagram illustrates a left outer join:



Right Outer Joins

A right outer join combines rows in the first (left) table with all rows in the second (right) table based on a common value. This join is similar to a left outer join, but the table order is reversed. The results of this join always contain every row in the second table, even if the join condition doesn't find any matching row in the first table.

An explicit right outer join uses the `RIGHT OUTER JOIN` and `ON` keywords. Here's an example of an explicit right outer join:

```

1 | SELECT customer.entityid, customer.email, employee.entityid
2 | FROM customer
3 | RIGHT OUTER JOIN employee ON customer.salesrep = employee.id

```

The `OUTER` keyword is optional, so this explicit join is equivalent to the following:

```

1 | SELECT customer.entityid, customer.email, employee.entityid

```

```

2 | FROM customer
3 | RIGHT JOIN employee ON customer.salesrep = employee.id

```

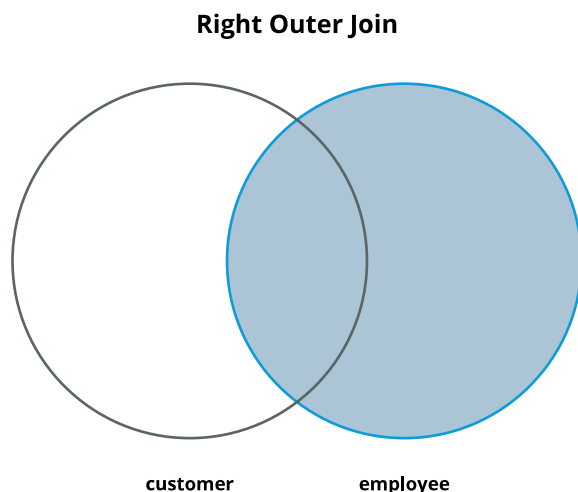
There's no implicit version of a right outer join in SuiteQL.

Using the example database tables, these joins produce six result rows:

customer.entityid	customer.email	employee.entityid
Tall Manufacturing	tallmanufacturing@example.com	Allie Anderson
Vision Corporation	visioncorporation@example.com	Allie Anderson
NULL	NULL	Billy Brown
Fantastic Laptops	fantasticlaptops@example.com	Carol Connors
NULL	NULL	David Davis
Big Computers	bigcomputers@example.com	Eugene Evans

In this example right outer join, rows for Billy Brown and David Davis are included in the results even though there are no matching values for those rows in the customer table.

The following diagram illustrates a right outer join:



Full Outer Joins

A full outer join combines the effects of a left outer join and a right outer join. A full outer join combines all rows in the first (left) table with all rows in the second (right) table based on a common value. The results of this join always contain every row in the first table and every row in the second table. Depending on the relationship between the tables, the results may include the same table row more than one time.

An explicit full outer join uses the `FULL OUTER JOIN` and `ON` keywords. Here's an example of an explicit full outer join:

```

1 | SELECT customer.entityid, customer.email, employee.entityid
2 | FROM customer
3 | FULL OUTER JOIN employee ON customer.salesrep = employee.id

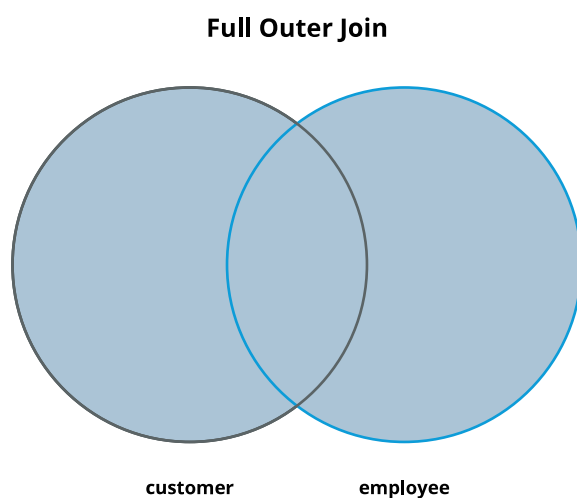
```


There's no implicit version of a full outer join in SuiteQL.

Using the example database tables, this join produces seven result rows:

customer.entityid	customer.email	employee.entityid
Big Computers	bigcomputers@example.com	Eugene Evans
Fantastic Laptops	fantasticlaptops@example.com	Carol Connors
PennyPack Systems	pennypacksystems@example.com	NULL
Tall Manufacturing	tallmanufacturing@example.com	Allie Anderson
Vision Corporation	visioncorporation@example.com	Allie Anderson
NULL	NULL	Billy Brown
NULL	NULL	David Davis

The following diagram illustrates a full outer join:



SuiteQL Syntax and Examples

SuiteQL supports the syntax for both SQL-92 and Oracle SQL. Also, SuiteQL supports specific SQL functions.

Note that you can't use both syntaxes in the same query. To learn about SQL-92, Oracle SQL, and SQL functions that you can use, see the following links:

- [SQL-92 Language Reference](#)
- [Oracle Database SQL Language Reference](#)



Important: Although SuiteQL supports ANSI SQL-92, there's a risk of critical performance issues, such as time outs that aren't operationally remediable, that may occur if a SuiteQL is converted to SQL. You should use Oracle SQL syntax when creating SuiteQL queries to avoid issues. For more information, see [SuiteQL Performance and Best Practices](#).

The following sections show you how to create queries using SuiteQL and the analytics data source.

Simple Queries

This section demonstrates how to perform simple queries using SuiteQL.

Field Values from Records

This string queries for entityid, firstname, and lastname field values in all employee records. By default, field values use the RAW field context. For more information, see the help topic [query.FieldContext](#).

```
1 | SELECT employee.entityid AS entityidRAW, employee.firstname AS firstnameRAW, employee.lastname AS lastnameRAW FROM employee
```

Field Values with Conditions

This string queries for entityid field values in employee records where the expenselimit of the employee is greater than or equal to 5000.

```
1 | SELECT employee.entityid AS entityidRAW FROM employee WHERE employee.expenselimit >= 5000
```

This string queries for entityid field values in employee records where the expenselimit of the employee is greater than or equal to 5000 or the employee is a sales representative.

```
1 | SELECT employee.entityid AS entityidRAW FROM employee WHERE employee.expenselimit >= 5000 OR employee.issalesrep = 'T'
```

Joined Records

This string queries for paytype and payrate field values in employee records joined to employee earning records where the employee is the same on both records:

```
1 | SELECT paytype, payrate FROM employeeEarning earning, employee WHERE earning.employee = employee.id
```

Advanced Queries

The following strings demonstrate advanced queries using SuiteQL.

```
1 | /* set operations */
2 | SELECT * FROM transaction UNION SELECT * FROM transaction /* top n */
3 | SELECT TOP 10 * FROM transaction
4 | SELECT TOP 1 id FROM transaction UNION SELECT TOP 1 id FROM transaction /* select list */
5 | SELECT DISTINCT email FROM transaction
6 | SELECT (SELECT MAX(1) FROM transaction) AS one FROM transaction
7 | SELECT COUNT(DISTINCT 1+id) FROM transaction
8 | SELECT COALESCE(email, id, 'some value', 1 + 1) FROM transaction /* from */
9 | SELECT * FROM transaction, transactionLine
10 | SELECT * FROM (SELECT * FROM transaction) a INNER JOIN ((SELECT * FROM transaction UNION SELECT * FROM transaction) b INNER JOIN
    (SELECT * FROM transaction) c ON 1=1) ON 1=1
11 | SELECT * FROM (SELECT id, COUNT(*) cnt FROM transactionLine GROUP BY id) WHERE cnt > 2 /* where */
12 | SELECT * FROM transaction t WHERE id IN (SELECT id FROM transaction WHERE id = t.id UNION SELECT -1 FROM transaction)
13 | SELECT * FROM transaction WHERE 1 = (SELECT MAX(1) FROM transaction)
14 | SELECT * FROM transaction WHERE EXISTS(SELECT 1 FROM transaction)
15 | SELECT * FROM transaction WHERE id IN ((SELECT MAX(1) FROM transaction), 2+1) /* having */
16 | SELECT email, COUNT(*), MAX(create_date) FROM transaction GROUP BY email HAVING COUNT(*) > 2
```

SuiteQL Limitations and Exceptions

The following are known limitations and exceptions when using SuiteQL:

- You can create SuiteQL queries using the syntax for either SQL-92 or Oracle SQL, but you can't use both syntaxes in the same query. To learn about SuiteQL syntax and grammar, see [SuiteQL Syntax and Examples](#).
- Square brackets [] are not supported in SuiteQL. For more information, see [SuiteQL Syntax and Examples](#).
- Although SuiteQL supports ANSI SQL-92, there's a risk of critical performance issues, such as time outs that aren't operationally remediable, that may occur if a SuiteQL is converted to SQL. You should use Oracle SQL syntax when creating SuiteQL queries to avoid issues. For more information, see [SuiteQL Performance and Best Practices](#).
- The casing of record type names and field names in SuiteQL query results may not always be consistent and can change. For example, the current name of the account record type is account, but in query results, this name could appear as Account or ACCOUNT. These name changes may occur after a NetSuite release.

```
1 | SELECT * FROM account
```

```
1 | SELECT * FROM Account
```

```
1 | SELECT * FROM ACCOUNT
```

When you work with query results, make sure that you don't depend on name casing. You can convert these names to uppercase or lowercase to ensure consistent casing.

When you run SuiteQL queries using the N/query module, the query results may not reflect the casing used to build the query. For example, you can use aliases (the AS keyword) in your query string. However, the query results don't reflect the casing of the aliases you specify, and the aliases appear in lowercase. Consider the following:

```
1 | query.runSuiteQL({ query: 'SELECT id AS myID FROM transaction'
2 | }).asMappedResults();
```

This method runs the specified SuiteQL query string and returns the query results as mapped results. In the array of mapped results, the key appears as myid instead of myID. For more information, see the help topics [query.runSuiteQL\(options\)](#) and [ResultSet.asMappedResults\(\)](#).

When you run constructed queries (those you create using the [query.create\(options\)](#) method in the N/query module), you don't need to consider potential casing changes. For constructed queries, you should specify the query type using values from the [query.Type](#) enum. If you specify the query type as a string directly and don't use values from this enum, the query type is processed in a case-insensitive way, and your query will still run correctly.

- `BUILTIN.HIERARCHY(PARENT, 'LEVEL')` can sometimes produce inconsistent results if the depth of the parent field is directly connected to the base order. The level value will be taken from $n-1$ instead of n , where n represents the depth of the parent. For more information, see [SuiteQL Supported Built-in Functions](#).
- To mitigate the possibilities of a CSV injection, the following formatting is applied to all dataset and workbook results that are exported to CSV using SuiteScript and SuiteQL:
 - All text values are enclosed in double quotation marks
 - Values that begin with the characters -, +, =, @, tab, and EOL (end of line) have single apostrophes added in front of the value
- Listing application data from custom fields is not supported with Analytic APIs. For example, if you run a SuiteQL query on a custom transaction body field, it won't return any information from its corresponding subtabs in the UI. For more information about custom fields, see the help topic [Custom Fields](#).

- Sorting results based on a field that contains CLOB (character large object) values only evaluates the first 250 characters for each result. Results with more than 250 characters may not be sorted properly.
- Queries that use the transaction type (TranType) field in a condition for custom transaction status (TransactionStatus) records can sometimes return unexpected results. The following example attempts to retrieve a custom transaction type for a transaction status record, but can sometimes return inconsistent results.

```
1 | SELECT * FROM TransactionStatus WHERE TranType = 'CuTrSale'
```

As a workaround, consider using the LIKE operator in your query as in the following example.

```
1 | SELECT * FROM TransactionStatus WHERE TranType LIKE 'CuTrSale%'
```

SuiteQL Supported and Unsupported Functions

The following sections list the supported and unsupported functions most commonly used in queries when using SuiteQL.

- [Supported Functions](#)
- [Unsupported Functions](#)

Supported Functions

The following table lists and describes functions you can use when you run queries using SuiteQL.

Supported Function	Description
ABS	returns the absolute value of n
ACOS	returns the arc cosine of n
ADD_MONTHS	returns the date date plus integer months
APPROX_COUNT_DISTINCT	returns the approximate number of rows that contain distinct values of expr
ASCII	returns the decimal representation in the database character set of the first character of char
ASCIISTR	takes as its argument a string, or an expression that resolves to a string, in any character set and returns an ASCII version of the string in the database character set
ASIN	returns the arc sine of n
ATAN	returns the arc tangent of n
ATAN2	returns the arc tangent of n1 and n2
AVG	returns the average value of expr
BFILENAME	returns a BFILE locator that is associated with a physical LOB binary file on the server file system
BITAND	computes an AND operation on the bits of expr1 and expr2
CEIL	returns smallest integer greater than or equal to n
CHARTOROWID	converts a value from CHAR, VARCHAR2, NCHAR, or NVARCHAR2 datatype to ROWIDdatatype

Supported Function	Description
CHR	returns the character having the binary equivalent to n as a VARCHAR2 value
COALESCE	returns the first non-null expr in the expression list
COMPOSE	takes as its argument a string, or an expression that resolves to a string, in any datatype, and returns a Unicode string in its fully normalized form in the same character set as the input
CONCAT	concatenates char1 and char2 into one string
CORR	returns the coefficient of correlation of a set of number pairs
CORR_K	calculates the Kendall's tau-b correlation coefficient
CORR_S	calculates the Spearman's rho correlation coefficient
COS	returns the cosine of n (an angle expressed in radians)
COSH	returns the hyperbolic cosine of n
COUNT	returns the number of rows returned by the query
COVAR_POP	returns the population covariance of a set of number pairs
COVAR_SAMP	returns the sample covariance of a set of number pairs
CURRENT_DATE	returns the current date in the session time zone, in a value in the Gregorian calendar of datatype DATE
CURRENT_TIMESTAMP	returns the current date and time in the session time zone
DECODE	compares expr to each search value one by one. If expr is equal to a search, then the Oracle database returns the corresponding result
DECOMPOSE	takes as its argument a string in any datatype and returns a Unicode string after decomposition in the same character set as the input
DENSE_RANK	computes the rank of a row in an ordered group of rows and returns the rank as a NUMBER
EMPTY_BLOB	returns an empty LOB locator
EMPTY_CLOB	returns an empty LOB locator
EXP	returns e raised to the nth power
FLOOR	returns largest integer equal to or less than n
FROM_TZ	converts a timestamp value and a time zone to a TIMESTAMP WITH TIME ZONE value
GREATEST	returns the greatest of a list of one or more expressions
INITCAP	returns char, with the first letter of each word in uppercase, all other letters in lowercase
INSTR	searches string for substring
LAST_DAY	returns the date of the last day of the month that contains date
LEAST	returns the least of the list of exprs
LENGTH	returns the length of char

Supported Function	Description
LENGTH2	returns the length of the specified string, using UCS2 code points
LENGTH4	returns the length of the specified string, using UCS4 code points
LENGTHB	returns the length of the specified string, using bytes instead of characters
LENGTHC	returns the length of the specified string, using Unicode complete characters
LN	returns the natural logarithm of n, where n is greater than 0
LOCALTIMESTAMP	returns the current date and time in the session time zone in a value of datatype TIMESTAMP
LOG	computes the logarithm of an expression
LOWER	returns char, with all letters in lowercase
LPAD	returns expr1, left-padded to length n characters with the sequence of characters in expr2
LTRIM	removes from the left end of char all of the characters contained in set
MAX	returns the maximum value of expr
MEDIAN	is an inverse distribution function that assumes a continuous distribution model. It takes a numeric or datetime value and returns the middle value or an interpolated value that would be the middle value after the values are sorted
MIN	returns the minimum value of expr
MOD	returns the remainder of n2 divided by n1. Returns n2 if n1 is 0
MONTHS_BETWEEN	returns the number of months between date1 and date2
NANVL	useful only for floating-point numbers of type BINARY_FLOAT or BINARY_DOUBLE. This function is useful for mapping NaN values to NULL
NEW_TIME	returns the date and time in time zone timezone2 when date and time in time zone timezone1 are date
NEXT_DAY	returns the date of the first weekday named by char that is later than the date
NLSSORT	returns the string of bytes used to sort char
NLS_INITCAP	returns char, with the first letter of each word in uppercase, all other letters in lowercase
NLS_LOWER	returns char, with all letters in lowercase
NLS_UPPER	returns char, with all letters in uppercase
NULLIF	compares expr1 and expr2. If they are equal, then the function returns null. If they aren't equal, then the function returns expr1
NVL	lets you replace null (returned as a blank) with a string in the results of a query
NVL2	lets you determine the value returned by a query based on whether a specified expression is null or not null
ORA_HASH	computes a hash value for a given expression
POWER	returns n2 raised to the n1 power

Supported Function	Description
RANK	calculates the rank of a value in a set of values
REGEXP_INSTR	extends the functionality of the INSTR function by letting you search a string for a regular expression pattern
REGEXP_REPLACE	extends the functionality of the REPLACE function by letting you search a string for a regular expression pattern
REGEXP_SUBSTR	extends the functionality of the SUBSTR function by letting you search a string for a regular expression pattern
REMAINDER	returns the remainder of n2 divided by n1
REPLACE	returns char with every occurrence of search_string replaced with replacement_string
ROUND	returns n rounded to integer places to the right of the decimal point
ROW_NUMBER	analytic function that assigns a unique number to each row to which it is applied
RPAD	returns expr1, right-padded to length n characters with expr2, replicated as many times as necessary
RTRIM	removes from the right end of char all of the characters that appear in set
SIGN	returns the sign of n
SIN	returns the sine of n (an angle expressed in radians)
SINH	returns the hyperbolic sine of n
SOUNDEX	returns a character string containing the phonetic representation of char
SQRT	returns the square root of n
SUBSTR	returns a portion of char, beginning at character position, substring_length characters long
SUM	returns the sum of values of expr. You can use it as an aggregate or analytic function
SYS_EXTRACT_UTC	extracts the UTC from a datetime value with time zone offset or time zone region name
TAN	returns the tangent of n (an angle expressed in radians)
TANH	returns the hyperbolic tangent of n
TO_BINARY_DOUBLE	returns a double-precision floating-point number
TO_BINARY_FLOAT	returns a single-precision floating-point number
TO_CHAR	(number) converts n to a value of VARCHAR2 datatype
TO_CLOB	converts NCLOB values in a LOB column or other character strings to CLOB values
TO_DATE	converts char of CHAR, VARCHAR2, NCHAR, or NVARCHAR2 datatype to a value of DATE datatype
TO_MULTI_BYTE	TO_MULTI_BYTE returns char with all of its single-byte characters converted to their corresponding multibyte characters
TO_NCHAR	converts a character string, CHAR, VARCHAR2, CLOB, or NCLOB value to the national character set

Supported Function	Description
TO_NCLOB	converts CLOB values in a LOB column or other character strings to NCLOB values
TO_NUMBER	converts expr to a value of NUMBER datatype
TO_SINGLE_BYTE	returns char with all of its multibyte characters converted to their corresponding single-byte characters
TO_TIMESTAMP	converts char of CHAR, VARCHAR2, NCHAR, or NVARCHAR2 datatype to a value of TIMESTAMP datatype
TO_TIMESTAMP_TZ	converts char of CHAR, VARCHAR2, NCHAR, or NVARCHAR2 datatype to a value of TIMESTAMP WITH TIME ZONE datatype
TRANSLATE	returns expr with all occurrences of each character in from_string replaced by its corresponding character in to_string
TRUNC	(number) returns n1 truncated to n2 decimal places
TZ_OFFSET	returns the time zone offset corresponding to the argument based on the date the statement is executed
UNISTR	takes as its argument a text literal or an expression that resolves to character data and returns it in the national character set
UPPER	returns char, with all letters in uppercase
VSIZE	returns the number of bytes in the internal representation of expr
WIDTH_BUCKET	lets you construct equiwidth histograms, in which the histogram range is divided into intervals that have identical size

For a list of supported built-in functions, see [SuiteQL Supported Built-in Functions](#).

Unsupported Functions

The following table shows some examples of functions that are not supported in SuiteQL. This list is not exhaustive and does not include all unsupported functions.

Unsupported Function	Alternative Function (If Available)
BIT_LENGTH	—
BIT_XOR_AGG	—
CEILING	CEIL
CHAR	—
CHARINDEX	INSTR
CHAR_LENGTH	LENGTH
CHARACTER_LENGTH	LENGTH
CONVERT	—
COT	—
DATEDIFF	—
LCASE	LOWER

Unsupported Function	Alternative Function (If Available)
LEFT	SUBSTR
LISTAGG	—
LOCATE	INSTR
POSITION	INSTR
REPEAT	—
RIGHT	SUBSTR
SUBSTRING	SUBSTR
UCASE	UPPER

SuiteQL Supported Built-in Functions

You can use built-in functions to perform certain operations in SuiteQL queries. These functions extend the capabilities that are provided by the SQL-92 specification. For example, you can use the `CONSOLIDATE` built-in function to convert a currency amount stored in a field to a target currency.

To use a built-in function, you must add `BUILTIN.` before the function name. For example, to call the `CONSOLIDATE` built-in function, you must use `BUILTIN.CONSOLIDATE` in your SuiteQL query.

The following list describes the built-in functions that are supported in SuiteQL and includes examples of each function. Some parameters are optional and are documented as such in the table, but all other parameters are required.



Note: Some parameters include descriptions that apply to the N/query module only.

- [BUILTIN.CF](#)
- [BUILTIN.CONSOLIDATE](#)
- [BUILTIN.CURRENCY](#)
- [BUILTIN.CURRENCY_CONVERT](#)
- [BUILTIN.DF](#)
- [BUILTIN.HIERARCHY](#)
- [BUILTIN.MNFILTER](#)
- [BUILTIN.NAMED_GROUP](#)
- [BUILTIN.PERIOD](#)
- [BUILTIN.RELATIVE_RANGES](#)

BUILTIN.CF

This function returns the criteria value of a field in a query. Some fields, such as `transaction.status`, use different SQL expressions to retrieve a value based on the defined context, similarly to composite keys. You should use this function when creating conditions on fields with combined values.

The following query returns a transaction status without the combined field value. An example output could return 'B' to represent a paid in full transaction.

```
1 | SELECT status FROM TRANSACTION
```

The following query uses the BUILTIN.CF function to return a transaction status with the combined field value. The result is a key value pair to represent the transaction type and status. An example output could return 'CustInv: B' to represent a paid in full invoice transaction.

```
1 | SELECT BUILTIN.CF(status) FROM TRANSACTION
```



Important: You must use this function on queries that use filters with composite key value fields to avoid potential result errors. See the syntax section below for an example of a filtered query on composite key value fields.

Parameters

Parameter	Required / Optional	Description	Acceptable Parameter Values
Field name	Required	The name of the field to return combine field values.	–

Syntax

```
1 | SELECT BUILTIN.CF(transaction.status) FROM Transaction
```

```
1 | SELECT status FROM Transaction WHERE BUILTIN.CF(status) = 'Opportunity:A'
```

BUILTIN.CONSolidATE

This function converts the currency amount in a field to a target currency.

Parameters



Important: To use the default value for any parameter (except original amount field name and view type), you can specify 'DEFAULT' as the parameter value.

Parameter	Required / Optional	Description	Acceptable Parameter Values
Original amount field name	Required	The field name to apply criteria context to.	–
View type	Required	Supported view types.	■ 'LEDGER' ■ 'INCOME'
Consolidation rate type	Required	Supported consolidation rate types.	■ 'DEFAULT' ■ 'STANDARD' ■ 'BUDGET'
Subsidiary rate type	Required	Supported rate types based on subsidiary.	■ 'DEFAULT' ■ 'CURRENT' ■ 'HISTORICAL' ■ 'AVERAGE'

Parameter	Required / Optional	Description	Acceptable Parameter Values
			SQL EXPRESSION
Target subsidiary	Required	The target subsidiary id.	- Number - SQL EXPRESSION
Period	Required	The accounting period.	- Number - SQL EXPRESSION
Book	Required	The accounting book.	'DEFAULT' - SQL EXPRESSION

Syntax

```
1 | BUILTIN.CONSolidate(sales.amount, 'INCOME', 'DEFAULT', 'DEFAULT', 1, NUMBER, 'DEFAULT')
```

BUILTIN.CURRENCY

This function returns a currency code.

Parameters

Parameter	Required / Optional	Description	Acceptable Parameter Values
Amount	Required	The field name to apply criteria context to.	- amount - SUM(amount) - BUILTIN.CONSolidate - SUM(BUILTIN.CONSolidate) - BUILTIN.CURRENCY_CONVERT - SUM(BUILTIN.CURRENCY_CONVERT)

Syntax

```
1 | BUILTIN.CURRENCY(sales.amount)
```

```
1 | BUILTIN.CURRENCY(BUILTIN.CONSolidate(sales.amount, 'INCOME', 'DEFAULT', 'DEFAULT', 1, 304, 'DEFAULT'))
```

BUILTIN.CURRENCY_CONVERT

This function converts a currency amount stored in a field to a target currency. The target currency is determined based on effective date of defined exchange rate. [Currency Exchange Rates](#).

Parameters

Parameter	Required / Optional	Description	Acceptable Parameter Values
Original amount field	Required	The original amount field to convert.	–

Parameter	Required / Optional	Description	Acceptable Parameter Values
Target currency	Optional	The currency to convert to. The default value is the default currency of the subsidiary.	- Number - Date
Date of the exchange rate	Optional	The effective date of exchange rate. The default value is today's date.	–

Syntax

```
1 | BUILTIN.CURRENCY_CONVERT(sales.amount)
```

```
1 | BUILTIN.CURRENCY_CONVERT(sales.amount, 1)
```

```
1 | BUILTIN.CURRENCY_CONVERT(sales.amount, 1, TO_DATE('2019-01-30', 'YYYY-MM-DD'))
```

BUILTIN.DF

Returns the display value of a field from a target record type without having to join the target record type explicitly.

Parameters

Parameter	Required / Optional	Description	Acceptable Parameter Values
Field name	Required	This field represents the relationship between the field and the source record type(for example, sales.item).	–

Syntax

```
1 | BUILTIN.DF(sales.item)
```

BUILTIN.HIERARCHY

Returns the full hierarchical path to a value. Values can be returned by id or display field.

Parameters

Parameter	Required / Optional	Description	Acceptable Parameter Values
Hierarchical field	Required	A field you can chose to display as hierarchies or child values. For	–

Parameter	Required / Optional	Description	Acceptable Parameter Values
		more information, see Hierarchical Fields .	
Expansion Type	Required	Specify the way a hierarchical field is displayed in results.	Parameter values must include single quotation marks: ■ 'DISPLAY' ■ 'DISPLAY_JOINED' ■ 'DISPLAY_SEPARATED' ■ 'IDENTIFIER' ■ 'IDENTIFIER_SEPARATED' ■ 'LEVEL' ■ 'SELF_DISPLAY' ■ 'SELF_IDENTIFIER'



Note: Hierarchy level can sometimes return varying results. For more information, see [SuiteQL Limitations and Exceptions](#).

Syntax

```
1 | BUILTIN.HIERARCHY(parent, 'IDENTIFIER_SEPARATED')
```

```
1 | SELECT id, parent,
2 |   BUILTIN.HIERARCHY(parent, 'IDENTIFIER_SEPARATED')
3 | FROM Item
```

BUILTIN.MNFILTER

This functions filters a record to specify a target value on a multiselect field.

Parameters

Parameter	Required / Optional	Description	Acceptable Parameter Values
Relationship field	required	Represents the selected value on a multiselect field.	–
Operator	required	Holds the values supported when filtering relationship fields.	■ 'MN_INCLUDE' ■ 'MN_INCLUDE_ALL' ■ 'MN_INCLUDE_EXACTLY' ■ 'MN_EXCLUDE' ■ 'MN_EXCLUDE_ALL' ■ 'MN_EXCLUDE_EXACTLY'
Display field or empty string	required	A display field should be used if you are filtering a display field value of a	■ 'DF'

Parameter	Required / Optional	Description	Acceptable Parameter Values
		relationship field. The empty string value should be used if filtering over foreign key values.	''
boolean value	required	A boolean value helps specify if the filter should be case sensitive.	- true - false
value	required	The list of values to be included in the filter.	–

Syntax

```
1 | SELECT BUILTIN_RESULT.TYPE_INTEGER("ROLE"."ID") AS "ID" /*{id#RAW}*/,FROM"ROLE"WHERE BUILTIN.MNFILTER("ROLE".accessprofiles, 'MN_IN
| CLUDE', '', 'TRUE', '-2') = 'T'
```

BUILTIN.NAMED_GROUP

This function returns filter options for department, class, location, subsidiary, and entity record types.

Parameters

Parameter	Required / Optional	Description	Accepted Parameter Values
Record type	required	The record to return filter options for.	- Department records - Class records - Location records - Subsidiary records - Entity records
Workgroup	required	Represents a way to filter a record with "me" or "mine".	'me' – Supported for Entity, Customer, Vendor, and Partner record types. 'mine' – Supported for Classification, Department, Location, and Subsidiary record types.

Syntax

```
1 | BUILTIN.NAMED_GROUP('employee', 'me')
```

```
1 | BUILTIN.NAMED_GROUP('subsidiary', 'mine')
```

```
1 | select *
2 | from Subsidiary
3 | where id in BUILTIN.NAMED_GROUP('subsidiary', 'mine')
```

BUILTIN.PERIOD

This function returns the contents of the IN predicate for a relative date range as a sub-selection.

Parameters

Parameter	Required / Optional	Description	Accepted Parameter Values
Range ID	Required	The supported date ranges in relative dates.	See RelativeDate.dateId Property values in the table on query.RelativeDateRange for list of supported values. For example, use 'LFY' to represent Last Fiscal Year in your queries.
Range type	Required	Represents supported period range filters.	'START' 'END'
Adjustment	Required	Represents supported adjustment period filters. For more information, see the help topic Adjustment Periods .	'NOT_LAST' 'ALL'
Operator	Required	The supported operators for filters.	'BETWEEN' 'NOT BETWEEN' '<' '<=' '>' '>='

Syntax

```
1 | BUILTIN.PERIOD('LFY', 'END', 'NOT_LAST', '>')
```

```
1 | BUILTIN.PERIOD('LFY', 'START', 'ALL', 'BETWEEN')
```

BUILTIN.RELATIVE_RANGES

This function returns dynamic calendar ranges for the filter options that apply to relative date fields (such as Last Fiscal Year and Current Week).

Parameters

Parameter	Required / Optional	Description	Accepted Parameter Values
Range ID	Required	The supported date ranges in relative dates.	See RelativeDate.dateId Property values in the table on query.RelativeDateRange for list of supported values. For example, use 'LFY' to represent Last Fiscal Year in your queries.
Range type	Required	Represents supported period range filters.	'START' 'END'

Parameter	Required / Optional	Description	Accepted Parameter Values
Date flag	Optional	Represents ways to present date values.	■ 'DATE' ■ 'DATETIME_AS_DATE' ■ 'DATE_FROM_GMT_TO_USER_TZ' ■ 'DATE_TO_USER_TZ' ■ 'ODBC_DATE_TO_GMT' ■ 'ODBC_TIMESTAMP_TO_GMT' ■ 'ODBC_TIMESTAMP_TO_GMT_AS_DATE' ■ 'TIMESTAMP_WITH_TIMEZONE' ■ 'TIMESTAMP_WITH_TIMEZONE_AS_DATE'

Syntax

```
1 | BUILTIN.RELATIVE_RANGES('LFYTD', 'START')
```

```
1 | BUILTIN.RELATIVE_RANGES('LFYTD', 'END', 'DATETIME_AS_DATE')
```

SuiteQL Performance and Best Practices

You have the ability to significantly influence the performance of a query by adhering to the guidelines and syntax best practices that help prevent issues that may occur. These guidelines apply even if a SuiteQL is later transformed to SQL by a process hidden to the user.

You should consider the following syntax requirements and performance best practices:

- [Syntax Requirements](#)
- [Performance Best Practices](#)

Syntax Requirements

As you create queries using SuiteQL and the Connect Service, consider the following syntax requirements to avoid errors in your queries:

- For string concatenation, you can't use the + operator. You should use the || operator instead. This restriction applies to both field and literal concatenation.
- You can't use more than 1000 arguments in a single IN clause.
- You can't use WITH clauses in your queries.
- You can't use date literals. You must encapsulate dates using the to_date() function.
- You can't use right outer joins in Oracle syntax. You can use ANSI syntax for outer joins. For example, the following Oracle SQL is not valid in SuiteQL:

```
1 | select a1.id from account a1, account a2 where a1.id (+) = a2.id
```

- SuiteQL supports the syntax for both ANSI and non-ANSI joins. However, you can't use both syntax types in the same query.
- Don't use quotation marks for field names in subselections. See the following examples:

- You should not use the following syntax with quotation marks:

```
1 | select * from ( select a.externalid "AccountId" from account a)
```

- You should use the following syntax without quotation marks:

```
1 | select * from ( select a.externalid AccountId from account a)
```

Performance Best Practices

Consider the following best practices:

- Avoid creating complex queries that include nested SELECT clauses. Use simple queries instead.
- Avoid creating queries that include "SELECT *". Include the specific fields that you want to query instead. See the following example:
 - Avoid using the following example, which includes "SELECT *":

```
1 | select * from transaction
```

- Use the following example, which includes the specific fields:

```
1 | select id, lastmodifieddate, tranid from transaction
```

- Avoid using the following field types in your queries when possible:
 - **Calculated fields** - Using calculated fields has an impact on performance. You can identify calculated fields by using the `oa_columns` table. Calculated fields are those that include "C" in the sixth position of the `oa_userdata` column. An example of a calculated field is `customer.oncredithold`. For more information about the `oa_columns` table, see the help topic [oa_columns](#).
 - **WLONGVARCHAR fields** - These fields include large values and this has an impact on performance. See the following examples of WLONGVARCHAR fields:
 - `item.featureddescription`
 - `item.storedetaileddescription`
 - `item.metataghtml`
- Use filters in your queries. Using filters helps reducing the amount of data retrieved. When adding filter, consider the following:
 - Using indexed fields as filters enhances the performance when running queries. The following fields show examples of indexed fields:
 - Primary keys, such as `id`
 - `lastmodifieddate`
 - Using filters is a good option when doing incremental loads. The following query shows an example of incremental loads. Filters help retrieving only those results changed during a specific period, instead of retrieving data from the whole table each time. In the following example results were retrieved for the 1st of January of 2024:

```
1 | select id, lastmodifieddate, tranid from transaction where lastmodifieddate >= to_date('2024-01-01 00:00:00', 'yyyy-mm-dd hh24:mi:ss')
```

- Use the same type of filter expressions when adding filters. If you include different types, they need to be converted to the same type and this can slow performance. For example, ensure that

you aren't comparing TO_TIMESTAMP against TO_DATE functions. You should change your query to compare either TO_TIMESTAMP to each other or TO_DATE functions to each other.

- Use batches in your queries. When you split a query into several batches, you avoid running queries that take long to return results. This reduces the impact of maintenance server restarts. The following examples show queries that you should avoid and that you can use.
 - When running queries that should return a large number of rows, you should split them into batches. The following examples show queries that you should avoid:

```
1 | select id, lastmodifieddate, tranid from transaction
```

- The following queries show how you can split your queries into several batches:
 - select id, lastmodifieddate, tranid from transaction where id < 10000000
 - select id, lastmodifieddate, tranid from transaction where id >= 10000000 and id < 20000000
 - select id, lastmodifieddate, tranid from transaction where id >= 20000000 and id < 30000000
 - select id, lastmodifieddate, tranid from transaction where id >= 30000000 and id < 40000000
 - select id, lastmodifieddate, tranid from transaction where id >= 40000000 and id < 50000000
 - select id, lastmodifieddate, tranid from transaction where id >= 50000000
- Avoid using too many joins. This can result in performance issues. Also, avoid using the same join in a specific table multiple times.
- Avoid using predicates with a logical disjunction (OR) of the arguments. Run a specific query for each predicate instead.
- Avoid using unnecessary sorting when possible. Run simple queries instead. In many cases, you can use inner joins. However, test your queries before you create a complex combination on top of your query.
- Using TOP clauses might have an impact on performance, as SuiteQL queries are run on a virtual schema and not on the actual database schema. Check out this example for improving performance:
 - The following query includes the TOP clause and all results are evaluated, which can slow performance.

```
1 | select top 100 id, lastmodifieddate, tranid from transaction
```

- The following query doesn't include the TOP clause and improves performance.

```
1 | select id, lastmodifieddate, tranid from transaction where id <= 100
```

- Evaluate the performance of the query after it is written. If the performance of the query is not ideal, you should experiment with query simplification or by replacing with semantically equivalent notation with better performance.

Custom Workbooks and Datasets

In SuiteAnalytics Workbook, you analyze your company data using two distinct objects: a dataset and a workbook.

Datasets are the basis for all workbooks and workbook visualizations in your account. In a dataset, you combine fields and filters to create a query, and the results become the source data for your workbook visualizations. A single dataset can be used in multiple workbooks and visualizations, so to avoid data issues, only dataset owners or users with the Analytics Administrator permission can edit or delete them. When you update a dataset, those changes automatically display in any workbook visualizations that use it. This means you don't need lots of datasets in your account, and it lets less experienced users build complex workbooks.

Workbooks are where you analyze the results of your dataset queries using different visualizations, such as tables, pivot tables, and charts. Every workbook visualization is based on a dataset or a set of linked datasets, and you can use different datasets for each visualization. You can access the dataset connected to a workbook visualization from the Dataset Panel. When you make a new visualization, you can base it on a new dataset or any existing datasets you have access to. You can also link two datasets to use as the basis for your visualizations, which lets you compare metrics from both datasets in one place.

To better understand the functionality and relationship between datasets and workbooks, see the following table:

Functionality	Dataset	Workbook
Add fields and join record types?	Yes You define the source data for your workbook visualizations by adding fields and joining record types in a dataset query.	— If you want to add fields or join record types to a workbook visualization, you must edit the underlying dataset.
Create visualizations?	— The dataset is only where you query your business data. To see the results, create a workbook visualization based on the dataset.	Yes The workbook is where you analyze your dataset query results using tables, pivot tables, and charts. You can use a different dataset for each visualization.
Automatically propagate edits to other workbook visualizations?	Yes A single dataset can be used in multiple workbooks and workbook visualizations. To avoid data errors, any edits you make to a dataset are automatically updated in those visualizations.	— Workbooks and visualizations in your account are independent. If you edit a visualization, your changes only affect that one visualization.
Currency conversion or consolidation?	Yes, but only with custom formula fields. Prior to 20.1, currency conversion or consolidation was applicable from the Data Grid in the Dataset Builder. As of 20.1, you can only apply currency conversion and consolidation to a field from within a workbook visualization unless you create a custom formula field. For more information, see Currency Conversion Using Custom Formula Fields .	Yes You can convert and consolidate field values to a specific currency right from the user interface in any workbook visualization. For more information, see Currency Conversion from the User Interface .
Sharing?	Yes You can share a dataset with other users in your account as its own object, or as part of a shared workbook. Recipients of a shared workbook cannot edit the underlying datasets but can edit and save	Yes Shared workbooks include all workbook visualizations and any datasets they're based on.

	their own versions of the datasets using different names. For more information, see Accessing and Sharing Workbooks and Datasets .	
Translatable text?	Yes You can translate the name, description, and fields of a dataset. For more information, see the help topic Manage Translations .	Yes You can translate the name, description, and fields of the workbook and its visualizations. For more information, see the help topic Manage Translations .
Conditional formatting?	— You can sort and filter dataset results but you can't use conditional formatting in them. Use workbook table views and pivot tables to highlight results.	Yes You can highlight or add icons to your table and pivot table results based on rules you set. For more information, see Conditional Formatting .
Dataset linking?	—	Yes. Link definitions are a part of the workbook where you create them, so you can use linked datasets for multiple visualizations, but only in that workbook. For more information, see Dataset Linking in SuiteAnalytics Workbook .

- For information about making a dataset, including how to join record types or create custom formula fields, see [Defining a Dataset](#).
- After you define your dataset, you can filter the results using criteria filters based on existing or custom values. For more information, see [Dataset Criteria Filters](#).
- For information about creating a workbook from a dataset, see [Creating a Workbook](#). You'll find steps for setting up visualizations and getting to your underlying datasets.
- To learn the difference between formula fields and calculated measures, see [Formula Fields and Calculated Measures](#).
- You can link two datasets in your workbooks to analyze metrics from both datasets in a pivot table or chart. For more information, see [Link Datasets in a Workbook](#).

Defining a Dataset

Datasets are the foundation for all workbook visualizations in your account. You can make a new dataset from the Analytics Home page or from the Dataset Panel in a workbook visualization.

In a dataset, you combine the fields of a root record type and any joined related record types to create a query. What you can access depends on your account features and permissions. You can join related record types based on common keys in your analytics data. If standard record types don't have what you need, you can add custom formula fields to your dataset.

In the Dataset Builder, your query results display in the Data Grid. When you create a new dataset, it automatically fills with fields from the root record type. Joinable record types and fields you can use display in the Records and Fields lists. Only fields you add to the Data Grid can be used for workbook visualizations like tables, pivots, and charts.

- To view a record type's fields, click the name of the record type to refresh the Fields list.
- To view additional related record types, click the arrow next to any record type in the Records list.
- To join a record type to a dataset, add fields from the record type to the Data Grid or use the field to create a criteria filter. For more information about dataset criteria filters, see [Dataset Criteria Filters](#).

- In workbook visualizations based on linked datasets, to view the fields within each dataset click the dataset name in the Dataset Panel.

You can join as many record types as you want in a dataset, but joining more than ten can hurt performance.

For more information about joining record types in a dataset, see [Guidelines for Joining Record Types in SuiteAnalytics Workbook](#).

For information about accessing specific record types and fields in NetSuite, download the following worksheet: [NetSuitePermissionsUsage.xls](#).

For information about editing a dataset that is used in a workbook visualization, see [Editing a Dataset](#).

To define a new dataset:

1. Click **New Dataset** on the Dataset subtab of the Analytics Home page or on the **Select a dataset to create a new workbook** page.
Alternatively, click **Connect Dataset** in the Dataset Panel of a workbook visualization, then click **New Dataset** on the **Select a dataset** page.
2. Select a root record type for the dataset. The record type you pick determines the record types and fields you can join to the dataset.
3. Add fields from the root record type to the Data Grid. Here are three ways you can do it:
 - Drag fields from the Fields list to the Data Grid.
 - Double-click field names in the Fields list.
 - Type a field name in the search bar above the Fields list, then drag or double-click it to add it to the Data Grid.
4. Add fields from related record types to the Data Grid.
 - a. Click the arrow next to any record type in the Records list to see more related record types.
 - b. Click a record type name to refresh the Fields list.
 - c. Double-click or drag any field to the Data Grid.



Note: Some record types have multiple versions, like entity job, customer, or vendor in a transaction dataset. Make sure you pick the one that fits your data.

5. (Optional) Click **Formulas** to see, create, and add custom formula fields to the Data Grid.
For step-by-step instructions on creating a custom formula field, see [Formula Fields](#).
6. Remove any fields you don't want to include in the dataset.
 - a. Click the Field Menu icon  in the column you want to remove from the Data Grid.
 - b. Select **Remove Column** from the dropdown list.



Important: If you create a workbook visualization from the current dataset and then remove a field from the Data Grid, it'll also be removed from any other workbook visualizations based on the same dataset. This can cause errors or unwanted results. Exercise caution when taking out fields used in several visualizations.

7. Click **Save** to save the dataset. Or, if you're making the dataset in a new workbook, click **Save & Close** to save it and open the workbook.
The Save Dataset As window will pop-up.
8. Type a name and description for the dataset, then click **Save**.

9. (Optional) To preview your dataset in a new workbook visualization before saving, click **Create New Workbook** or **Apply to workbook**. The workbook will open.

 **Note:** You must save the dataset before you can save the workbook it's connected to. Unsaved datasets show a warning icon  in the Dataset Panel of a workbook. For more information, see [Editing a Dataset](#).

10. (Optional) To export your dataset, click **Export** and pick one of the following options:

- Export data to CSV

 **Important:** To mitigate the possibility of a CSV injection, exported CSV files may have extra formatting. For more information, see [Known Limitations in SuiteAnalytics Workbook](#).

- Export dataset to SDF XML
- Export dataset to SuiteQL TXT

 **Note:** For information about the benefits of exporting dataset definitions to SuiteQL, see [Exporting Dataset Definitions to SuiteQL](#).

Editing a Dataset

Only a dataset owner or someone with the Analytics Administrator permission can save changes to an existing dataset. This helps avoid data discrepancies in visualizations that share the same dataset. If a dataset is shared with you, you can edit it but have to save it with a new name. Anyone with access to a dataset can edit and preview changes in a workbook to see how it affects their visualizations. This means you can have a single dataset for lots of different users and needs.

 **Note:** In workbooks with linked datasets, removing a field used to define a link can affect your results and may keep you from saving.

 **Note:** If your account uses multiple languages and you edit and save default text in a dataset, a new translation will be added to the Manage Translations page. For more information, see the help topic [Manage Translations](#).

To edit a dataset that you own:

1. Click the dataset or workbook name on the Analytics Home page. To quickly filter the available options, select **My Workbooks** or **My Datasets** in the dropdown list of either the Workbooks or Datasets subtabs.
2. Edit the dataset. For information, see [Defining a Dataset](#).
3. Click **Save**. Alternatively, if you accessed the dataset through a connected workbook, click **Save & Close** to save your changes and open the workbook.
4. (Optional) Click **Create New Workbook** or **Apply to workbook** to preview your changes in a workbook visualization before saving the dataset .
The workbook will appear with a warning icon  next to the dataset name, denoting an unsaved dataset.
5. If you are satisfied with your changes, click the arrow next to the dataset name and select **Save Changes**.
If you're not happy with your changes, click the arrow and select **Open Dataset** to continue editing, or **Discard Changes**.



Important: When you save changes to your datasets, they're automatically updated in any related workbook visualizations, even if someone else made them. This can seriously change other workbooks, so if you're not sure how your changes will affect things, try saving the dataset with a new name instead.

To edit a dataset that has been shared with you:

1. Click the dataset or workbook name on the Analytics Home page. To quickly filter the available options, select **Shared with me** in the dropdown list of either the Workbooks or Datasets subtabs.
2. Edit the dataset. For more information, see [Defining a Dataset](#).
3. Click **Save As**.
The Save Dataset As window appears.
4. Enter a name and description for the dataset in the fields provided, then click **Save**.
5. (Optional) Click **Create New Workbook** or **Apply to workbook** to preview your changes in a workbook.
The workbook will appear with a warning icon  next to the dataset name, denoting an unsaved dataset.
6. If you are satisfied with your changes, click the arrow next to the dataset name and select **Open Dataset** to save it using a new name.

Exporting Dataset Definitions to SuiteQL

You can export saved dataset definitions to a SuiteQL TXT file. This helps you understand how fields are joined so you can build SuiteQL queries. SuiteQL queries are available through the SuiteScript Analytic API, SuiteAnalytics Connect, and SuiteTalk REST Web Services.

To export a dataset definition to a SuiteQL TXT file, open your dataset, click **Export** and choose **Export dataset to SuiteQL TXT**. For more information, see [Defining a Dataset](#).



Note: For SuiteAnalytics Connect, this option applies to the NetSuite2.com data source only. The export option is only available for users that have the AnalyticsAdmin permission assigned to their roles. For more information about Permissions, see the help topic [Setting Permissions](#).

When you export your dataset definitions, you must consider the following:

- You can only export dataset definitions that are saved. Unsaved datasets won't show the export option..
- Exported datasets may include internal records or fields not available everywhere. Check the Records Catalog to see which ones are exposed. For information about the Records Catalog, see the help topic [Records Catalog Overview](#).
- Some dataset exports might be incomplete, so you may have to fill in criteria values manually. For example, in this query you'll need to replace "?" with a corresponding value:

```
1 | AND TransactionAccountingLine.posting = ?
```

- With SuiteAnalytics Connect, right outer joins aren't supported. For example, the following Oracle SQL doesn't work in SuiteQL:

```
1 | select a1.id from account a1, account a2 where a1.id (+) = a2.id
```

For information about syntax requirements for SuiteAnalytics Connect, see [SuiteQL Performance and Best Practices for Connect](#).

Formula Fields

In SuiteAnalytics Workbook, you can make custom formula fields to calculate values you can't get from standard record fields. You can add formula fields to your datasets in the Data Grid and use them for filters or in any workbook visualizations based on the dataset. If you try to link datasets but the fields don't match, you can use formulas to cast the values to the needed types.

Custom formula field values update every time you refresh the dataset or connected workbook visualizations. Right now, you can't make custom formula fields based on other formula fields.

To make a custom formula field, combine NetSuite fields and SQL formula functions. You can also customize fields by using formula context values like `#converted`. When you set up a custom formula field, you'll need to pick an output type based on what the formula uses. Workbook supports these output types:

- Boolean- Returns True, False, or NULL values
- Date- Returns day, month, and year values
- Datetime- Returns day, month, year, hour, minute, and second values
- Float- Returns values that include a decimal, such as currency values
- Integer- Returns values that are in whole numbers only, such count values
- String- Returns fixed string character values, such as names
- Clobtext- Returns long unicode character strings, such as item descriptions
- Percent- Returns numeric values as percentages
- Duration- Returns hour and minute values, such as hours worked

Knowing SQL helps you get the most out of the supported SQL functions when defining a new formula. To read a description of each SQL function currently supported in the formula builder, click the Functions subtab of the Formula Field window. From the Functions subtab, you can also add a sample formula definition for each of the listed functions. If you're not sure which function to use, try out this feature in your formula definitions.

For more information about SQL formula functions, see the help topic [SQL Expressions](#).

For more information about using formula fields, see [Creating Formula Fields in SuiteAnalytics Workbook](#).

For sample formulas, see [Basic Sample Formula Fields](#) and [Advanced Sample Formula Fields](#).

In your workbooks, you can also make calculated measures to use in your pivot tables and charts. But using formula fields or calculated measures affects what results you get. To learn the differences between formula fields and calculated measures, see [Formula Fields and Calculated Measures](#).

Creating Formula Fields in SuiteAnalytics Workbook

You can customize formula fields by using SQL functions and context values. After you create formula fields, you can use them in datasets and workbook visualizations. When working with custom formula fields, keep these points in mind:

- [Creating a Custom Formula Field in SuiteAnalytics Workbook](#)
- [Creating Criteria Filters Using Formula Fields](#)
- [Using Context Values in Formula Fields](#)
- [Adding Formula Fields to a Workbook Visualization](#)

Creating a Custom Formula Field in SuiteAnalytics Workbook

In the following example, you'll create a formula field to convert data values to another format. The Formula Field window only shows fields from the root record type and any related fields added to the Data Grid.



Important: To do arithmetic calculations or view totals for fields with values in more than one currency, you must convert or consolidate the values to a specific currency. In the Dataset Builder, you can only do this using custom formula fields. For more information, see [Currency Conversion Using Custom Formula Fields](#).

To create a custom formula field in SuiteAnalytics Workbook:

1. Click **Formulas** above the Records list on the Dataset Builder.
2. In the Fields list, click **New Formula**.

The Formula Field window appears.

- DATA GRID -	
Transaction	trandisplayname
Date	trandate
Memo	memo
Type	type

3. Enter a Formula Field Name.
For this example, enter **Formatted Date**.
4. Choose an Output Type for the formula field values.
For this example, select **String**.

Formula Field

FORMULA FIELD NAME* ⚠️
Type in new field name

FORMULA*
1 Type in your formula

CHOOSE OUTPUT TYPE*
STRING
BOOLEAN
CLOBTEXT
DATE
DATETIME
DURATION
FLOAT
INTEGER
PERCENT
STRING

Need help with defining the formula?

Fields Functions

CHOOSE SOURCE
Show all available fields

Search through records, fields, and field IDs 🔍

- DATA GRID -	
Transaction	trandisplayname
Date	trandate
Memo	memo
Type	type

Save Cancel

Note: Each output type only works with certain fields and formula functions. If you select an incompatible output type for the formula, the formula is invalidated and the Formula Builder highlights the error in the definition. To view details about the validation error and for information about how to correct the definition, click the error icon ⚠️.

- In the Formula field, enter the field IDs and SQL functions you want to use in the formula. You can also double-click field IDs or formula functions from the Functions or Fields tabs to add them.

For this example, select the TO_CHAR and replace the content in the brackets with **{trandate}**, **'Month DD, YYYY'**.

Formula Field

FORMULA FIELD NAME*
Formatted Date

CHOOSE OUTPUT TYPE*
STRING

FORMULA*

```
1 TO_CHAR({trandate}, 'Month DD, YYYY')
```

Need help with defining the formula? [Validate](#)

Fields Functions

CHOOSE SOURCE
Show all available fields

Search through records, fields, and field IDs

- DATA GRID -	
Transaction	trandisplayname
Date	trandate
Memo	memo
Type	type

[Save](#) [Cancel](#)

- To validate the formula, click **Validate**.

Note: If there are errors with the formula, click the error icon  to view the details.

- Click **Apply** to add the formula field to your dataset.

The field appears in the Field lists on the Dataset Builder when you click **Formulas**. If you add the field to the Data Grid, you can use it in any workbook visualizations that use this dataset.

Creating Criteria Filters Using Formula Fields

In this example, you'll use the Criteria Builder to make a filter with a formula field.

To create criteria filters using formula fields:

- Create a valid formula field on the Dataset Builder.
- Double-click or drag the field from the Fields list or the Data Grid to the Criteria Builder.
- Set your filter conditions in the Filter window. For more details, see [Filter Types](#).

Using Context Values in Formula Fields

You can use different context values in fields to set how values are shown. For example, you can show a field's name instead of its ID.

The following example shows a formula definition where *type* is the field and *display* is the context value that defines how results are displayed. Context values are placed after the number sign # within the same parentheses containing the field name in a formula definition.

Example: {type#display}

Before you start working with context values, keep these things in mind:

- [Supported Context Values](#)
- [Editing Context Values](#)

Supported Context Values

SuiteAnalytics Workbook supports the following context values:

Context Value	Description
#currency_consolidated	Shows consolidated currency amounts in the currency of the lowest-level parent subsidiary you can access. The exchange rate for the consolidation is based on the rate in effect when you run the query or visualization. For more information, see Currency in Datasets and Workbooks. Formula example: ■ ID of the field used for this example: amount ■ Formula output type: FLOAT ■ SQL function required: TO_NUMBER() ■ Formula definition: TO_NUMBER({amount#currency_consolidated})
#converted	Shows converted currency amounts in your account's base currency. The exchange rate for the conversion is based on the rate in effect when you run the query or visualization. For more information, see Currency in Datasets and Workbooks. Formula example: ■ ID of the field used for this example: amount ■ Formula output type: FLOAT ■ SQL function required: TO_NUMBER() ■ Formula definition: TO_NUMBER({amount#converted})
#converted[1]	Shows the currency amounts in the currency ID specified in the brackets [1]. The exchange rate is based on the run date for your query or visualization. For more information, see Currency in Datasets and Workbooks.  Note: The currency ID 1 corresponds to the base currency. Formula example: ■ ID of the field used this example: amount ■ Formula output type: FLOAT ■ SQL function required: TO_NUMBER() ■ Formula definition: TO_NUMBER({amount#converted[1]})
#converted[1;yyyy-mm-dd]	Shows the currency amounts in the currency specified in the brackets [1], using the exchange rate in effect on the date you specify (use YYYY-MM-DD format). By default, the currency ID is set to 1. For more information, see Currency in Datasets and Workbooks.

Context Value	Description
	 Note: The currency ID 1 corresponds to the base currency. Formula example: ■ ID of the field used for this example: amount ■ Formula output type: FLOAT ■ SQL function required: TO_NUMBER() ■ Formula definition: TO_NUMBER({amount#converted[1;2020-03-10]})
#display	Shows field values in a more user-friendly way. For the type field, you'll see labels like bill, cash refund, or inventory adjustment. Formula example: ■ ID of the field used for this example: type ■ Formula output type: INTEGER ■ SQL function required: none ■ Formula definition: {type#display}
#raw	Shows raw field values, like internal IDs. For the type field, the raw value shows the transaction type ID.  Note: The ID is a unique identifier that doesn't change. If a field name changes, you can still find it using the ID instead of the name. For example, you'd get CashRfnd instead of cash refund. Formula example: ■ ID of the field used for this example: type ■ Formula output type: INTEGER ■ SQL function required: none ■ Formula definition: {type#raw}
#hierarchy	Shows more user-friendly field values for hierarchical fields. This context value is similar to #display but applies to hierarchical fields. Formula example: ■ ID of the field used for this example: subsidiary ■ Formula output type: STRING ■ SQL function required: none ■ Formula definition: {subsidiary#hierarchy}
#hierarchy_identifier	Shows more user-friendly field values for hierarchical fields. This value is similar to #display but applies to hierarchical fields Formula example: ■ ID of the field used for this example: subsidiary ■ Formula output type: FLOAT ■ SQL function required: TO_NUMBER() ■ Formula definition: TO_NUMBER({subsidiary#hierarchy_identifier})

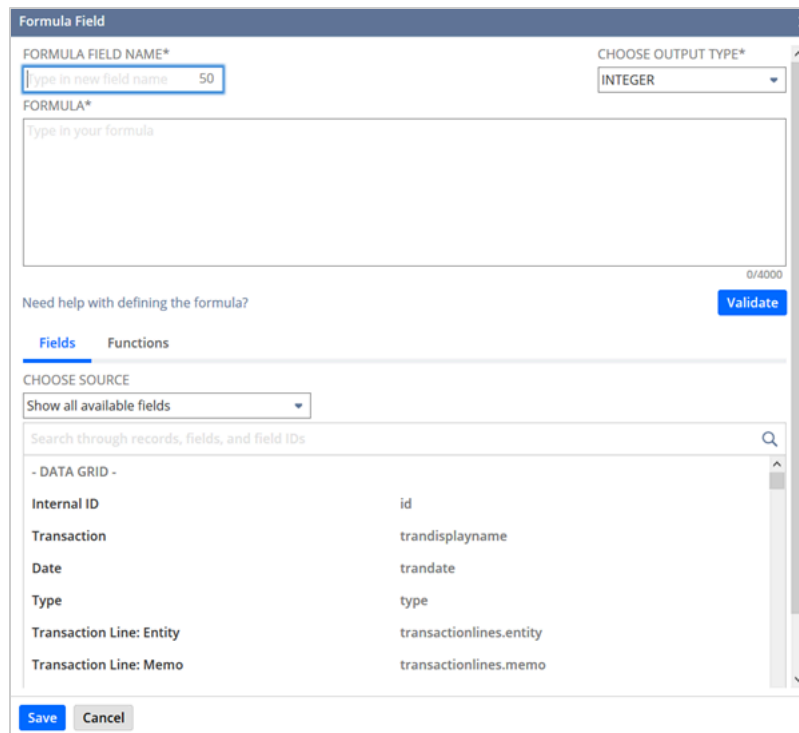
Editing Context Values

In this example, you can change the context value of the type field. By default, the type field displays names, but if you change the context value, it displays the ID instead.

Note: This example is applicable for any record type that includes the type field. The following example is the transaction record type, available in the Transaction Detail dataset template.

To edit the context value in a formula

1. Click **Formulas** above the Records list on the Dataset Builder.
2. In the Fields list, click **New Formula**.



The Formula Field window appears.

3. Enter a Formula Field Name.
For example, enter **Type (ID)**.
4. Choose an Output Type for the formula field values.
For the purposes of this procedure, select **String**.
5. Double-click fields from the Fields subtab to add them to the FORMULA text box.
For the purposes of this procedure, double-click Type.
The formula will appear in the **Formula** field. In this example, {type#display}.
6. In the **Formula** field, replace what's after the # sign with the value you want.
For this example, remove #display and enter #raw.
7. To validate the formula, click **Validate**.

Note: If there are errors with the formula, click the error icon (warning) to view the details.

8. Click **Apply** to add the formula field to the dataset.

The field appears in the Field lists on the Dataset Builder when you click **Formulas**. If you add it to the Data Grid, you can use it in any workbook visualization based on that dataset. In those visualizations, the type field shows the ID of the values.

Adding Formula Fields to a Workbook Visualization

In this example, you add a formula field to a workbook and then place it on a table, pivot, or chart of the workbook.

To add formula fields to a workbook visualization:

1. Create a valid formula field on the Dataset Builder.
2. Add the field to the Data Grid so that it is included in any workbook visualizations based on the dataset.
3. On the Table, Pivot, or Chart tabs of the workbook, drag the field from the Dataset Panel to a section of the Layout panel.

Basic Sample Formula Fields

The SuiteAnalytics Workbook Formula Builder lets you create custom formula fields for these output types:

- Boolean- Returns True, False, or NULL values
- Date- Returns day, month, and year values
- Datetime- Returns day, month, year, hour, minute, and second values
- Float- Returns values that include a decimal, such as currency values
- Integer- Returns values that are in whole numbers only, such count values
- String- Returns fixed string character values, such as names
- Clobtext- Returns long unicode character strings, such as item descriptions
- Percent- Returns numeric values as percentages
- Duration- Returns hour and minute values, such as hours worked

See the table below for some basic formulas you can create for each output type:

Output Type	Base Record	Formula	Description
BOOLEAN	Transaction	<pre>1 CASE WHEN {transaction^transaction.daysoverduesearch} > 0 THEN 'T' ELSE 'F'END</pre>	Identifies overdue transactions
DATE	Employee	<pre>1 CASE WHEN {lastreviewdate} IS NULL 2 THEN {hiredate} 3 ELSE {lastreviewdate} 4 END</pre>	Displays the last review date of an employee's salary
DATETIME	Any	<pre>1 CURRENT_TIMESTAMP</pre>	Displays a time stamp for each result
FLOAT	Transaction	<pre>1 {foreigntotal}*-1</pre>	Changes the - sign applied to foreign amounts
INTEGER	Revenue Recognition Plan	<pre>1 CEIL(MONTHS_BETWEEN(CURRENT_DATE, {plannedrev</pre>	Displays the number of months since the revenue plan was created

Output Type	Base Record	Formula	Description
		<code>enue.plannedperiod.end date)))</code>	
STRING	Transaction	<pre> 1 CASE WHEN TRUNC(CURREN T_DATE)-{duedate} <= 7 THEN 'Week 1 (Current)' WHEN TRUNC(CURREN T_DATE)-{duedate} <= 14 THEN 'Week 2' ELSE '...' 2 END </pre>	Displays the number of weeks a transaction is overdue
CLOBTEXT	Item	<pre> 1 {storedetaileddescrip tion} </pre>	Displays a detailed description of the item, which can include HTML code. This can be displayed to web store users.
PERCENT	Transaction	<pre> 1 CASE WHEN {transaction lines.quantity} > 0 2 THEN {transac tionlines.quantity billed}/{transaction lines.quantity} 3 ELSE 0 4 END </pre>	Displays how much of the billing process is done, as a percentage
DURATION	Time Tracking	<pre> 1 CASE WHEN {isbillable} = 'F' THEN {hours} else 0 END </pre>	Displays non-billable time

For more advanced formulas, see [Advanced Sample Formula Fields](#).

Advanced Sample Formula Fields

The SuiteAnalytics Workbook Formula Builder lets you to create custom formula fields using any functions in the [SuiteAnalyticsWorkbookFormulaFunctions.xls](#) spreadsheet. You can add sample formulas for any of these functions directly from the Functions subtab in the Formula Builder using the **Add the example to the formula** link. See the sections below for more sample formulas you can create using the available functions:

- [Calculating Duration Values with TO_NUMBER and TO_NCHAR](#)
- [Combining CONCAT and other Functions to Calculate String Values](#)
- [Casting Field Values using TO_NUMBER and TO_NCHAR](#)
- [Currency Consolidation and Conversion Using Custom Formula Fields](#)
- [Conditional Evaluations Using CASE WHEN](#)
- [Calculating Amounts for Relative Date Ranges](#)
- [Calculating Specific Dates](#)

For more sample formulas, see the help topic [Search Formula Examples and Tips](#).

For information about using custom formula fields for currency conversion or consolidation, see [Currency Conversion Using Custom Formula Fields](#).

Calculating Duration Values with TO_NUMBER and TO_NCHAR

Duration values are incompatible with the TO_NUMBER function unless they are first converted to a variable format using the TO_NCHAR function. So if you want to make a custom formula field that shows duration values as a number, use this formula:


```
1 | TO_NUMBER(TO_NCHAR({field_ID}))
```

For example, let's say you want to see the time elapsed for each case in your account, in hours.

CREATION DATE/TIME	TIME ELAPSED	PRIORITY	ASSIGNED TO	STATUS	SUBJECT
6/4/2010 12:33 pm	83649:00	Medium	Cameron Murdock	Re-Opened	Access to Mana
6/16/2010 5:27 pm	1774:00	Medium	Edwin Goldwasser	Closed	Additional Trai
8/5/2010 12:55 pm	578:00		Aubrey Pober	Closed	Auctions going
11/10/2010 11:24 am	81369:00	Medium	German Support Rep	Not Started	Auftrag nicht er
9/3/2010 12:37 pm	81752:00	Medium	suite analytics	Escalated	Auto Update
11/10/2010 11:29 am	81369:00	Medium	German Support Rep	Escalated	Can't addieren
7/20/2010 11:31 am	82834:00	High	suite analytics	Escalated	Can't import pr
6/17/2010 6:25 am	1186:00	Medium	Cameron Murdock	Closed	Changing Your
6/18/2010 1:50 pm	1653:00		suite analytics	Closed	Config Problem
6/29/2010 11:26 am	1513:00	High	suite analytics	Closed	Data Configura
6/17/2010 6:21 am	83367:00	Medium	Edwin Goldwasser	Escalated	Enhancement t

In this case, use this formula with an output type of Float:

```
1 | TO_NUMBER(TO_NCHAR({timeelapsed}))
```

Formula Field

FORMULA FIELD NAME*
Time Elapsed (Hours)

CHOOSE OUTPUT TYPE*
FLOAT

FORMULA*
TO_NUMBER(TO_NCHAR({timeelapsed}))

Need help with defining the formula?
Validate

Fields Functions

CHOOSE SOURCE
Show all available fields

Search through records, fields, and field IDs

- DATA GRID -	
Creation Date/Time	datecreated
Time Elapsed	timeelapsed
Priority	priority
Assigned To	assigned
Status	status
Subject	title
Company	company
Type	category

Save Cancel

CREATION DATE/TIME	TIME ELAPSED	TIME ELAPSED (HOURS)	PRIORITY	ASSIGNED TO
6/4/2010 12:33 pm	83663:00	83,663.00	Medium	Cameron Murdock
6/16/2010 5:27 pm	1774:00	1,774.00	Medium	Edwin Goldwasser
8/5/2010 12:55 pm	578:00	578.00	Medium	Aubrey Pober
11/10/2010 11:24 am	81383:00	81,383.00	Medium	German Support Rep
9/3/2010 12:37 pm	81766:00	81,766.00	Medium	suite analytics
11/10/2010 11:29 am	81383:00	81,383.00	Medium	German Support Rep
7/20/2010 11:31 am	82848:00	82,848.00	High	suite analytics
6/17/2010 6:25 am	1186:00	1,186.00	Medium	Cameron Murdock
6/18/2010 1:50 pm	1653:00	1,653.00	Medium	suite analytics
6/29/2010 11:26 am	1513:00	1,513.00	High	suite analytics
6/17/2010 6:21 am	83381:00	83,381.00	Medium	Edwin Goldwasser

Combining CONCAT and other Functions to Calculate String Values

You can combine the CONCAT function with other functions to calculate complex string values. For example, to show duration values in days and hours, first normalize the values with TO_NUMBER and TO_NCHAR, then use this formula. The output type is STRING:

```
1 CONCAT(CONCAT('Days: ', TO_NCHAR(FLOOR(TO_NUMBER(TO_NCHAR({timeelapsed})/24))))),
2 CONCAT(' Hours: ', TO_NCHAR(MOD(TO_NUMBER(TO_NCHAR({timeelapsed})), 24)))
```

Formula Field

FORMULA FIELD NAME*
Time Elapsed (Days a...)

CHOOSE OUTPUT TYPE*
STRING

FORMULA*
CONCAT(CONCAT('Days: ', TO_NCHAR(FLOOR(TO_NUMBER(TO_NCHAR({timeelapsed})/24))))),
CONCAT(' Hours: ', TO_NCHAR(MOD(TO_NUMBER(TO_NCHAR({timeelapsed})), 24)))

Need help with defining the formula? [Validate](#)

Fields Functions

CHOOSE SOURCE
Show all available fields

Search through records, fields, and field IDs

- DATA GRID -

Creation Date/Time	datecreated
Time Elapsed	timeelapsed
Priority	priority
Assigned To	assigned
Status	status
Subject	title
Company	company
Type	category

[Save](#) [Cancel](#)

The screenshot shows the NetSuite Case Dataset interface. On the left, there's a sidebar with a search bar and a list of formulas. The main area displays a table with the following columns: CREATION DATE/TIME, TIME ELAPSED, TIME ELAPSED (DAYS AND HOURS), PRIORITY, and ASSIGNED TO. A red box highlights the 'TIME ELAPSED (DAYS AND HOURS)' column. The table contains 10 rows of data. At the bottom, it says 'Total row count: 101'.

CREATION DATE/TIME	TIME ELAPSED	TIME ELAPSED (DAYS AND HOURS)	PRIORITY	ASSIGNED TO
6/4/2010 12:33 pm	83664:00	Days: 3486 Hours: 0	Medium	Cameron Murdock
6/16/2010 5:27 pm	1774:00	Days: 73 Hours: 22	Medium	Edwin Goldwasser
8/5/2010 12:55 pm	578:00	Days: 24 Hours: 2	Medium	Aubrey Pober
11/10/2010 11:24 am	81384:00	Days: 3391 Hours: 0	Medium	German Support Rep
9/3/2010 12:37 pm	81767:00	Days: 3406 Hours: 23	Medium	suite analytics
11/10/2010 11:29 am	81384:00	Days: 3391 Hours: 0	Medium	German Support Rep
7/20/2010 11:31 am	82849:00	Days: 3452 Hours: 1	High	suite analytics
6/17/2010 6:25 am	1186:00	Days: 49 Hours: 10	Medium	Cameron Murdock
6/18/2010 1:50 pm	1653:00	Days: 68 Hours: 21	Medium	suite analytics
6/29/2010 11:26 am	1513:00	Days: 63 Hours: 1	High	suite analytics
6/17/2010 6:21 am	83382:00	Days: 3474 Hours: 6	Medium	Edwin Goldwasser

Try using different formula functions with the CONCAT function to calculate other string values.

Casting Field Values using TO_NUMBER and TO_NCHAR

Fields in NetSuite store values in different data types, such as STRING, INTEGER, and FLOAT.

- String- text values, such as a customer name
- Integer- positive or negative whole numbers that do not contain a decimal, such as counts
- Float- numeric values that can contain a decimal, such as amounts

You may need to change a field to another data type before using it in a custom formula or running calculations. For example, if item price is an INTEGER and cost is a FLOAT, use this formula to get the difference:

```
1 | TO_NUMBER({item<pricing.unitprice}) - TO_NUMBER({cost})
```

Formula Field

FORMULA FIELD NAME*
Possible Profit

CHOOSE OUTPUT TYPE*
FLOAT

FORMULA* ✓
TO_NUMBER({item<pricing.unitprice}) - TO_NUMBER({cost})

55/4000

Need help with defining the formula? [Validate](#)

Fields Functions

CHOOSE SOURCE
Show all available fields

Search through records, fields, and field IDs

- DATA GRID -

Item Name/Number	itemid
------------------	--------

- ITEM -

Item Name/Number	itemid
[Missing Label:hashOfLinkedBoms]	hashoflinkedboms
Allocation Price	vsoeprice
Allocation Type	vsoesopgroup
Alternate Source Item	alternatedemandsourceitem
Amortization Period	amortizationperiod

[Save](#) [Cancel](#)

Item Dataset

Search for records and fields...

Formulas

Item

Amortization Template

Asset Account Account

Bed Size

Bed Size 3 Bed Sizes

Billing Schedule

Class

Color

Colors

Consumption Units Units

Correlated Items

Customer Return Variance A...

Default WT Code Withholdin...

Deferred Expense Account

Department

Dropship Expense Account A...

Exchange Rate Variance Acc...

Expense/COGS Account Acco...

Field [Asset/Sales Tax Item] ...

Field [Inventory Item] on IP...

Possible profit

New Formula

Drop fields here to add criteria.

ITEM NAME/NUMBER	POSSIBLE PROFIT
Armless Dining Chair	18.00
Armless Dining Chair	18.00
Armless Dining Chair	18.00
Armless Dining Chair	18.00
Armless Dining Chair	18.00
Armless Dining Chair	7.80
Armless Dining Chair	7.80
Armless Dining Chair	7.80
Armless Dining Chair	7.80
Armless Dining Chair	7.80
Armless Dining Chair	7.80
Armless Dining Chair	7.80
Armless Dining Chair	7.80
Armless Dining Chair	7.80
Armless Dining Chair	7.80

Total row count: 675

1 of 2

Use the formulas in the table below if you need to change a field to a different data type:

Data Type	To cast to STRING	To cast to INTEGER	To cast to FLOAT
INTEGER	TO_CHAR({field_ID})	—	Automatic casting
FLOAT	TO_CHAR({field_ID})	CEIL({field_ID}) FLOOR({field_ID}) ROUND({field_ID})	—

		TRUNC({field_ID})	
BOOLEAN	Automatic casting, returns 'T' or 'F'	CASE field WHEN 'T' THEN 1 ELSE 0 END	CASE field WHEN 'T' THEN 1 ELSE 0 END
DATE	TO_CHAR({field_ID})	—	—
DATETIME	TO_CHAR({field_ID})	—	—
PERCENT	TO_CHAR({field_ID}) Returns values between 0 and 1	—	TO_NUMBER({field_ID})
DURATION	TO_CHAR({field_ID}) Returns FLOAT values, not in the hours:minutes format	—	TO_NUMBER({field_ID})
STRING	—	TO_NUMBER({field_ID}) Not preferred	TO_NUMBER({field_ID})
CURRENCY	TO_CHAR({field_ID}) Returns values without the currency symbol	—	TO_NUMBER({field_ID})
CURRENCY_HIGH_PRECISION	TO_CHAR({field_ID}) Returns values without the currency symbol	—	TO_NUMBER({field_ID})

Currency Consolidation and Conversion Using Custom Formula Fields

You can consolidate or convert values in multiple currencies using the context values `#currency_consolidated` or `#converted`. In the Dataset Builder, you'll need this if you want to see totals or run calculations on fields with multiple currencies. In workbook visualizations, any field with values in more than one currency will have conversion options right in the interface. For more information, see [Currency in Datasets and Workbooks](#).

In Workbook, currency consolidation is only applicable to amount fields from the transaction accounting line record type. Currency conversion is only applicable to amount fields from the transaction and transaction line record types.

For example, to consolidate values from the amount net field in the transaction accounting line record type, use this formula:

```
1 | TO_NUMBER({transactionlines.accountingimpact.netamount#currency_consolidated})
```

To convert the values from the amount paid field on the transaction record type, use this definition:

```
1 | TO_NUMBER({foreignamountpaid#converted})
```

With `#consolidated`, you can also choose the specific currency code and exchange rate date if needed. For more information, see [Currency Conversion Using Custom Formula Fields](#).

You always need to use `TO_NUMBER` with `#currency_consolidated` or `#converted`. That's because the current Formula Builder doesn't support the `CURRENCY` output type, so `TO_NUMBER` changes results to `FLOAT`. For more information about using `TO_NUMBER` to cast currency values, see [Casting Field Values using TO_NUMBER and TO_NCHAR](#).

Conditional Evaluations Using CASE WHEN

You can perform conditional evaluations by making a formula field with the CASE WHEN function. In these formulas, if values meet the conditions in WHEN you get the result from THEN; all others use the result from ELSE.

For example, to find out which season a transaction occurred, use this formula. The output type is STRING:

```
1 CASE
2   WHEN EXTRACT(Month FROM {trandate})= 12 THEN 'winter'
3   WHEN EXTRACT(Month FROM {trandate})= 6 THEN 'summer'
4   ELSE 'it was fall or spring' END
```

Formula Field

×

FORMULA FIELD NAME*

Season

CHOOSE OUTPUT TYPE*

STRING

FORMULA*✓

```
1 CASE
2   WHEN EXTRACT(Month FROM {trandate})= 12 THEN 'winter'
3   WHEN EXTRACT(Month FROM {trandate})= 6 THEN 'summer'
4   ELSE 'it was fall or spring'
5   END
```

Need help with defining the formula?

Validate

Fields

Functions

Search functions

Q

- MOST RECENTLY USED -

- ALL -

ABS

ACOS

ADD_MONTHS

ASCI

ASCIISTR

ABS(n)

Example

ABS(-2.57)

[Add the example to the formula](#)

The ABS function calculates the absolute value of an expression. Because the absolute value of a real number is its numeric value without regard to its sign, this function always returns a positive value. For example, 3 is the absolute value of both 3 and -3.

Save

Cancel

TRANSACTION	DATE	SEASON	MEMO	TYPE	TOTAL AMOUNT (TRANSACTION CURRENCY)
Check #2074	12/5/2004	winter	WE00001	Check	-\$3
Check #2028	12/15/2002	winter	451-9935	Check	-\$1
Payment #12	12/8/2002	winter		Payment	\$1
Check #2053	12/15/2003	winter	451-9935	Check	-\$1
Check #2050	12/5/2003	winter	WE00001	Check	-\$3
Check #2027	12/5/2002	winter	WE00001	Check	-\$3
Check #2077	12/15/2004	winter	451-9935	Check	-\$1
Check #2065	6/15/2004	summer	451-9935	Check	-\$1
Check #2062	6/5/2004	summer	WE00001	Check	-\$3
Check #2041	6/15/2003	summer	451-9935	Check	-\$1

In Workbook, you can nest different CASE WHEN statements within the same formula.

Calculating Amounts for Relative Date Ranges

You can create formula fields to calculate amounts for relative date ranges. This includes transaction sales amounts, quantities, and other numerical values. Depending on the date range, here are some formulas you can use. The output type for each of these formulas is FLOAT:



Important: If you're working with values in multiple currencies, make sure you consolidate or convert the field values to a single currency in your formula. For more information, see [Currency Consolidation and Conversion Using Custom Formula Fields](#).



Note: These samples work for formula fields in datasets based on the transaction record type. If you're working on a dataset based on the sales (ordered) or sales (invoiced) analytical record types, replace {foreigntotal} with {amountnet}. If you'd like to calculate other values, like quantity sold, replace {foreigntotal} with the right field ID.

- Amount Last Year to Date
- Amount This Year to Date
- Amount This Year
- Amount Last Year
- Amount This Quarter to Date
- Amount Last Quarter
- Amount This Month to Date
- Amount Last Month to Date

After you've created your formulas, you can use them to make a calculated measure and analyze period-over-period variances. For more information, see [Year over Year Growth](#).

Amount Last Year to Date

Use this formula to get transaction amounts from all of last year up to today:

```
1 CASE WHEN (ADD_MONTHS({trandate}, 12) <= CURRENT_DATE) AND (EXTRACT(YEAR FROM {trandate}) + 1 = EXTRACT(YEAR FROM CURRENT_DATE))
   THEN TO_NUMBER({foreigntotal})
```

```
2 | END
```

Amount This Year to Date

Use this formula to get transaction amounts from the start of this year up to today:

```
1 | CASE WHEN {trandate} <= CURRENT_DATE AND EXTRACT(YEAR FROM {trandate}) = EXTRACT(YEAR FROM CURRENT_DATE) THEN TO_NUMBER({foreigntotal})
2 | END
```

Amount This Year

Use this formula to get the total amounts for transactions created this year:

```
1 | CASE WHEN EXTRACT(YEAR FROM {trandate}) = EXTRACT(YEAR FROM CURRENT_DATE) THEN {foreigntotal}
2 | END
```

Amount Last Year

Use this formula to get the total amounts for transactions created last year:

```
1 | CASE WHEN EXTRACT(YEAR FROM {trandate}) + 1 = EXTRACT(YEAR FROM CURRENT_DATE) THEN TO_NUMBER({foreigntotal})
2 | END
```

Amount This Quarter to Date

Use this formula to get amounts for transactions created during this quarter:

```
1 | CASE WHEN TO_CHAR({trandate}, 'Q') = TO_CHAR(CURRENT_DATE, 'Q') AND EXTRACT(YEAR FROM {trandate}) = EXTRACT(YEAR FROM CURRENT_DATE)
   THEN TO_NUMBER({foreigntotal})
2 | END
```

Amount Last Quarter

Use this formula to get transaction amounts from last quarter:

```
1 | CASE
2 |   WHEN TO_NUMBER(TO_CHAR(CURRENT_DATE, 'Q')) > 1 AND TO_NUMBER(TO_CHAR({trandate}, 'Q')) = TO_NUMBER(TO_CHAR(CURRENT_DATE, 'Q'))-1 AND
   EXTRACT(YEAR FROM {trandate}) = EXTRACT(YEAR FROM CURRENT_DATE)
3 |   THEN TO_NUMBER({foreigntotal})
4 |
5 |   WHEN TO_NUMBER(TO_CHAR(CURRENT_DATE, 'Q')) = 1 AND TO_NUMBER(TO_CHAR({trandate}, 'Q')) = 4 AND EXTRACT(YEAR FROM {trandate})+1 =
   EXTRACT(YEAR FROM CURRENT_DATE)
6 |   THEN TO_NUMBER({foreigntotal})
7 | END
```

Amount This Month to Date

Use this formula to get amounts for transactions from the start of this month up to today:

```
1 | CASE
2 |   WHEN {trandate} <= CURRENT_DATE AND EXTRACT(Month FROM {trandate}) = EXTRACT(Month FROM CURRENT_DATE) AND EXTRACT(YEAR FROM
   {trandate}) = EXTRACT(YEAR FROM CURRENT_DATE)
3 |   THEN TO_NUMBER({foreigntotal})
```



```
4 | END
```

Amount Last Month to Date

Use this formula to get transaction amounts from all of last month up to today:

```
1 | CASE
2 |   WHEN (ADD_MONTHS({trandate}, 1) <= CURRENT_DATE) AND EXTRACT(Month FROM ADD_MONTHS({trandate}, 1)) = EXTRACT(Month FROM CURREN
   T_DATE) AND EXTRACT(YEAR FROM ADD_MONTHS({trandate}, 1)) = EXTRACT(YEAR FROM CURRENT_DATE)
3 |   THEN TO_NUMBER({foreigntotal})
4 | END
```

Calculating Specific Dates

Use these formulas to calculate specific dates. The output type for each of these formulas is DATE:

- [First Day of This Month](#)
- [Last Day of This Month](#)
- [First Day of Last Month](#)
- [Last Day of Last Month](#)

First Day of This Month

Use this formula to get the first day of the month:

```
1 | TRUNC(LAST_DAY(CURRENT_DATE)-1, 'MONTH')
```

Last Day of This Month

Use this formula to get the last day of the month:

```
1 | LAST_DAY(CURRENT_DATE)
```

First Day of Last Month

Use this formula to get the first day of the last month:

```
1 | TRUNC(LAST_DAY(ADD_MONTHS(CURRENT_DATE,-1)), 'MONTH')
```

Last Day of Last Month

Use this formula to get the last day of the last month:

```
1 | LAST_DAY(ADD_MONTHS(CURRENT_DATE,-1))
```

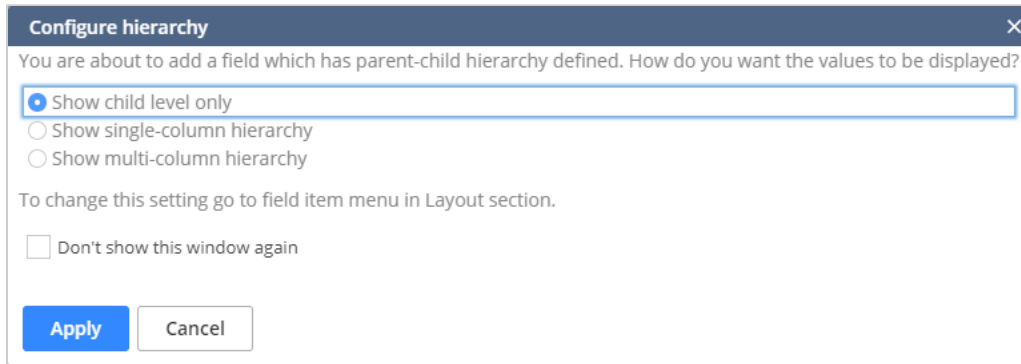
Hierarchical Fields

SuiteAnalytics Workbook supports hierarchical fields. Hierarchical fields work like standard record fields, but you can choose to show field values as hierarchies or child values. You can also set filter conditions at the parent or child level of the hierarchy.

Display Options for Hierarchical Fields

When you add a hierarchical field to a pivot table or chart, you'll be asked to pick how you'd like to show the field values.

You have the following options:



Configure hierarchy [X]

You are about to add a field which has parent-child hierarchy defined. How do you want the values to be displayed?

☒ Show child level only

☐ Show single-column hierarchy

☐ Show multi-column hierarchy

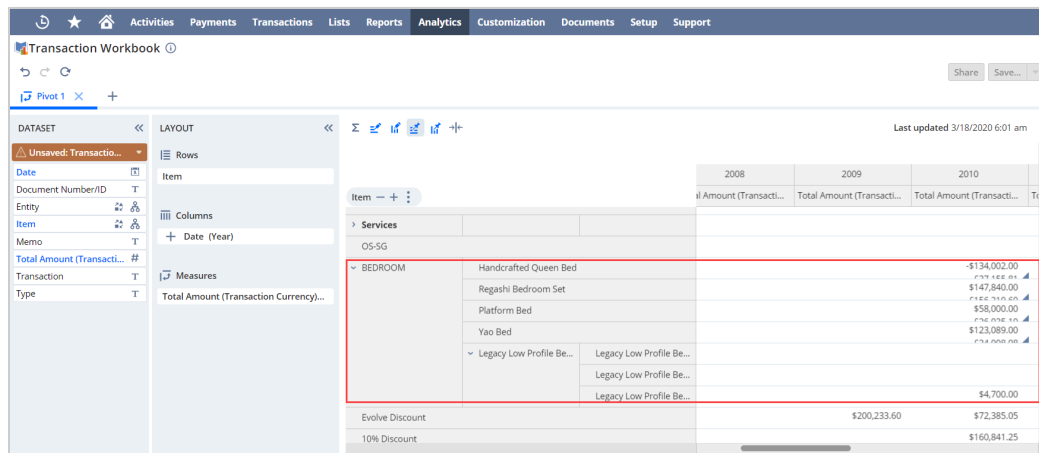
To change this setting go to field item menu in Layout section.

☐ Don't show this window again

Apply **Cancel**

- Select **Show child level only** to display only the child-level values for the field.
- Select **Show single-column hierarchy** to display the full hierarchy values for the field in one column or chart measure.
- Select **Show multi-column hierarchy** to display each hierarchy level in a separate table column or chart measurement. If you select this and enable totals or grand totals for the pivot table, you'll see totals for each hierarchy level.

For example, in the table below, the Item field uses a multi-column hierarchy. The table then calculates totals for each bedroom item and for the bedroom parent level.



The screenshot shows the SuiteAnalytics Workbook interface. The Dataset panel on the left lists fields: Date, Document Number/ID, Entry, Item, Memo, Total Amount (Transaction...), Transaction, and Type. The Layout panel shows Rows (Item) and Columns (Date (Year)). The Measures panel shows Total Amount (Transaction Currency...). The main table displays data for the years 2008, 2009, and 2010. The Item field is expanded to show a multi-column hierarchy for the BEDROOM category, including Handcrafted Queen Bed, Regashi Bedroom Set, Platform Bed, Yao Bed, and Legacy Low Profile Be... (repeated three times). The table also shows totals for the BEDROOM category and for the entire dataset.

	2008	2009	2010	Total
Item				
Services				
OS-SG				
BEDROOM				
Handcrafted Queen Bed				
Regashi Bedroom Set				
Platform Bed				
Yao Bed				
Legacy Low Profile Be...				
Legacy Low Profile Be...				
Legacy Low Profile Be...				
Evolve Discount		\$200,233.60		\$72,385.05
10% Discount				\$160,841.25

To change the hierarchy display for a field after adding it to a pivot table or chart, click the Field Menu icon [⋮] in the Layout panel and select a new display option.

The screenshot shows the SuiteAnalytics Workbook interface. On the left, the 'DATASET' pane lists fields like Date, Document Number/ID, Entity, Item, Memo, Total Amount (Transaction...), Transaction, and Type. The 'LAYOUT' pane shows a pivot table with 'Item' in the Rows area and 'Total Amount (Transaction Currency)...' in the Measures area. A filter menu is open for the 'Item' field, showing options like 'Bottom 10 Item', 'Filter Item by...', 'Show Grand Total', 'Show Hierarchy Sub-totals', 'Rename...', 'Display: Child Level Only', 'Single-Column Hierarchy', 'Multi-Column Hierarchy' (which is selected and highlighted with a red box), and 'Remove'. The pivot table displays data for the years 2008, 2009, and 2010, with columns for 'Total Amount (Transaction...)'.

Filter Conditions for Hierarchical Fields

When you create a filter condition based on a hierarchical field, you'll see an extra **Show Hierarchy** option in the Filter window.

The screenshot shows the 'Filter: Class' dialog box. It has a 'Values' section with 'Field Values' and 'Custom Values' tabs. The 'Field Values' tab is active, showing a list of values: '<NO VALUE>', 'Repeat Business', 'New Business', 'Corporate Sales', 'Repeat Customer', 'Repeat Customer', 'New Customer', 'New Customer', 'Retail Store Business', and 'Web Store Business'. The 'Custom Values' tab is empty, showing 'No items selected'. At the bottom, there is a 'Show Hierarchy' option with a radio button and a red box around it. The 'OK', 'Cancel', and 'Reset' buttons are at the bottom.

When you turn on this option, the Filter window shows only the parent values for the field you picked. If you set a filter at the parent level, you'll only see records with that parent in your pivot table or chart. For example, say your dataset has two records with a class of New Customer, but only one has a parent class

of Web Store Business. If you set a filter to exclude records with a parent value of Web Store Business, one New Customer record will be removed from your workbook visualization.

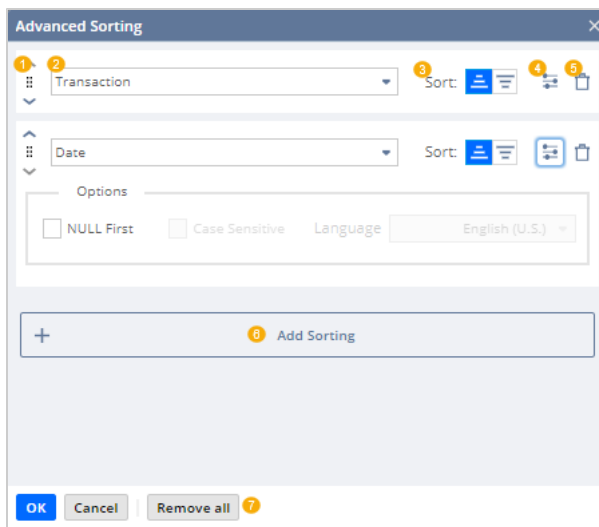
Advanced Sorting Options

You can sort the values in the Data Grid using the Advanced Sorting window. Sorting options you set in the Data Grid won't affect any workbook visualizations, and should only be used to help make the grid easier to read. To filter data in your dataset, use the Criteria Builder above the Data Grid to make criteria filters. For more information, see [Dataset Criteria Filters](#).

To open the Advanced Sorting window, click the Field Menu icon  next to a column in the Data Grid and select **Add Sort...** from the menu.

In the Advanced Sorting window, you can add sorting conditions for any Data Grid field. After adding each condition, choose the order in which the conditions should be applied.

The following image shows the parts of the Advanced Sorting window:



1	Up/ Down Arrows — Click the arrows to change the order of the sorting conditions, or drag the entire row to the location you want to move it to.
2	Field list — Select the field you want to create a sorting condition for.
3	Sort — Sort the result set in ascending  or descending  order, based on the selected field's values.
4	Options — Click this icon to view the following options specific to the selected sorting condition: ■ NULL First — If checked, blank values for the selected field are listed first in the corresponding column in the Data Grid. ■ Case Sensitive — If checked, values beginning with capital letters are listed first in the corresponding column in the Data Grid. ■ Language — Sort the result set based on the alphabetical order for the selected language.
5	Delete — Remove the selected sorting condition.
6	Add Sorting — Add another sorting condition for a different field.
7	OK — Apply the selected sorting conditions to the Data Grid. Cancel — Close the Advanced Sorting window without making any changes to the existing sorting conditions.

Remove All — Remove all sorting conditions from the Advanced Sorting window.

Joining Record Types in a Dataset

With SuiteAnalytics Workbook, you can join multiple record types in the same dataset, even ones that are more than one join away from your root record type, so you can pull data from a wider range of sources.

By default, the Dataset Builder shows you all the record types you can join. To complete a join, add fields from a record type to the Data Grid or make a criteria filter with the Criteria Builder. If you're not sure which record type to join, search for the field you want to add using the Records and Fields list search function. For more information, see [Locating Fields in Workbook](#) and [Dataset Builder](#).

If joining record types is new to you, there are a few things you should keep in mind, like how one-to-many relationships work and the order and direction of your joins. This is especially important if you're working with a transaction dataset and want to join transaction line and transaction accounting line record types.

For more information, see [Guidelines for Joining Record Types in SuiteAnalytics Workbook](#).

Locating Fields in Workbook

If you are not sure which record type you want to join to a dataset, you can browse or search the Records and Fields list in the Dataset Builder. To get an overview of the Dataset Builder, see [Dataset Builder](#).

The record types and fields you can see are based on your permissions and the role you use to log in. If you don't see a record type or field you think you should have access to, contact your administrator.

By default, record types and fields are listed alphabetically, except in a transaction dataset. In a transaction dataset, the transaction line and transaction accounting line types are listed at the top, since they're the most commonly joined record types.

You can search for fields by their label or internal ID. Searches work across record types up to three joins away from your dataset's root. If you'd like to search more, click **Show more results**.

If you're working in an existing dataset and aren't sure which record type a field belongs to, point to the field in the Data Grid or criteria filter, click the field menu icon, and select **Show Location in Field List**. The record type will be highlighted in the Records List.

Linking Datasets

You can link two datasets if each one has at least one field with shared data, like a date. Then you can use the linked datasets to build one workbook visualization and compare data from both.

Link definitions are part of the workbook they're created in and can't be shared on their own, but you can still share each linked dataset by itself. If someone shares a workbook with you that uses linked datasets, you can use them to make new visualizations in that workbook. For more information, see [Dataset Linking in SuiteAnalytics Workbook](#).

Dataset Criteria Filters

In the Dataset Builder above the Data Grid, you can use the Criteria Builder to filter your dataset results. Unlike filters you add inside workbook visualizations, criteria filters remove whole records and fields from

your dataset. This affects all workbook visualizations based on that dataset. For example, if you create a criteria filter to remove invoice records from a dataset, workbook visualizations based on that dataset won't include invoice data anymore. Use caution when adding new criteria filters to a dataset used in multiple workbook visualizations.

Criteria filters use AND, OR logic in the Criteria Builder, and you can group filters so multiple conditions are treated as one. You can edit, delete, or move criteria filters to change the order they're evaluated in. Any fields used in a criteria filter are highlighted in the Fields list, but won't appear at the top of the list like fields added to the Data Grid. If you're not sure which record type a field in the Criteria Builder belongs to, click the Field menu icon next to the criteria filter and click **Show Location in Field List**.

When you make a new criteria filter, the Data Grid updates automatically. But to update results in a chart or pivot table based on the dataset, you'll need to refresh it manually.



Important: If you create a criteria filter on a joined record type that has a one-to-many or many-to-many relationship with your dataset's root record type, you can get duplicate data. For more information, see [Data Duplication Based on Record Joins](#).

If you're trying to filter results from a pivot table or chart using linked datasets, be sure to apply the filter to the matching field in each dataset. If you don't, you might see blank rows in your results. For more information, see [Create Visualizations Based on Linked Datasets](#).

To create a dataset criteria filter:

1. In the Dataset Builder, drag a field from the Fields list or Data Grid to the Criteria Builder.
The Filter window will pop up.
2. In the Filter window, select the filter conditions you want for the field. You might see up to four filter types, depending on the field you choose. For more information, see [Filter Types](#).
3. (Optional) If you want to group your criteria filters, click **New Group**.
Grouped filters are treated as one item in the list and are evaluated based on their order in the Criteria Builder. For more information, see [Grouping Filters](#).
4. Repeat steps 1–3 for every filter or group you want to add.
By default, criteria filters use the AND operator. To change it, click **AND** and select **OR** in the popup window.
5. (Optional) If you're working on the dataset in a workbook, click **Apply to workbook** to preview your changes.
6. Click Save to save your changes.
If you don't own the dataset, click **Save As** to save it with a new name.

Filter Types

When you add a field from the Fields list or the Data Grid to the Criteria Builder, the Filter window pops up. Use the Filter window to set up filter conditions for a field. The filter types you can use depend on the field's values, and each type shows up as a separate tab in the Filter window.



Important: You can also open the Filter window from the Pivot tab in a workbook, but filters you set there only affect the pivot table, not the dataset or other visualizations. Only filters set up from the Criteria Builder above the Data Grid can change a dataset.

In each tab of the Filter window, you can set values and expressions for the filter. You can only make one filter type per field at a time, unless you use a group of filters. For more information, see [Grouping Filters](#). The available values and expressions in each Filter window tab depend on the field you pick.

- Values: Create a filter condition based on existing values or dates in the source data, or custom values
- Ranges/ Date Ranges: Create a filter based on a range of values or dates from the source data
- Relative Conditions/ Relative Dates: Create a filter that's relative to existing values or dates in the data
- Conditions/ Specific Dates: Create a filter using custom values or dates and an expression



Note: For hierarchical fields, you can set filters at the parent or child level. For more information, see [Hierarchical Fields](#).

If you're using a NetSuite OneWorld account and make a criteria filter on a field with multiple currencies, you'll see a Consolidate Field box in the Filter window. Check the Consolidate Field box to apply the filter based on the consolidated field values. For more information, see [Currency in Datasets and Workbooks](#).

Grouping Filters

In the Criteria Builder, you can click **New Group** to create a group of criteria filters.

Grouped criteria filters work like parenthetical expressions in Saved Search. Use them to link filter conditions with AND and OR, so they're checked as a single item in your criteria list.

For example, if you set up these criteria filters, SuiteAnalytics Workbook evaluates the whole group (Sales Rep is A Wolfe-admin OR Status is Open) before the Start Date filter and after the Date filter.

The screenshot shows the SuiteAnalytics Workbook interface for a 'Transaction Dataset'. On the left, there's a sidebar with 'Formulas' and 'Transaction' categories. The main area displays a table of transaction data. A red box highlights the 'Criteria summary' section, which shows a filter group: 'Date • AND • (Sales Rep • OR • Status) • AND • Start Date'. Below this, the table lists transactions with columns: DATE, MEMO, ENTITY, DOCUMENT NUMBER/ID, TRANSACTION, TYPE, and TOTAL AMOUNT (TRANSACTION). The table contains 7 rows of data.

DATE	MEMO	ENTITY	DOCUMENT NUMBER/ID	TRANSACTION	TYPE	TOTAL AMOUNT (TRANSACTION)
11/6/2010	Paid upfront	Gilchrist Pty Ltd	0117	Invoice #0117	Invoice	
8/15/2010		USD Coding	2001	Sales Order #2001	Sales Order	
6/30/2010		Motiva Mobility	2005	Sales Order #2005	Sales Order	
7/15/2010		Fabre Technology	2006	Sales Order #2006	Sales Order	
10/22/2010		Venture Capital Inc.	2007	Sales Order #2007	Sales Order	
1/1/2010		Jupiter Technology	2014	Sales Order #2014	Sales Order	
11/6/2010	Paid upfront	Gilchrist Pty Ltd	2075	Sales Order #2075	Sales Order	

You can also create a group of filters inside another group. If you do, SuiteAnalytics Workbook checks the conditions from the inside out.

For example, in this group of filters, (Status is Sales Order: Closed AND Sales Rep is Franco Africa) is checked before (Status is Bill: Open OR Sales Rep is). And since the whole group comes before the Start Date filter, the grouped filters are checked first.

The screenshot shows the SuiteAnalytics Workbook interface for a 'Transaction Dataset'. On the left, there's a sidebar with 'Formulas' and 'Transaction' categories. The main area displays a table of transaction data. A red box highlights the 'Criteria summary' section, which shows a filter group: '(Status is Bill : Open • OR • Sales Rep is <NO VALUE> • OR • (Status is Sales Order : Closed • AND • Sales Rep is Franco Africa)) • AND • Start Date is not empty'. Below this, the table lists transactions with columns: DATE, MEMO, SALES REP, ENTITY, DOCUMENT NUMBER/ID, TRANSACTION, and TYPE. The table is currently empty.

DATE	MEMO	SALES REP	ENTITY	DOCUMENT NUMBER/ID	TRANSACTION	TYPE
------	------	-----------	--------	--------------------	-------------	------

Creating a Workbook

A workbook lets you analyze dataset results by making visualizations like tables, pivot tables, and charts.

Every workbook visualization is based on a dataset, but one dataset can be used in different workbooks and visualizations. You can also use a different dataset for each visualization in a workbook. You can open the dataset right from a workbook visualization, but the dataset is still its own object with its own creator and access controls. Any changes to a dataset are automatically applied to all connected visualizations. This keeps your data accurate, but your workbooks might change unexpectedly if you don't own the underlying dataset.

You can only make visualizations with fields already added to the dataset. If you don't see a field you want, you'll need to add it to the dataset or ask the dataset owner to do it for you. For more information about defining a dataset, see [Defining a Dataset](#).

In workbooks with linked datasets, link definitions are part of the workbook. You can use these linked datasets for lots of visualizations, but only in that workbook. You can't share the link definition by itself.

The main workbook visualizations you can create are table views, pivot tables, and charts:

- Table Views let you see and analyze your dataset results in a simple table. You can't use linked datasets to make a table view. For more information, see [Workbook Table Views](#).

 **Note:** By default, every workbook template has a Table View based on the dataset. For more information about workbook and dataset templates, see [Workbook and Dataset Templates](#).

- Pivot tables let you break down your dataset results to analyze different subsets of data—like the number of unfulfilled orders per customer. With linked datasets, you can use pivot tables to look at metrics from two different levels. For more information, see [Workbook Pivot Tables](#).
- Charts let you create visualizations of your dataset results, so you can analyze them quickly. You can also build charts with linked datasets. For more information, see [Workbook Charts](#).

After you create your workbook visualizations, you can rearrange, rename, or delete workbook tabs as needed.

 **Note:** If you have multiple languages in your account and you edit and save default text in a workbook or visualization, a new translation collection gets added to the Manage Translations page. For more information, see the help topic [Manage Translations](#).

To create a new workbook or workbook visualization using a new dataset:

1. From the Analytics Home page on the Workbooks subtab, click **New Workbook**.
2. On the **Select a dataset to create a new workbook** page, click **New Dataset**.
3. Select a record type for your dataset.
The Dataset Builder will appear with some fields already added to the Data Grid.
4. Select the fields and criteria filters you want in your dataset. For more information, see [Defining a Dataset](#) and [Dataset Criteria Filters](#).
5. When you're satisfied with your results, click **Apply to workbook**.
6. Select a workbook visualization to get started.
7. Edit the layout and formatting for each workbook visualization you add.
For more information, see [Workbook Table Views](#), [Workbook Pivot Tables](#), or [Workbook Charts](#).
8. Click **Save...**
The Save Workbook As window appears.

 **Important:** You can't save a workbook if it has visualizations based on unsaved datasets. If there are unsaved datasets, click the menu icon next to the dataset name in the Dataset Panel and select **Save changes as** or **Discard changes**. Or, if you want to keep editing the dataset, click **Open Dataset**.

9. Enter a name and description in the fields, then click **Save**.
10. (Optional) Click **Share** to share the workbook with other users in your account.
The Share Workbook window will open.
11. Select the roles or user IDs you want to share the workbook with, then click **Share**.
For more information about sharing workbooks, see [Accessing and Sharing Workbooks and Datasets](#).

12. (Optional) To change the order of your workbook visualizations, drag the workbook tab to a new spot.

To create a new workbook or workbook visualization based on an existing dataset:

1. From the Analytics Home page on the Workbooks subtab, click **New Workbook**.
Or, if you're already in a dataset, click **Create New Workbook** and proceed to step 3.
2. Select a dataset on the **Select a dataset to create a new workbook** page.
3. Select a workbook visualization to get started.



Important: Remember, you can only make visualizations with fields already in the selected dataset. If you don't own the dataset and want to add fields or filters, you'll need to save it with a new name. If you keep using the original, the creator can make changes that might affect your visualization. Use caution when making visualizations from datasets you don't own.

4. Edit the layout and formatting for each workbook visualization that you add.
For more information, see [Workbook Table Views](#), [Workbook Pivot Tables](#), or [Workbook Charts](#).
5. Click **Save...**
The Save Workbook As window appears.
6. Enter a name and description for the workbook in the fields, then click **Save**.
7. (Optional) click **Share** to share the workbook with other users.
The Share Workbook window appears.
8. Select the roles or user IDs that you want to share the workbook with, then click **Share**.
For more information about sharing workbooks, see [Accessing and Sharing Workbooks and Datasets](#).
9. (Optional) To change the order of your workbook visualizations, drag the workbook tab to a new spot.

To create a new workbook visualization within an existing workbook:

1. On the Analytics Home page under the Workbooks subtab, click the name of the workbook you want to add your visualization to.
2. Click the Add icon **+**.
3. Select the visualization type you want to add.
By default, it's based on the dataset currently selected in the Dataset Panel. If you want to use a different dataset, click **Connect Dataset**.
4. Edit the layout and format for the visualization.
For more information, see [Workbook Table Views](#), [Workbook Pivot Tables](#), or [Workbook Charts](#).
5. (Optional) To change the order of your workbook visualizations, drag the workbook tab to a new spot.

Workbook Table Views

Workbook table views let you explore your dataset results without changing the source data for any workbook visualizations based on that dataset. You don't need complex setup to use table views, it lets

you see your data without making a layout or custom formulas. You can't use linked datasets to build a table view.

You can only create table views with fields you picked in the underlying dataset, as shown in the Dataset Panel. By default, every workbook template comes with a Table View based on the dataset. By default, all workbook templates include a Table View based on the underlying dataset. For more information about workbook and dataset templates, see [Workbook and Dataset Templates](#).

To make a table view, drag fields from the Dataset Panel to the Table Viewer. After you add fields, you can filter and sort them as needed. When you add a new field or sort/filter, the results update automatically. You can also customize your table view by renaming columns or applying conditional formatting. You can create as many table views as you want using different datasets in the same workbook. You can also export your table view data to a CSV file, and add table views to the Analytics portlet on any NetSuite dashboard.



Important: Sorting, filtering, conditional formatting, and other customizations you add to a table view won't affect the dataset or other workbook visualizations. To change the dataset, click the dataset name in the Dataset Panel. If you don't own the dataset, contact the owner.

To create a table view:

1. Click the add icon  from anywhere within the workbook and select **Table**.



Important: By default, the table uses the most recently connected dataset in the workbook. To change it, open the dataset selector in the Dataset Panel and pick another dataset. Or, if the dataset you want isn't already in this workbook, click **Connect Dataset** to select one from your account or create a new one.

2. On the Table tab, drag fields from the Dataset Panel to the Table Viewer. There are two ways to add fields:
 - You can drag fields one by one.
 - You can add all fields at a time by clicking the **Add all fields to the table** button.

Fields added to the Table Viewer are highlighted in blue on the Dataset Panel.



Note: If you add hierarchical fields to your table, you'll be prompted to pick how to display field values. For more information, see [Hierarchical Fields](#).

3. Customize the appearance of your table:
 - a. Change column order by dragging them or by clicking the Field Menu icon and choosing a move option.
 - b. Rename columns by clicking the Field Menu icon in the column header you want to rename, then selecting **Rename**.



Note: This only changes the column name in the current table view, the name won't change in the Dataset Panel, underlying dataset, or other visualizations.

- c. Remove columns by clicking the Field Menu icon in the column header you want to remove, then selecting **Remove Column**.
4. Sort your table data:
 - a. Click the Field Menu icon next to the field you want to sort.
 - b. Select **Ascending** or **Descending** for default sorting, or select **Add Sort...** to apply advanced sorting.

 **Note:** The advanced sorting options for a table view are the same as those for a dataset. For more information, see [Advanced Sorting Options](#).

5. Filter your table data:
 - a. Click the Field Menu icon next to the field you want to filter and select **Filter...**
The Filter window appears.
 - b. Pick the filter conditions or values, then click **Apply**. For more information about the available filtering options, see [Value-based Filters](#).
The Table Viewer is automatically updated.
6. Apply conditional formatting to your table data.

 **Important:** Conditional formatting doesn't tell apart values in different currencies. To highlight them correctly, convert everything to a single currency first, then apply the formatting. For more information, see [Currency in Datasets and Workbooks](#).

- a. Click the Field Menu icon in the column you want to highlight and point to **Conditional Formatting**.

 **Note:** Conditional formatting is currently not available for fields containing DATE values.

- b. Click **Manage Conditional Formatting**, operators, values, and colors or icons for your rule. You can click the add icon  to set up more rules, using different colors or icons for the same field. For more information, see [Conditional Formatting](#).

 **Note:** If you're formatting percentage values, use decimal format for the rule. For example, instead of **greater or equal to 20%**, set it as **greater or equal to 0.2**.

- c. Click **Apply**.
 - d. Repeat steps A-C for each column that you want to highlight.
7. (Optional) Click the Export icon to save a CSV file of your completed table.

 **Note:** To export your table view, you must have the Export Lists permission.

After you create your table view, you can rename it or delete the tab. Hover over the Tab menu icon  and select the **Rename Tab** or **Delete Tab** options, or double-click the tab name to rename it.

Table View Portlets

You can see your table view data on your home dashboard by adding the Analytics portlet. This lets you quickly see workbook data and jump right into your workbook. Seeing data right on your dashboard means you'll spend more time learning from it and less time tracking it down.

You can pick how you want to see your table view portlet by setting how many rows show at one time. You can also give the Analytics portlet a custom title, by default it shows the workbook and pivot table names. Conditional formatting from your table views appears in the portlet, too. For more information, see [Conditional Formatting](#).

In your table view portlets, you can do the following:

- View the column summary – When you pick a column in your table view, you can also see a summary of results in the table footer. What's shown depends on the field type and could include the count, count (distinct), average, and more.

- Sort and filter your results – You get the same sorting and filtering options as you have in your workbook table view.

My Sales (Invoiced) Table View Portlet

DATE	TRANSACTION	TRANSACTION TYPE	AMOUNT (NET)	QUANTITY	CLASS
1/4/2002	Cash Sale #1001	Cash Sale	\$695.00	1	Repeat Customer
1/10/2002	Cash Sale #1002	Cash Sale	\$2,299.00	1	Repeat Customer
1/10/2002	Cash Sale #1002	Cash Sale	\$169.95	1	Repeat Customer
1/14/2002	Cash Sale #1003	Cash Sale	\$12.95	1	New Customer
1/14/2002	Cash Sale #1003	Cash Sale	\$59.95	1	New Customer
1/14/2002	Cash Sale #1003	Cash Sale	\$10.95	1	New Customer
1/18/2002	Invoice #1001	Invoice	\$71.92	2	New Customer
1/18/2002	Invoice #1001	Invoice	\$269.96	1	New Customer
1/18/2002	Invoice #1001	Invoice	\$1,798.20	2	New Customer
1/18/2002	Invoice #1001	Invoice	\$404.96	1	New Customer

Summary of all data: Count: 53 Count (Distinct): 24

1 of 2

As with charts and pivot tables, you can get to your workbooks through table view portlets.

For more information about how to add and set up the Analytics portlet, see the help topic [Workbooks](#).

Workbook Pivot Tables

The Pivot tab lets you pivot your dataset results to look at different data subsets. In each table, you can add multiple fields for each dimension, create measures and calculated measures, or set up table-specific filters to customize your results. You can also change how your table looks by formatting numbers, renaming rows and columns, adding totals and grand totals, or using sorting and conditional formatting. You can add your pivot tables to the Analytics portlet on any NetSuite dashboard, and export them as a CSV file. If your table is linked to other datasets, you can compare data at different levels based on the fields you pick. For more information, see [Dataset Linking in SuiteAnalytics Workbook](#).

Note: The following formatting won't be exported to CSV: percentages, currency symbols, totals, and layout options like conditional formatting and compact mode.

You can only make a pivot table with fields included in the underlying datasets. On the Pivot tab, you'll see them in the Dataset Panel on the left. In pivot tables with linked datasets, a link icon next to a field shows it's a common key. For more information about common keys, see [Common Keys](#).

The Pivot Table Viewer on the right shows your table based on the fields you set as rows, columns, and measures in the Layout Panel. Every time you update the layout, click the Refresh icon  to update the table. You'll also need to refresh the table if the datasets change.

To create a pivot table:

- Click the add icon  from anywhere within the workbook and select **Pivot**.

 **Important:** By default, the pivot table uses the most recently connected dataset in the workbook. To switch datasets, open the dataset selector in the Dataset Panel and pick a different one. Or, if the dataset you want isn't in this workbook yet, click **Connect Dataset** to select from your account or create a new one.

2. On the Pivot tab, drag fields from the Dataset Panel to the Rows, Columns, or Measures tabs in the Layout panel. Or, drag fields straight into the Pivot Table Viewer.
3. (Optional) To create a calculated measure, click **Create Calculated Measure** in the Layout Panel. You can also click the Field Menu icon  next to a measure in the Pivot Table Viewer and select **Create Calculated Measure**. Calculated measures are displayed with a calculator icon . For more information, see [Calculated Measures](#).

 **Note:** If you add hierarchical fields to the table, you'll be prompted to pick how to show the field values. Depending on where you add the field and which display type you pick, you can also add more subtotals to the pivot table for each hierarchy level. For more information, see [Hierarchical Fields](#).

4. (Optional) If you want to build your pivot table with two datasets, click the Dataset Menu icon and select **Create Dataset Link**. Depending on the common keys you pick and where you place them in the table, your results will be aggregated. For more information, see [Create Visualizations Based on Linked Datasets](#).
5. Select the summary type and formatting options for any date or numeric fields you add to the pivot table.
 1. Click the Field Menu icon  next to the field you want to format in the Layout panel.
 2. Select a summary type from the popup window.
The summary options vary depending on the type of field you select.
 3. (Optional) Select **Currency...** to view the currency consolidation or conversion options for any fields with values in multiple currencies.
For more information, see [Currency Conversion from the User Interface](#).
 4. (Optional) Click **Format...** to customize the numeric values for a field.
For more information about numeric formatting options, see [Customizing Numeric Values](#).
6. Add totals and grand totals to the pivot table.
 1. Click the Totaling icon .
 2. In the Totaling window, choose where you want totals or grand totals for each field. If you have multiple fields to total in the rows or columns, check **Set Individually** to pick where each field's totals will show up on the pivot table.
 3. Click **OK**.
7. Click the Refresh icon  to generate the pivot table.
8. Filter the data displayed in the pivot table.

 **Note:** Filter conditions created on the Pivot tab only impact the data displayed in the pivot table. No changes are made to the dataset that the workbook is based on.

1. Click the Field Menu icon  next to the field you want to filter. The options you'll see depend on whether the field is set as a column, row, or measure. If you click the Field Menu icon from the Fields List or Layout Panel, that can also change what options you get:
 - Top 10: display only the top 10 rows or columns based on the measures defined for the table.

- Bottom 10: display only the bottom 10 rows or columns based on the measures defined for the table.
- Filter [Field Name] by...: enables you to define a custom measure-based filter for the selected row or column.
- Filter [Field Name]: enables you to define a custom value-based filter based on specific values within the table results.
- Add as Filter...: enables you to define a custom value-based filter based on specific values within the table results.
- Filter Date: enables you to define a custom date filter based on date ranges that you choose.

2. The results in the table are updated automatically.

For more information, see [Workbook Visualization Filters](#).



Important: If you build your pivot table with linked datasets, be sure to apply filters to the matching field in each dataset. If you don't, you might see cells in your results that you meant to filter out. For more information, see [Create Visualizations Based on Linked Datasets](#).

9. Apply conditional formatting to your pivot table measures.



Important: If the measure field you pick has values in multiple currencies, you'll need to convert or consolidate them before adding conditional formatting. For more information, see [Currency in Datasets and Workbooks](#).

- a. Click the Field Menu icon next to the measure you want to highlight in the Layout Panel and go to **Conditional Formatting**.
- b. Click **Manage Conditional Formatting**, then pick the operators, values, and colors or icons for your rule. You can click the add icon **+** to make more rules, using different colors or icons for the same measure or column. For more information, see [Conditional Formatting](#).



Note: If you're formatting percentage values, use decimal format for your rule. For example, rather than **greater or equal to 20%**, the rule should be defined as **greater or equal to 0.2**.

- c. Check **Apply to subtotals and grand totals** if you want to apply your rules to the measure subtotals and grand totals.
- d. Click **Apply**.
- e. Repeat steps A-D for every measure you want to highlight.

10. (Optional) Click the Export icon to save your pivot table as a CSV file.

After you make your pivot table, you can rename it or delete the tab. Hover over the Tab menu icon  and select the **Rename Tab** or **Delete Tab**, or double-click the tab name to rename it.

Pivot Table Customization

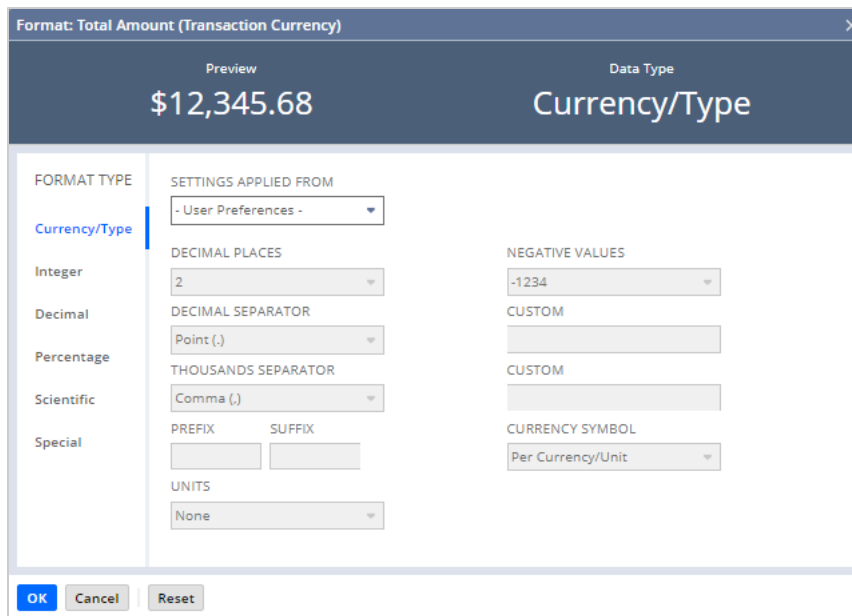
SuiteAnalytics Workbook enables you to customize many aspects of your pivot tables. Click the links below for more information:

- [Customizing Numeric Values](#)
- [Grouping Pivot Table Fields](#)

- [Compact and Expanded Mode](#)
- [Additional Pivot Table Customization Options](#)

Customizing Numeric Values

You can customize how numbers display in your pivot table with the Format window. To access it, click the Field Menu icon  next to a field in the Fields list or Layout panel and select **Format...** from the list.



The dialog box titled "Format: Total Amount (Transaction Currency)" shows a preview of "\$12,345.68" and the data type "Currency/Type". The left sidebar lists format types: Currency/Type (selected), Integer, Decimal, Percentage, Scientific, and Special. The main area contains settings for "Settings Applied From" (User Preferences), Decimal Places (2), Decimal Separator (Point (.)), Thousands Separator (Comma (,)), Prefix, Suffix, Units (None), Negative Values (-1234), Custom (empty), Custom (empty), Currency Symbol (Per Currency/Unit), and OK, Cancel, and Reset buttons.

The following options are available. After you make a selection, the Preview field shows your changes. To apply them, click **OK** and refresh the table. To make your own format, click the Special tab and enter your syntax.

- **Settings Applied From:** Lets you set numeric formatting based on a specific language. To set every setting manually, select **Custom**. To use the formatting from your NetSuite personal preferences, select **User Preferences**. For more information about personal preferences, see the help topic [Setting Personal Preferences](#).
- **Decimal Places:** The number of decimal places to show for each value.
 - Select Per Currency/Unit to use the currency or unit's decimal places.
 - Select Unbound for no limits on the number of decimal places.
- **Negative Values:** The format for negative values.
- **Decimal Separator:** The punctuation for the decimal place. Select **Custom** to set your own.
- **Thousands Separator:** The punctuation that splits up thousands. Select **Custom** to set your own.
- **Prefix:** Lets you add something before each value.
- **Suffix:** Lets you add something after each value.
- **Currency Symbol:** Lets you choose where the currency symbol shows up for dollar values.
- **Units:** Lets you shorten values by thousands, millions, or billions.

Custom Number Formatting

can set custom number formats for pivot tables in the **special** tab. Define your own syntax, or choose from the Predefined Values.

The pivot table below shows numbers without no formatting in the first column. In the second column, the numbers use this custom format: [0000ff]"gain "00.00;[ff0000](##,##00.00)" loss";[ff00ff]"nothing "0.00" zero"

Entity	Amount (Sum)	Amount (Sum)
D&G Consulting	\$1,898.00	gain 01.90k
Best Computer Services	\$0.00	nothing 0.00 zero
Eala Land Holdings	-\$338,972.26	(338,972.26) loss
Clemens Fritz	-\$10,330.68	(10,330.68) loss
Waste Management Sys...	-\$3,730,682.45	(3,730,682.45) loss
Code Development	\$155,520.00	gain 155.52k
Denmark 5	\$0.00	nothing 0.00 zero

The available symbols for custom numbering syntax are:

Symbol	Name	Application
#	Number Sign	Displays significant digits
0	Zero	Displays non-significant zeros
" "	Double Quotation Marks	Displays text inside quotation marks. You can place this text before or after numbers (positive, negative, or zero).
+	Plus Sign	Typically used for positive numbers
()	Parentheses	Typically used for negative numbers
-	Minus Sign	Typically used for negative numbers
,	Comma	Thousands separator, or to scale a number by 1000
.	Period	Decimal separator
;	Semicolon	Separates parts of the syntax. When you make custom number formats, you can use up to three sections for positive, negative, and zero values, in that order.
[]	Brackets	A hexadecimal number (without #) in brackets sets the color of that format section. For example, [0000ff] is blue.

Grouping Pivot Table Fields

Grouping fields in the pivot table Layout panel changes how detailed your table data is. If you set more than one field as a row or column, you can show subsets of data in your table.

For example, in the transaction table below, only the sales rep field is set as a row. The pivot table only shows total transactions for each sales rep.

Transaction Workbook

Dataset: Unsaved: Transaction... | Layout: Rows: Sales Rep, Columns: + Date (Year), Measures: Total Amount (Transaction Currency)...

	2007	2008	2009	2010	2011	2012
Total Amount (Transacti...	\$100.00		\$5,600.00	-\$1,081,945.27	-\$90,705.01	-\$501...
- No Sales Rep -		\$1,440.00	\$157,125.00	\$9,488,058.28	\$398,140.00	
suite analytics				€5,443,295.23		
Clemens Fritz						
Lewis Hamilton				€696,416.00		
Franco Africa				\$8107.00	\$82,120.00	PHP83,161
Dennis Sørensen				€2,751.28		
Jensen Button				€138,953.00		
Antonio F.					-PHP112.00	
Mini Saunders				-\$1,002.80		
Jeff Shen			-\$748.22	-\$370.33		
Kim Huffman				\$822,673.75		
Malin Concannon				\$726,680.00		
				\$50,404.13		

If you also define the entity field as a row however, the table displays the total transactions per sales representative and customer.

Transaction Workbook

Dataset: Unsaved: Transaction... | Layout: Rows: Sales Rep, Entity, Columns: + Date (Year), Measures: Total Amount (Transaction Currency)...

	2007	2008	2009	2010	2011
Total Amount (Transacti...					
June Peiris				\$694,150.00	
Monrovia Database				\$700.00	
Golden Technology				\$263,390.00	
Motiva Mobility				\$4,775.00	
Bay Media Research				\$5,000.00	
Baxter Venture Capital				\$29,337.38	
Gentry Inc.				\$29,070.00	
Cara Systems				\$14,320.00	
Sales Search Tool				\$4,512.00	\$3,121
Flexis Search Tool				\$999.00	
McGurk Prospect Min...				\$300,000.00	
John Smith				\$1,200.00	\$3,601
ABC Co.					
Steen Solutions					

Try grouping different fields in your pivot table layouts to analyze different subsets of data.

Compact and Expanded Mode

By default, if you group fields or use multicolumn hierarchical fields, your pivot table expands horizontally, which can make it difficult to view all your data.

To display the table data vertically, click the Compact Mode icon .

Additional Pivot Table Customization Options

To customize your pivot tables even more, SuiteAnalytics Workbook has these options:

- To keep column or row headers visible while you scroll, click the Freeze Column  or Freeze Row  icons.
- To highlight rows or columns when you point to them, click the Highlight Column  or Highlight Row  icons.
- To rename a column or row, click the Field Menu icon  and select Rename... from the menu.
- To resize a column or row, drag the boundary until it's the size you want.
- To auto-fit a column's width to its values, double-click the column header.

- To show more or fewer values for a row, click the   icons in the column header.
- To expand or minimize the number of values displayed for a row, click the  icon in the row header.

Pivot-based Portlets

You can view your pivot-based workbook data on your home dashboard by adding the Analytics portlet. This lets you quickly see workbook data and jump right into your workbook. Having the data right on your dashboard means you'll spend more time learning from it and less time gathering it.

You can also pick how you want to see your pivot-based workbook by setting how many rows show at one time. You can give the Analytics portlet a custom title. By default, it shows the workbook and pivot table names. Conditional formatting from your pivot table also appears in the portlet.

For more information about adding and setting up the Analytics portlet, see the help topic [Workbooks](#).

		Date (Quarter)			
		Q1	Q2	Q3	Q4
Location	Sales Rep	Total Amount (Transacti...	Total Amount (Transacti...	Total Amount (Transacti...	Total Amount (Transacti...
New Jersey	Theodore Hosch	\$3,447,915.21	\$211,153.79	\$184,710.00	\$184,710.00
	Jon Baker	\$2,232,170.42	\$41,855.85	\$877,851.06	\$1,746,732.96
London	Luke Duke	\$6,949.44	\$1,007,700.00		
	Poncho Ponchereli	\$37,564.29	\$22,297.52	\$804.00	\$4,561.19
	Mathew Christner	\$136,712.84	\$996.45		
Vancouver	Jessie Barto	\$36,743.20	\$816.80		

Workbook Charts

The Chart tab lets you create visualizations of your dataset results. You can make different types of charts and add any of them to the Analytics portlet on any NetSuite dashboard. You can print or export your charts to an SVG file. You can also customize how your chart looks by adding a title and subtitle, or renaming the axes. Filtering is supported too, so you can show only the most relevant values in each chart. If you want to compare metrics from two datasets in the same chart, you can build it using linked datasets. For more information, see [Link Datasets in a Workbook](#).

The Chart tab shows all fields from the dataset in the Dataset Panel on the left. In charts using linked datasets, a link icon next to a field means it's a common key. For more information about common keys, see [Common Keys](#).

Completed charts appear in the Chart Viewer on the right, based on the fields you choose for the x-axis, series, and measures in the Layout Panel. Whenever you update the layout or the dataset changes, be sure to click the Refresh icon  so the chart updates.

To create a chart:

1. Click the add icon  from anywhere within the workbook and select **Chart**.

 **Important:** By default, the chart uses the most recently connected dataset in the workbook. To change it, open the dataset selector in the Dataset Panel and pick a new dataset. Or, if the one you want isn't in the workbook yet, click **Connect Dataset** to select one from your account or create a new one.

2. Drag fields from the Dataset Panel to the X-Axis, Series, or Measures tabs in the Layout panel.

3. (Optional) In the Layout Panel, click **Create Calculated Measure** to create new measures using fields from the Dataset Panel. Calculated measures are displayed with a calculator icon . For more information, see [Calculated Measures](#).
4. (Optional) If you want to build your chart with two datasets, click the Dataset Menu icon and select **Create Dataset Link**. Depending on the common keys you select and where you place them in the layout, your results will be aggregated. For more information, see [Create Visualizations Based on Linked Datasets](#).
5. Select the summary type and format options for any date or number fields you add to the chart.
 - a. Click the Field Menu icon next to the field you want to format in the Layout panel.
 - b. Select a summary type from the popup window.
 - c. (Optional) Click **Format...** to customize the numeric values for a field.
For more information about numeric formatting options, see [Customizing Numeric Values](#).
 - d. (Optional) Select **Currency...** to view the currency consolidation or conversion options for fields with multiple currencies.
For more information, see [Currency Conversion from the User Interface](#).
6. In the Layout panel, select the chart type you want from the popup.
For information about each chart type, see the [Chart Types](#) topic.
7. (Optional) Click the Properties subtab in the Layout panel to add a title and subtitle to the chart, or to rename each axis.
8. Click the Refresh icon  to generate your chart.
9. (Optional) Filter the data displayed in the chart:

 **Note:** Filters you add on the Chart tab only affect the chart itself. They don't change the main dataset or other visualizations.

- a. Click the Field Menu icon  next to the field you want to filter. The options you see depend on whether the field is set as the X-Axis, Series, or Measure, or where you click the Field Menu icon (Fields List or Layout Panel):
 - Top 10: display only the top 10 results based on the measures defined for the chart.
 - Bottom 10: display only the bottom 10 results based on the measures defined for the chart.
 - Filter [Field Name] by...: enables you to define a custom measure-based filter for the selected chart.
 - Filter [Field Name]: enables you to define a custom value-based filter based on specific values within the chart results.
 - Add as Filter...: enables you to define a custom value-based filter based on specific values within the chart results.
- b. The results in the table are updated automatically.
For more information, see [Workbook Visualization Filters](#).



Important: If you build your chart with linked datasets, make sure you apply filters to the matching field in each dataset. If you don't, your results might show values you meant to filter out. For more information, see [Create Visualizations Based on Linked Datasets](#).

After you create your chart, you can rename it or delete the chart tab. Hover over the Tab menu icon  and select **Rename Tab** or **Delete Tab**, or double-click the tab name to rename it.

Chart Types

There are lots of chart types you can use to visualize your data. To choose one from the Chart tab or Analytics portlet, click the chart type dropdown and pick from the popup window. Changing the chart type in the Analytics portlet doesn't change the chart type in the workbook. Chart types help you compare values and spot trends. When picking a chart type, think about the data you're showing and what you want to highlight. Here are your options:

- Column Chart – Lets you compare values across categories or over time. For example, sales by week, month, quarter, or year. You can also use them to show negative values and see the range between highest and lowest categories.
- Bar Chart – Lets you compare values like a column chart. Use this style when data labels are long, like regions or sales reps, since it's easier to read.
- Area Chart – Lets you show trends and changes over time, like item stock, employee count, or account balance.
- Line Chart – Great for showing trends when you have lots of data points (like more than 20).
- Stacked Column Chart – Shows how data breaks down, with each category's part of the whole. For example, use it to see the percent of base price vs. discount price per customer.
- Stacked Bar Chart – Shows data in a bar chart that highlights the breakdown, not only comparisons.
- Stacked Area Chart – Shows data in an area chart, so you can see how each item adds up to the total over time.

To see some examples, see the following charts:

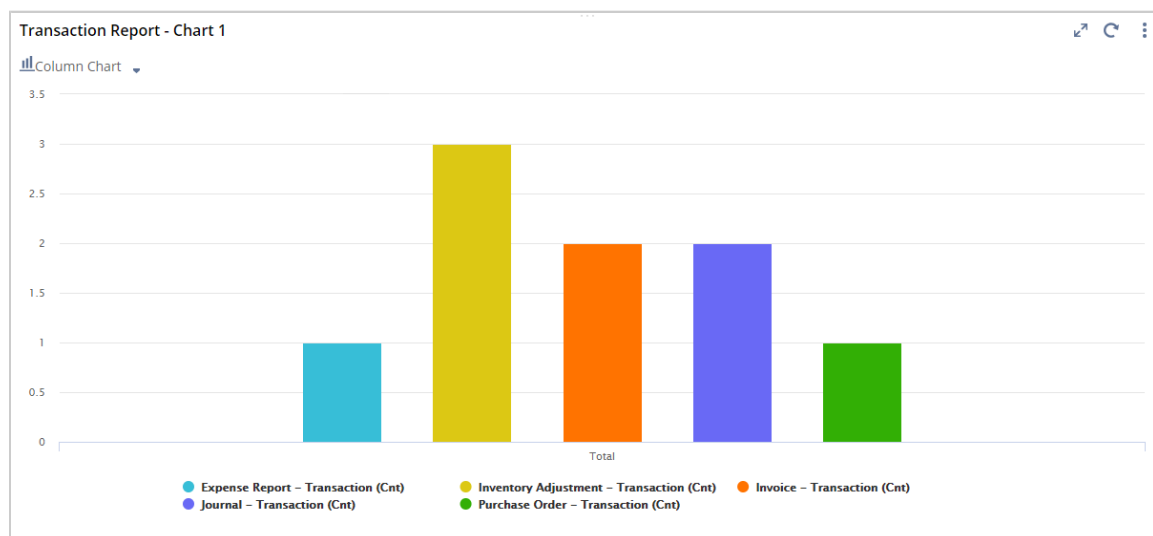
- [Open Sales Orders Lines Charts](#)
- [Number of Fulfillments Chart](#)

Chart-based Portlets

You can see your chart data on your home dashboard by adding the Analytics portlet. This lets you quickly view your chart and jump into your workbook. Having the data right on your dashboard means you'll spend more time learning from it and less time looking for it.

You can also pick how you want to see your chart by choosing layout options like portlet size and chart type. You can give the Analytics portlet a custom title. By default it shows the workbook and chart names.

For more information about how to add and set up the Analytics portlet, see the help topic [Workbooks](#).



Workbook Visualization Filters

After you set up a table view, pivot table, or chart, you can filter the data using a few types of filters: value-based, condition-based, measure-based, or date filters.

In table views and pivot tables, value-based filters check every row and column and remove those that don't have your selected value. In charts, any columns, bars, or lines that don't have the value you picked are removed too.

In pivot tables, measure-based filters check row and column totals and remove anything that doesn't match your filter. In charts, columns, bars, or lines with totals that don't meet your filter are removed. You can use measure-based filters on measure fields (ones that need sum, count, average, etc.), so they're only for charts and pivot tables.

In any visualization, you can make filters based on dates or conditions. For date fields, filter using specific or dynamic dates. For number fields, you can add condition-based filters for any range of numbers.

You can add as many value-based filters as you want to a visualization, but only one measure-based or date filter. If you use both, value-based filters are always applied first.

Filters you add to a visualization only affect what you see in that visualization, they don't change the dataset it's using. To filter the dataset, make criteria filters in the Dataset Builder. Also, if your NetSuite account has values in different currencies or is a OneWorld account with multiple subsidiaries, you'll need to convert or consolidate those values before you can use them in a measure-based filter.

For more information about filtering a dataset, see [Dataset Criteria Filters](#).



Important: If you are working in a visualization based on linked datasets, you must apply dataset criteria filters to the matching field in each dataset. If you do not, your results might include values that you wanted to filter. For more information, see [Create Visualizations Based on Linked Datasets](#).

For more information converting or consolidating values in a workbook visualization, see [Currency in Datasets and Workbooks](#).

To learn how to use the workbook visualization filters, see the following:

- [Available Filters for Aggregate and Non-Aggregate Fields](#)
- [Value-based Filters](#)
- [Condition-based Filters](#)
- [Measure-based Filters](#)
- [Date Filters](#)

Available Filters for Aggregate and Non-Aggregate Fields

When you're working with filters, each field shows different options based on the type of field you pick (like numeric or date fields) and how you use them in your workbook visualizations. In charts and pivot tables, you can use some fields in two different ways:

- Use fields as aggregate fields – If you drag fields to the **Measure** section, the visualization shows results as an aggregate field. Therefore, results are grouped together to display a single summary result for several value entries.
- Use fields as non-aggregate fields – If you drag fields to the **Rows** or **Columns** sections (for pivot tables) or the **X-Axis** and **Color Stack** sections (for charts), each aggregate field shows a result for each value entry.

Whether you use fields as aggregate or non-aggregate values affects how your workbook visualizations show results. For example, if you want to see the sales amount broken by sales rep, your workbook visualization can display results in different ways. If you use the Amount (Net) field as an aggregate (or

measure) field, the visualization shows a single result for all sales transactions by a specific sales rep, based on the aggregate function you choose. For example, if you pick the sum function, you'll see a single value for all the sales transactions. In this case, the field displays 10,000 USD. If you use the Amount (Net) field as a non-aggregate field, the visualization shows a result for each sales transaction of that sales rep. In this case, the visualization displays the following field results: transaction A (9,000 USD), transaction B (600 USD), and transaction C (400 USD).

Value-based Filters

Value-based filters let you filter the data in your workbook visualizations using specific values from any field you've added to the Layout panel or Table Viewer. In pivot tables, value-based filters apply to every row and column, and remove any that don't contain the selected value. In charts, any columns, bars, or lines that don't contain the selected value are removed from the results.

To apply a value-based filter to a workbook visualization:

1. Click the Field Menu icon  next to any field in the Layout panel or Viewer and select **Filter [Field Name]...**
The Filter window appears.
2. In the Filter window, select the field values and operator to use in the filter condition.
3. (Optional) In the search box, you can enter the value you're looking for.
4. Click **OK**.
The results update, and the selected filter criteria shows above the Viewer.

Condition-based Filters

You can use condition-based filters on numeric fields to show results that match a specific numeric range. To do this, you can pick the operator you want to use (like greater than, equal to, and so on), then enter the numeric value. For example, you can see opportunity amounts by sales rep, and show only results with a won probability above 80%. To see these results in your workbook visualization, you can add a condition-based filter to the probability (%) field, and set it to "greater" and "80%".

 **Note:** This option is available for percentage, integer and float fields that do not return currency values. If you want to use this option on currency fields, customize your field by using a TO_NUMBER formula function and changing the field's output type. For more information about how to customize a currency field, see [Advanced Sample Formula Fields](#).

To apply a condition-based filter to a workbook visualization:

1. Click the Field Menu icon  next to a field label and select **Filter [Field Name]...**
The Filter window appears.
2. In the Filter window, select the operator you want to use, then enter the numeric value.
3. Click **OK**.
The results update, and the selected filter criteria shows above the Viewer.

Measure-based Filters

In pivot tables, measure-based filters let you filter out entire columns or rows using conditions applied to the totals shown, based on the measures defined for the table. In charts, any columns, bars, or lines with total values that don't meet the selected criteria are removed from the chart.

For example, say you've set up a pivot table and want to see only customers with over thirty thousand dollars in total transactions:

The screenshot shows the SuiteAnalytics Workbook interface. The 'DATASET' pane on the left has 'Transaction' selected. The 'LAYOUT' pane shows 'Entity' as the row field and 'Date (Year)' as the column field. The main table displays a pivot table with 'Grand Total' as the measure. The data is organized by entity (rows) and year (columns: 2021, 2009, 2010, 2008, 2003). The entities listed include Thomas, Jan; Philips Medical Systems; Acme Medical Supply; Lumenis Inc.; Medical Resource USA; Ed Baumann; Scott Blosser; Sharon Sepulveda; The Lab Depot, Inc.; Jon Baker; Internal Revenue Service-Income.PICA,Med; Samuel Pober; ACME Shipping Co.; Montana Property Management; Poncho Poncherelli; Montana Tax Man; and Unknown, All Unallocated Products.

To do this, you can apply a measure-based filter to the entity field, like **Entity by Total Amount (Transaction Currency) (Sum) greater 30,000.00**.

The screenshot shows the SuiteAnalytics Workbook interface with a measure-based filter applied. The filter is 'Entity by Total Amount (Transaction Currency) (Sum) greater 30,000.00'. The pivot table displays data for entities where the total amount is greater than 30,000.00. The entities listed include Andrew Goldwasser; Justin Garcia; Garrett Garrett, Eric; Aaron Rosewall-Godley Scherba and Salinas Medical ...; Abdullah Bhupathiraju; Andrew Kuykendall; Billings Dental Clinic; Twin Bridges Family Medicine; Joshua Woodward; Chris Lipscomb; 31 Test Company; Lou Liang, MD; 59 Gerbera Unlimited; Abe Simpson Abe Si...; and A Company 40L.

You can also use a measure-based filter on a dimension to show the top or bottom items you want, for example, **Entity by Total Amount (Transaction Currency) (Sum) TOP 10**.

The screenshot shows the SuiteAnalytics Workbook interface with a measure-based filter applied. The filter is 'Entity by Total Amount (Transaction Currency) (Sum) TOP 10'. The pivot table displays the top 10 entities by total amount. The entities listed include Craig Koozer, DDS; Abe Lincoln; 40 Test Cart; Julie Kelly; AAKASH CHEMICAL; Craig Erickson; Cesar Petras; Cameron Sardella; Clayton Przybyla; and Jebidiah Palm.

In both cases, you apply a filter to the Entity rows based on their total amount (transaction currency) totals.

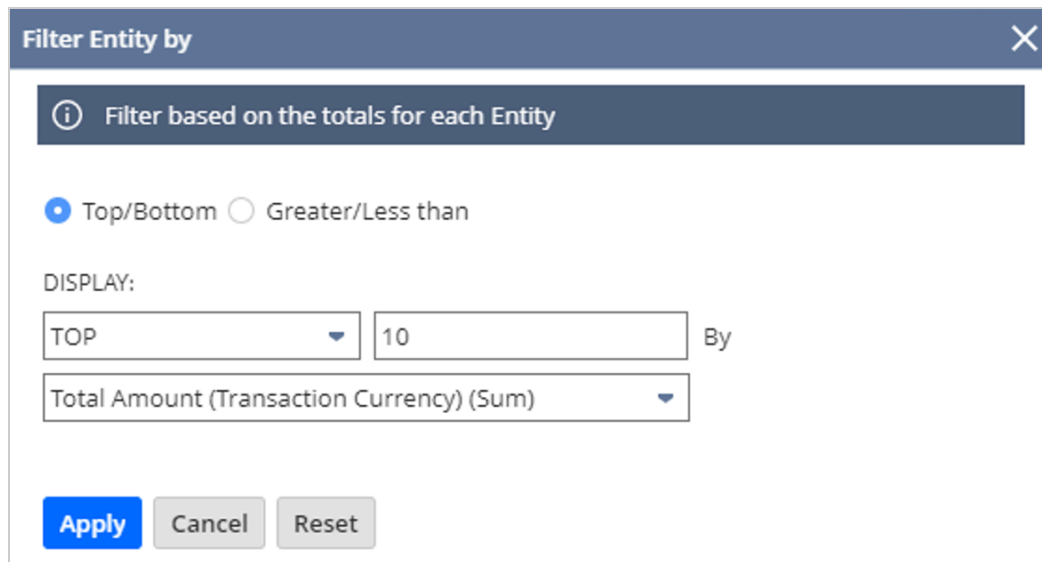
Note: If you're in a NetSuite account with values in multiple currencies, or a NetSuite OneWorld account with multiple subsidiaries, you need to convert or consolidate these values before you can use them in a measure-based filter condition. For more information about currency conversion in SuiteAnalytics Workbook, see [Currency in Datasets and Workbooks](#).

Measure-based filters are available in measure fields (also called aggregate fields), and only apply to charts and pivot tables. You can apply measure-based filters from either a dimension or a measure field.

To apply a measure-based filter to your pivot table or chart from a dimension field:

1. Click the Field Menu icon  next to a dimension field in the Layout panel or Viewer.
2. Select **Filter Top/Bottom**.

The Filter window appears.

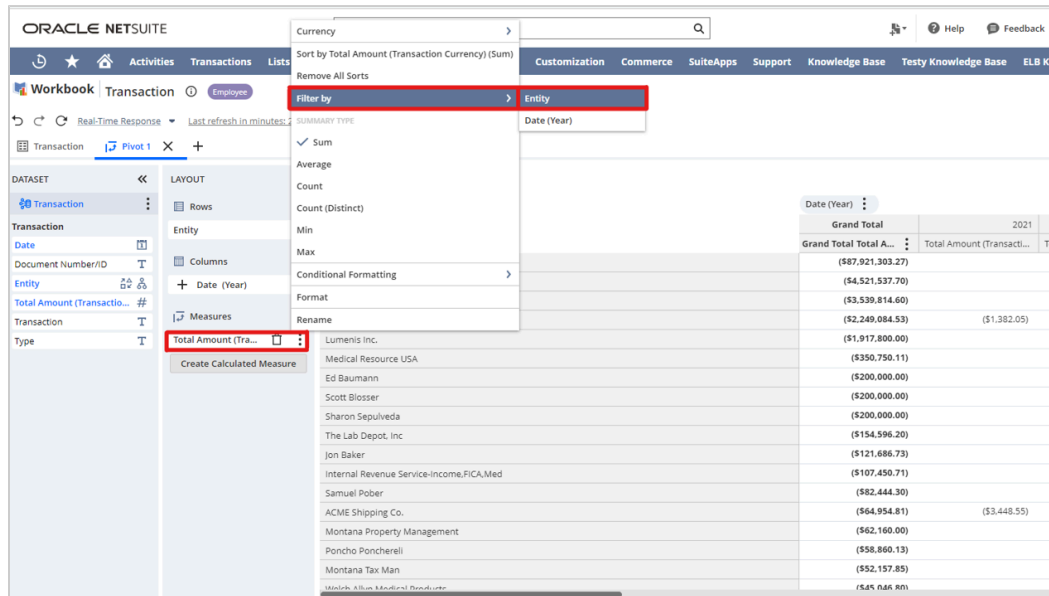


3. In the Filter window, select **Top/Bottom** (default option) or **Greater/Less than**, then select the measure and set the conditions for the filter.
4. Click **Apply**.

The table or chart automatically updates the results.

To apply a measure-based filter to your pivot table or chart from a measure field:

1. Click the Field Menu icon  next to a measure field in the Layout panel. In pivot tables, you can also access the filter by clicking the Field Menu icon  next to the **Grand Total** field in the Viewer, or from any measure column if there are no dimension fields displayed in the columns section.
2. Select **Filter by** and click a dimension.



The filter window appears.

Filter Entity by

Filter based on the totals for each Entity

☐ Top/Bottom
☒ Greater/Less than

Total Amount (Transaction Currency) (Sum)

greater

Enter value

Apply
Cancel
Reset

- In the Filter window, select **Greater/Less than** (default option) or **Top/Bottom**, then select the measure and set the conditions for the filter.
- Click **Apply**.

The table or chart automatically updates the results.

Date Filters

Date filters let you filter your results by a specific date, custom date, or dynamic date range you choose. For example, you can select a dynamic date like **today**, which updates when the workbook runs. You can only apply one date filter per workbook visualization.

Note: Date filters are available in date fields and work with all workbook visualizations (charts, pivot tables, and table views).

To apply a date filter to a workbook visualization:

1. Click the Field Menu icon  in the Dataset Panel or Layout Panel next to a date field and select **Filter Date> Advanced...**
2. In the Filter Date (Advanced...) window, use the **Specific Dates** tab to enter a custom date or the **Date Ranges** tab to select a dynamic date. Use the Values tab to select specific dates from the underlying dataset query results.
3. Click **Apply**.

The workbook visualization automatically updates the results.

The results are updated and the selected filter criteria is displayed above the Viewer.

Calculated Measures

A calculated measure is something you can create in pivot tables and charts using arithmetic operations with base measures. For example, you can use the following definition to calculate profit percentages:

$\% \text{ Profit} = \text{Est. Gross Profit (Sum)} / \text{Amount (Sum)}$

Formula fields are built at the dataset level and evaluate results row by row, while calculated measures are created at the workbook level using aggregated results. For example, if you create a formula field to calculate sales amounts, each row in the dataset shows a single value based on the formula. If you use that formula field as a measure in a pivot table or chart, the results are simply an aggregate of the formula field values. In contrast, calculated measures use the aggregated results of base measures and arithmetic operations to show results like year-over-year variances and profit percentages. You can't get these types of results using formula fields alone. For more information about the difference between formula fields and calculated measures, see [Formula Fields and Calculated Measures](#).

You can create as many calculated measures as you need using fields from the connected dataset, including custom formula fields. In visualizations based on linked datasets, you can even combine fields from both datasets to make a calculated measure. However, you can only use a calculated measure in the pivot tables or charts where you create it.

To create your calculated measures, see the following:

- [Working with Calculated Measures](#)
- [Supported Calculations](#)
- [Examples of Calculated Measures](#)

Working with Calculated Measures

Before you create a calculated measure, you should think about the fields and operators you want to include. To learn about the type of fields and basic operators that you can use in your calculations, see [Supported Calculations](#).

You can define your calculated measure using fields from the dataset connected to your workbook visualization. In visualizations based on linked datasets, this means you can use any combination of fields

from either dataset. If you don't see the fields you want to use, edit the connected datasets or contact the dataset owner if you're not the original author. For more information, see [Editing a Dataset](#).

After you've created a calculated measure, you can use it to sort and filter your results. Calculated measures support all sorting and filtering options available for base measures. You can also change the format of a calculated measure's numbers, add totals and grand totals, and apply conditional formatting. For more information about these options, see [Workbook Visualization Filters](#), [Pivot Table Customization](#), and [Conditional Formatting](#).

To open the calculated measure editor, click **Create Calculated Measure** in the Layout panel of any pivot table or chart. You can create as many calculated measures as you need, but you can only use them in the workbook visualization where you made them. If you want to use the same calculated measure in other visualizations, copy and paste the measure definition.

To learn how to work with calculated measures, see the following:

- [Creating Calculated Measures](#)
- [Calculated Measure Editor](#)

Creating Calculated Measures

When you create a calculated measure, you define the fields and basic operators for your calculation. All calculated measures need a name and a valid definition. As you create your calculated measure, a message at the bottom of the editor tells you if the definition is valid. If the definition isn't valid, an error icon appears next to the line with validation errors. Hover over the icon to see how to fix it.

After you create your measure, you can edit or delete it from the Layout panel of the chart or pivot table.

To create a calculated measure:

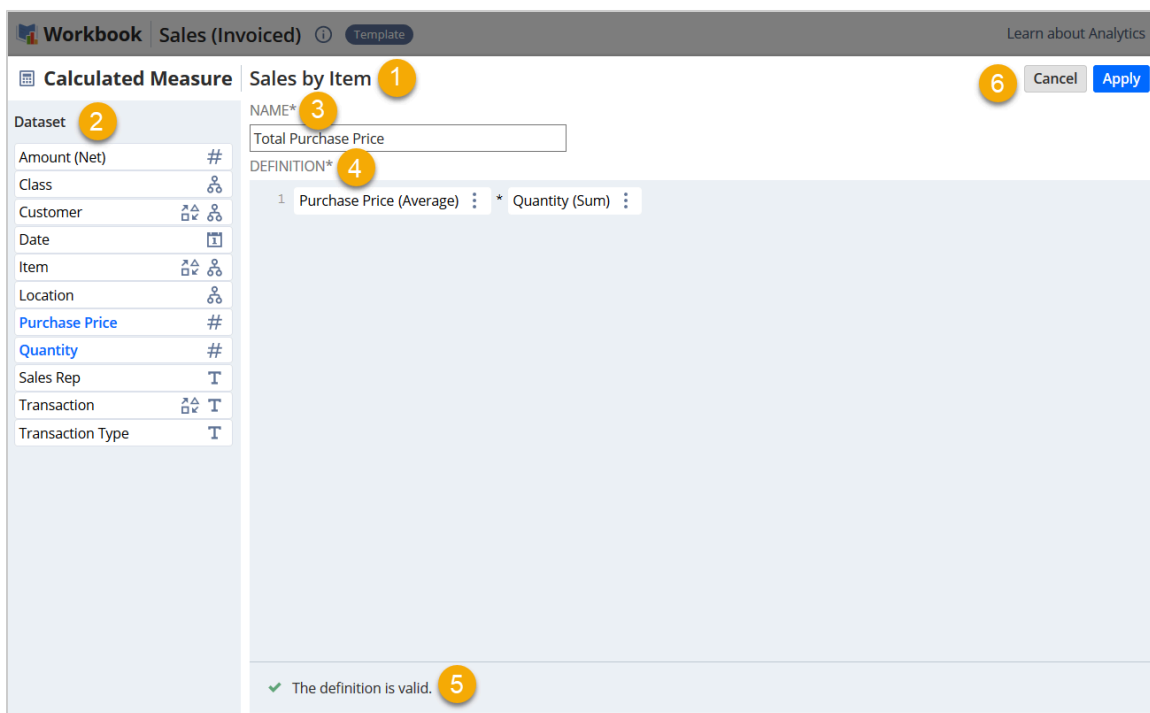
1. On a pivot table or chart, click **Create Calculated Measure** from the Layout panel. Alternatively, from within a pivot table, click the Field Menu icon  next to the field that you want to use in your calculated measure and select **Create Calculated Measure**.
The calculated measure editor appears with instructions on how to set up your calculations.
2. In the Name field, enter the name of your calculated measure.
3. Add fields to the editor:
 - Drag fields from the Dataset Panel to the Definition field. In visualizations based on linked datasets, you can switch between the fields within each dataset by clicking the dataset name.
 - In the Definition field, start typing the field names.
Fields you use in your calculated measure are highlighted in blue in the Dataset panel.
4. Add calculation operators:
 - In the Definition field, type a basic operator such as plus (+), minus (-), multiplied by (*), divided by (/), or parenthesis ().
 - Press Ctrl + Space to display all options and select an operator.
5. (Optional) To make changes to one of the fields used in the definition, click the Field Menu icon . The following options appear:
 - If you want to change the aggregate function of the field, select a different function from the list.
 - If you are working with fields that have values in multiple currencies, select **Currency...** to view the consolidation and conversion options. For more information, see [Currency Conversion in Workbook](#).

6. (Optional) To remove a field from the definition, place the cursor next to the field and press Backspace, or click the Field Menu icon and select **Remove**.
7. Click **Apply** to add the calculated measure to your pivot table or chart.
Your calculated measure appears in the list of measures in the Layout panel with a Calculator icon .
8. On your pivot table or chart, click the Refresh icon  to see the results of the calculated measure in the viewer.

Calculated Measure Editor

The Calculated Measure editor is where you set up new calculated measures.

The elements of the editor are identified in the image below:



1	Workbook Visualization Name - Displays the name of the visualization where you're creating the calculated measure.
2	Dataset Panel - Contains all fields from the connected dataset. In visualizations using linked datasets, both are shown so you can switch between them. To see more info about a field, hover over it and click the Information icon  on the right.
3	Calculated Measure Name - Enter the name of your calculated measure.
4	Definition Editor - Create your calculations using fields in the Dataset Panel and basic operators. To create your calculations, you can do the following: ■ Type a field name in the editor or drag a field from the Dataset panel to the editor. ■ Type a basic operator such as plus (+), minus (-), multiplied by (*), divided by (/), and parenthesis (). ■ (Optional) To change the aggregate function for a field, click the Field Menu icon  next to its name. ■ (Optional) To see all options, press Ctrl + Space and select the fields and operators you want to add.

5	Validation Message - Displays a message that tells you if the definition is valid or has errors. If it's not valid, you'll see an error icon next to the line with issues. Hover over it to see how to fix the definition.
6	Editor Menu - You can do the following: - Click Apply to leave the editor and view the calculated measure in your workbook visualization. You can only apply it if the name and definition are valid. - Click Cancel to go back to your workbook visualization without saving your changes.

Supported Calculations

In your calculations, you can use different basic operators depending on the type of data in each field you use in the measure: numeric, single currency, multiple currency, or duration.

To see which measures and operators you can use in calculated measures, check the supported calculations for each data type.

- Numeric values
- Single currency values
- Multiple currency values
- Duration values

Numeric values

In calculated measures, numeric values are percentages, integers, or floats. They don't return currency results.

This table shows which measures and basic operators you can use with calculated measures that return numeric values.

Data Type	Plus (+) Minus (-)	Multiplied by (*)	Divided by (/)
Single currency values	✓	✓	—
Multiple currency values	—	✓	—
Duration values	—	✓	—
Numeric values	✓	✓	✓

Note: If the calculation you want isn't supported for multiple currency values but works for single currency, try consolidating or converting the field to a single currency field. For more information, see [Currency Consolidation in Workbook](#).

Returned format for percentage values

When you combine percentage values with integers or floats in your calculation, the result shows up as a decimal value.

For example, if a field with the percentage value **200%** is divided by the integer value **10**, the operation calculates **2.0** divided by **10**. Therefore, the result returns the decimal value **0.2**.

$$0.2 = 2.0 / 10$$

Note: The result that you obtain isn't a percentage value. The decimal value **0.2** is the same as **20%**. If you'd rather see results as percentages in your pivot tables and charts, you can change the format to percentages. For more information see, [Pivot Table Customization](#).

Single currency values

Single currency values are amounts that return values in one currency, like dollars.

This table shows which measures and basic operators you can use with calculated measures that return single currency values.

Data Type	Plus (+) Minus (-)	Multiplied by (*)	Divided by (/)
Numeric values	✓	✓	✓
Multiple currency values	✓	—	—
Duration values	—	—	—
Single currency values	✓	—	✓

Multiple currency values

Multiple currency values are amounts that return values in different currencies, like dollars and British pounds.

This table shows which measures and basic operators you can use with measures that return multiple currency values.

Data Type	Plus (+) Minus (-)	Multiplied by (*)	Divided by (/)
Numeric values	—	✓	✓
Single currency values	✓	—	—
Duration values	—	—	—
Multiple currency values	✓	—	—

Note: If the calculation that you want to use is not supported for multiple currency values but is supported for single currency values, consider consolidating or converting the selected field to a single currency field. For more information, see [Currency Consolidation in Workbook](#).

Returned values for multiple currency calculations

The results that you obtain depend on how the measure definition is built:

- Calculations with numeric values** - When multiple currency values are multiplied by numeric values, the calculation returns a result for each currency. For example, if a field with the numeric value **100** is multiplied by a field with the multiple currency values **10 USD** and **20 EUR**, the calculation returns a result for each currency: **1000 USD** and **2000 EUR**.

$$1000 \text{ USD} = 100 * 10 \text{ USD}$$

2000 EUR = 100 * 20 EUR

Note: This also applies to multiple currency values divided by numeric values.

- **Calculations with single currency values** - When you use multiple currency values and single currency values in a measure definition, the calculation only returns a result for the values that share the same currency. For example, a field with the multiple currency values **100 USD** and **200 EUR** plus a field with the single currency value **20 EUR** returns **100 USD** and **220 EUR**.

100 USD = 100 USD

220 EUR = 200 EUR + 20 EUR

Note: This also applies when you calculate multiple currency values minus single currency values.

Duration values

Duration values are those that return values in units of time such as hours, minutes, and seconds.

This table shows the type of measures and basic operators that you can use with calculated measures that return duration values.

Data Type	Plus (+) Minus (-)	Multiplied by (*)	Divided by (/)
Numeric values	—	✓	✓
Single currency values	—	—	—
Multiple currency values	—	—	—
Duration values	✓	—	✓

Examples of Calculated Measures

The following examples show how you can use calculated measures in your pivot tables and charts.

Year over Year Growth

You can create a calculated measure to compare year-over-year differences in your sales reps' performance. Here's how you can create a pivot table with a calculated measure that shows this difference.

Step	Description
Step 1: Create two formula fields using the Sales (Ordered) dataset	To calculate year-over-year variance, you'll need to split the sales amount into two fields in the dataset. You can create these two formula fields: ■ Amount LYTD - The Amount Last Year to Date field shows sales amounts from all of last year up to today. ■ Amount TYTD - The Amount This Year to Date field shows sales amounts for transactions from the start of this year to today. To copy and paste the formulas, see Calculating Amounts for Relative Date Ranges.

Step	Description
Step 2: Create your pivot table	Set the following fields in the Layout panel: - Rows: Sales Rep - Columns: None - Measures: Amount TYTD (Sum) and Amount LYTD (Sum) For information about how to create pivot tables, see Workbook Pivot Tables.
Step 3: Create your calculated measure	Use the following definition: $(\text{Amount TYTD (Sum)} - \text{Amount LYTD (Sum)}) / \text{Amount LYTD (Sum)}$ For information about how to create calculated measures, see Working with Calculated Measures.
Step 4: Format the results	You can also customize your pivot table results, add totals and grand totals, or set the decimal values to percentage values. For more information, see Pivot Table Customization.

After you create the calculated measure, the pivot table displays the difference over time.

The screenshot shows the SuiteAnalytics Workbook interface. The 'DATASET' panel on the left lists fields like Amount, Amount (Net), Amount LYTD*, Amount TYTD*, Class, Customer, Date, Department, Est. Gross Profit, Item, Location, Quantity, Sales Channel, Sales Rep, and Transaction Type. The 'LAYOUT' panel in the center shows 'Sales Rep' in the Rows section and 'Amount TYTD* (Sum)', 'Amount LYTD* (Sum)', and 'YoY Variance %' in the Measures section. The 'YoY Variance %' measure is highlighted with a red box. The main view displays a pivot table with columns for Sales Rep, Amount TYTD* (Sum), Amount LYTD* (Sum), and YoY Variance %. The 'YoY Variance %' column is also highlighted with a red box. The table includes individual sales reps and a Grand Total row.

Sales Rep	Amount TYTD* (Sum)	Amount LYTD* (Sum)	YoY Variance %
Susan Miller	51,520.00	31,540.00	63.35 %
Jasmin Coke	68,130.00	50,434.00	35.09 %
Bill Clark	26,752.00	18,216.00	46.86 %
Timothy Little	13,604.00	37,760.00	-63.97 %
Brent Raines	298,940.00	87,229.00	242.71 %
Alex Fabre	41,240.00	56,760.00	-27.34 %
Evaline Mann	30,619.00	21,016.00	45.69 %
Bob Sales	100,130.00	60,424.00	65.71 %
Grand Total	630,935.00	363,379.00	73.63 %

For more information about comparing sales amounts over time using calculated measures, see the following video:

[Calculated Measures and Conditional Formatting](#)

Conditional Formatting

Conditional formatting lets you highlight your workbook visualization results based on rules you set. Right now, you can only apply conditional formatting to table views and pivot tables if the fields have NUMBER or STRING values. This includes standard, custom, formula, and calculated measure fields. Fields with DATE values aren't supported.

Conditional formatting does not differentiate between values in multiple currencies. For example, if you set a rule to highlight results over 10 dollars, any value over 10 euros will be highlighted too. To highlight

results accurately when using multiple currencies, convert or consolidate the values before applying conditional formatting. For more information, see [Currency in Datasets and Workbooks](#).

To use conditional formatting on percentage fields, you have to use decimal values in the rule. For example, if you want to highlight results greater than 20%, use 0.2 in your rule.

Conditional formatting doesn't currently let you filter results by color or icon. If you apply conditional formatting to a numeric field, you can sort the highlighted results in ascending or descending order to mimic filtering.

Any conditional formatting you use on a table view or pivot table is automatically applied to Analytics portlets based on that visualization. To turn off conditional formatting in the portlet, edit the visualization in its workbook. For more information, see the help topic [Opening Workbooks from the Analytics Portlet](#).

- [Conditional Formatting in Table Views](#)
- [Conditional Formatting in Pivot Tables](#)

Conditional Formatting in Table Views

In table views, you can highlight any column that you add to the table if it contains NUMBER or STRING values.

To apply conditional formatting to a table view:

1. Add the field you want to highlight to your table.
2. Click the Field Menu icon in the column header and go to **Conditional Formatting**.
3. Click **Manage Conditional Formatting**.
4. Select the operators, values, and color or icon for the rule. You can click the add icon **+** to create multiple rules and apply different colors or icons to the same column.
5. Click **Apply**.
6. Repeat steps 1–5 for each column you want to highlight.

Conditional Formatting in Pivot Tables

In pivot tables, you can highlight measures, subtotals, and grand totals, but only for individual cells, not entire rows.

To apply conditional formatting to a pivot table:

1. Add the field you want to highlight to the measures section of the Layout panel.
2. Click the Field Menu icon next to the field in the Layout panel and point to **Conditional Formatting**.
3. Click **Manage Conditional Formatting**.
4. Select the operators, values, and color or icon for the rule. You can click the add icon **+** to create multiple rules and apply different colors or icons to the same measure.
5. Check **Apply to subtotals and grand totals** if you want to apply your rules to the measure subtotals and grand totals.

6. Click **Apply**.
7. Repeat steps 1–6 for each measure you want to highlight.

Dataset Linking in SuiteAnalytics Workbook

Watch the following help video for an introduction to dataset linking in SuiteAnalytics Workbook:

 [Working with Linked Datasets in SuiteAnalytics Workbook](#)

You can link two datasets if each one has at least one field with common data, like a date. You can do this even if the datasets are based on record types without predefined relationships in the analytics data source.

For example, let's say you want to compare sales rep quotas to their real sales. You'd create two datasets: one based on the sales (invoiced) record type and the other on the quota record type. These record types have fields like date and department in common, but you can't join them in a dataset because there are no predefined relationships in the analytics data source. Even if you could join them, the one-to-many relationship between sales reps and transactions would create unwanted duplicate results.

This happens because datasets don't aggregate data, so you see all possible results. This is intentional—datasets are meant to be source data for visualizations like pivot tables and charts, where you set the aggregation. For more information, see [Data Duplication Based on Record Joins](#).

In contrast, with linked datasets, aggregation happens right before your results are generated, based on the fields you set as common keys and how you use them in your visualization. For example, with sales quotas and real sales, you can link datasets using the sales rep, accounting period, and posting period fields. Since quotas are created monthly, if you use the accounting period from the quota dataset as a pivot table dimension, your results are aggregated by month before showing up in the visualization.

For more information about linking datasets in SuiteAnalytics Workbook, see the following topics:

- Data aggregation isn't the only difference between linking datasets in a workbook and joining record types in a dataset. To learn more, see [Joining Record Types Versus Linking Datasets](#).
- To link datasets in a workbook, you need to set common keys between each dataset as part of the link definition. Common keys need to have the same values and use the same data type, but you can match fields with different types if you cast their values using a custom formula. To learn more about creating links and defining common keys, see [Link Datasets in a Workbook](#).
- After you link two datasets in a workbook, you can use them to create pivot tables and charts. Table views aren't supported yet. For more information, see [Create Visualizations Based on Linked Datasets](#).

Joining Record Types Versus Linking Datasets

In SuiteAnalytics Workbook, joins within a dataset happen through SQL queries running in the background. Specifically, a left outer join runs for each record type you add to a dataset. This means the results include all rows from the primary record type, plus any that match from the joined record type. No aggregation happens, so you might see too many or too few results depending on the relationship between record types and the join order. For more information, see [Guidelines for Joining Record Types in SuiteAnalytics Workbook](#).

In contrast, linked datasets work more like an SQL union. In a union, two queries run to give one set of results, defined by fields that share the same data type in each query. In Workbook, this means each

linked dataset runs as a background query, and then the results are aggregated and shown based on the common keys you set in the link.

To sum up: joins in a dataset let you combine and analyze fields from several record types in one set of results. Linked datasets let you analyze and aggregate results from two queries as if they were one. For more information about linking datasets and common keys, see [Link Datasets in a Workbook](#).

To learn more about the differences between joining record types in a dataset and linking datasets in a workbook, see the following table:

	Joining Record Types	Linking Dataset
SQL execution	Left outer join. Results include all fields from the primary record, plus any matches from the joined records.	Union. The results from two datasets show up in one visualization, including only distinct values, and aggregation is based on the fields you set as common keys when you create the link. For more information, see Common Keys .
Number of data sources	Unlimited. You can join as many record types as you want, but too many joins can cause performance issues.	Two datasets.
Data duplication	Data duplication can happen depending on the relationship between record types and the join order. For more information, see Data Duplication Based on Record Joins .	No. Results only contain distinct values from each dataset, so there is no data duplication.
Aggregation	No. This is intentional, since datasets are meant to be used as raw data for your visualizations.	Yes. Aggregation occurs automatically based on the fields you define as common keys and where you place them in the visualization.
Sharing	Yes. After you join record types in a dataset, you can save and share it with others in your account.	Yes, but only as part of the workbook they're linked in.
Requirements	- Access to record types - Predefined relationship between record types in the analytics data source. For more information, see <Analytics Data Source>.	- Access to datasets and underlying record types. - At least two datasets connected to the workbook. - At least one common key. For more information, see Common Keys.

Link Datasets in a Workbook

To link datasets in a workbook, you'll need to connect at least two datasets and set up at least one set of common keys between them. Common keys are fields that contain the same data in each dataset and use the same data type. For example, the sales rep field in both the sales (invoiced) and transaction record types has sales rep names as text, so you can use these as common keys for your link.

When you create a link, common keys are automatically set for fields with matching names and data types. If you want to set your own common keys, you can use any fields from either dataset if they're the same data type. Either date, text, numeric, or polymorphic. If the fields you want to use don't have the same data type, you can use a custom formula field (like TO_NCHAR) to cast their values. For more information, see [Common Keys](#).

To avoid errors, when you pick a field from the first dataset, any fields with a different data type in the second dataset won't be available. Also, after you've created a link, you can't unlink the datasets if they're used anywhere in the workbook. Linked datasets can also only be shared as part of the workbook they're in. You can't share them separately.



Important: If two fields in a dataset have the same ID but different join paths, you might run into issues when using them as common keys. To avoid errors, try deleting one of the fields before creating the link.

To link datasets in a workbook:

1. In the Dataset Panel of a workbook, click **Connect Dataset**.
2. Select a dataset from the list or click **New Dataset** if you want to define a new one. For more information see [Defining a Dataset](#).
3. After you have at least two datasets connected to the workbook, click the Dataset Menu icon next to a dataset in the Dataset Panel and select **Create Dataset Link**.
4. On the Dataset Linking page, double click or drag and drop the fields you want to define as common keys to the Common Keys area of the page.



Important: Any fields you plan on using as dimensions in your pivot table or chart need to be defined as common keys for your results to render properly. For more information, see [Common Keys](#).

5. Click **Link Datasets**.

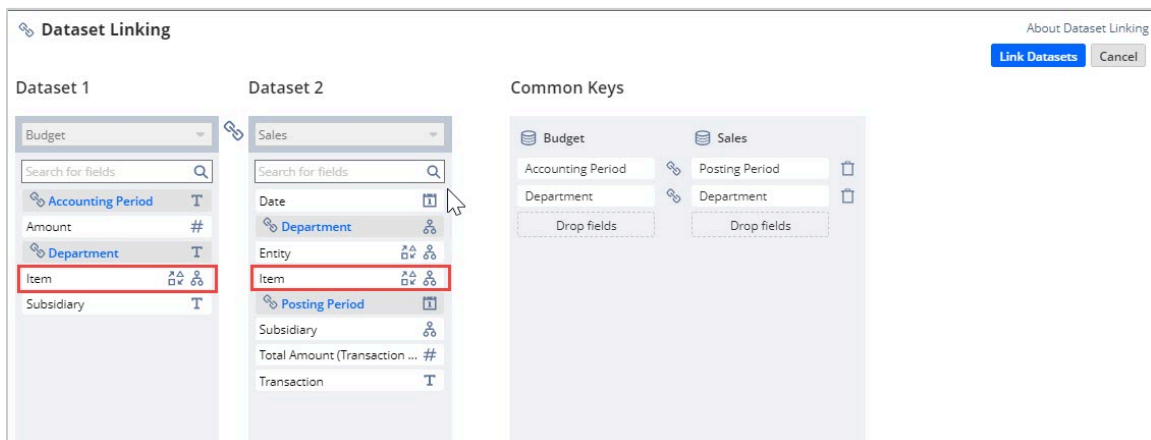
After you link your datasets, you can use them to build a pivot table or chart in a workbook. For more information, see [Create Visualizations Based on Linked Datasets](#).

You can also build calculated measures using fields from either of the linked datasets. For more information, see [Calculated Measures](#).

Common Keys

Common keys are fields with the same data in each dataset that you use to define a dataset link. To make sure results display and aggregate correctly in visualizations with linked datasets, any field you want to use as a dimension must also be set as a common key. If you don't do this, adding the field as a dimension might give you blank cells. This happens because without a matching key field, instead of showing only matching results from both datasets, all possible values from one dataset are included.

For example, say you have a workbook with linked budget and sales datasets. Posting period, accounting period, and department fields from each dataset are set as common keys for the link. There's also an item field in each dataset, but it's not set as a common key.



When you add the posting period field to the table rows and refresh, the results are generated as if an SQL union was run in the background—each dataset runs as a separate query, and you see results for matching posting period and accounting period values. Since posting periods are monthly, the results are also aggregated by month before they're displayed.

Posting Period	Sales (Sum)	Sales budget (Sum)
Jan 2021	\$77,353.00	75,000.00
Feb 2021	\$80,109.00	75,000.00
Mar 2021	\$83,022.00	75,000.00

The same occurs when you add the department field from either dataset to the pivot table, with matching results being displayed.

Posting Period	Marketing			Sales		
	Sales (Sum)	Sales budget (Sum)	Variance	Sales (Sum)	Sales budget (Sum)	Variance
Jan 2021	\$9,493.00	15,000.00	\$5,507.00	\$67,860.00	60,000.00	\$7,860.00
Feb 2021	\$62,989.00	15,000.00	\$47,989.00	\$17,120.00	60,000.00	\$42,880.00
Mar 2021	\$65,902.00	15,000.00	\$50,902.00	\$17,120.00	60,000.00	\$42,880.00
Grand Total	\$138,384.00	45,000.00	\$93,384.00	\$102,100.00	180,000.00	\$77,900.00

If you add the item field from the sales dataset, you'll see blank cells in the results. This happens because the union doesn't recognize the item field in the budget dataset, since it wasn't set as a common key.

Posting Period	Compaq Evo n800c			HP Compaq d230		
	Sales (Sum)	Sales budget (Sum)	Variance	Sales (Sum)	Sales budget (Sum)	Variance
Jan 2021	\$9,493.00			\$21,540.00		
Feb 2021	\$62,989.00					
Mar 2021	\$65,902.00					
Grand Total	\$138,384.00			\$21,540.00		

To avoid this, when you create a new link, Workbook automatically sets common keys using fields with matching names and data types. If the fields you want as common keys don't have the same data type, you can make a custom formula field (like with TO_NCHAR) to cast the field values to another data type. For more information, see [Advanced Sample Formula Fields](#).

After you've set your common keys, use caution when editing the related datasets. If you delete a field that's a common key, it can cause issues in your workbooks and keep you from saving. Also, if you filter a dataset by a field that's a common key, be sure to apply the same filter to its matching key field in the other dataset. If you don't, only results from one dataset will be filtered. Plus, any calculated measures you've made will only show up at the total and grand total levels.

Create Visualizations Based on Linked Datasets

You can use linked datasets to make pivot tables and charts in a workbook. Table views don't support data aggregation, so you can't use linked datasets for them.

As with visualizations based on single datasets, you build ones with linked datasets by assigning fields as dimensions and measures in the Layout panel. Any fields set as common keys in the link are marked with a chain link icon in the Dataset panel. For results to render and aggregate right, any fields you use as a dimension should also be set as a common key. If not, you might see blank cells. For more information, see [Common Keys](#).

Also, if you filter a field that's a common key, make sure you use the same filter on its matching key field. If you don't, only results from one dataset will be filtered.

c

To create a visualization based on linked datasets:

1. Click the **Add** icon from anywhere within the workbook.
2. Select the type of visualization you want to create.

 **Note:** Table Views do not support linked datasets.

3. Drag fields from the Dataset Panel to the Layout Panel. To switch between datasets, click the dataset name

 **Important:** Any fields you assign as dimensions should also be set as common keys in the link. This controls how your results are aggregated and prevents empty cells. For more information, see [Common Keys](#).

4. (Optional) Create a calculated measure using fields from either dataset. For more information, see [Calculated Measures](#).
5. (Optional) Filter your results. For more information, see [Workbook Visualization Filters](#).

 **Important:** If you want to filter your results with a dataset filter and use a field that's a common key, be sure to apply the same filter to its matching key in the other dataset. If you don't, only results from one dataset will be filtered. Plus, any calculated measures you've made will only show up at the total and grand total levels.

6. Click the **Refresh** icon to generate the results.

Currency in Datasets and Workbooks

If you're in a NetSuite account with transactions in different currencies, or a NetSuite OneWorld account with multiple subsidiaries, amount fields in your datasets and workbooks might show data in multiple currencies. To do math or see totals for these fields, you'll need to consolidate or convert the values

to a single currency. This is especially important if you're working with transaction, transaction line, or transaction accounting line record types.

- In transaction accounting lines, amount fields store monetary values in the base currency of the subsidiary assigned to the transaction. The amount fields in this record type can only be consolidated, and are usually for accounting purposes, like tracking financial results for your whole company and all subsidiaries. In Workbook, currency consolidation converts a field's values to the currency of the lowest-level subsidiary that's a common parent to all subsidiaries you have access to. This works the same way as currency consolidation in NetSuite saved searches and reports. By default, if you use amount fields from the transaction accounting line record type in workbook table views, pivot tables, or charts, currency consolidation is enabled.
- In transaction and transaction line records, amount fields store monetary values in the transaction currency. These values can only be converted, and they're usually for operational purposes, like analyzing total sales for a specific item. Currency conversion is another Workbook tool that lets you translate a field's values from one currency to another in your account.

You can apply currency consolidation or conversion to applicable fields right from the user interface in any workbook visualization. To consolidate or convert field values in a dataset, you'll need to use a custom formula field.

For more information about amount fields in each transaction record type, see the video  [Transaction Data Model in 2020.1](#).

For more information about currency consolidation and conversion in Workbook, see the following topics:

- [Currency Consolidation in Workbook](#)
- [Currency Conversion in Workbook](#)

Currency Consolidation in Workbook

Currency consolidation in Workbook combines a field's values into the currency of the lowest-level subsidiary that's a common parent to all the subsidiaries you have access to. The rate type used for consolidation comes from the general rate type set up for each account in your chart of accounts: Historical, Average, or Current. This matches the Per-Account option for exchange rates in transaction saved searches. For more information, see the help topic [Consolidated Exchange Rate Types for Transaction Searches](#).

 **Note:** In Workbook, if **Current** is selected for an account, you can't pick the exact period or date for the exchange rate. The rate is based on when you run the workbook visualization that contains the field. For more information, see the help topic [Consolidated Exchange Rates](#).

In Workbook, you can only use currency consolidation on amount fields from the transaction accounting line record type. Like saved searches and reports, you can only use consolidation if you're in a NetSuite OneWorld account with several subsidiaries. To make sure the right rates are used for consolidation, check that your chart of accounts and consolidated exchange rates are set up correctly. You should also check that your currency exchange rates are set up right, since they're part of some consolidated exchange rate calculations. For more information about currency exchange rates, see the [Currency Exchange Rates List Page](#).

Currency Consolidation Using Custom Formula Fields

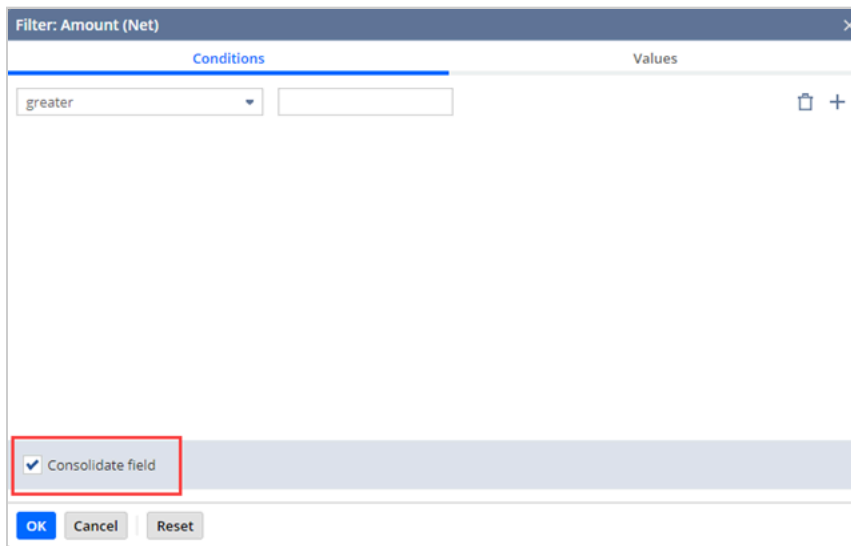
To use currency consolidation in a dataset, you'll need to create a custom formula field using the context value `#currency_consolidated`. For example, to consolidate the amount net field from the transaction accounting line record type, use this formula:


```
1 | TO_NUMBER({transactionlines.accountingimpact.netamount#currency_consolidated})
```

Currency Consolidation from the User Interface

In workbook visualizations, amount fields from the transaction accounting line record type have currency consolidation on by default. If you turn it off, you can turn it back on by clicking the Field Menu icon  next to a transaction accounting line field in the Layout Panel or column header, then go to **Currency** and click **Apply Consolidation**. If you're creating a calculated measure that has fields with values in different currencies, you can apply consolidation in the measure definition too. For more information, see [Calculated Measures](#).

In the Dataset Builder, the Filter window also has a Consolidate field box for when you make criteria filters based on transaction accounting line amount fields.



When you check the Consolidate field box, the field's values are consolidated before your filter conditions are applied. The consolidation rate type comes from the general rate type set for each account in your chart of accounts: Historical, Average, or Current.



Important: The Consolidate field box only works with fields from the transaction accounting line record type. To make criteria filters based on converted values from other record types, you'll need to create a custom formula field that applies currency conversion, then build your filter conditions using that field. For more information, see [Currency Conversion Using Custom Formula Fields](#).

Currency Conversion in Workbook

Currency conversion in Workbook lets you convert field values to any currency you've set up in your account. To use currency conversion in a dataset or workbook, you need the Multiple Currencies feature enabled in your account. Also, make sure your currency records and exchange rates are set up right for accurate conversion. For more information, see the help topic [Currency Exchange Rates](#).

Currency Conversion Using Custom Formula Fields

In the Dataset Builder, you can create custom formula fields with the context value `#converted` to change field values that are in different currencies.

For example, to show remaining balances on invoices no matter the original currency, you can create a custom formula field like this:

```
1 | TO_NUMBER({foreignamountpaid#converted}) - TO_NUMBER({foreignamountunpaid#converted})
```

The function TO_NUMBER is needed because the current Formula Builder doesn't support the CURRENCY output type, so TO_NUMBER converts these values to FLOAT. In single currency accounts, you could use this formula without the #converted context value:

```
1 | TO_NUMBER({foreignamountpaid}) - TO_NUMBER({foreignamountunpaid})
```

Depending on your formula, you can convert field values to your account's base currency or a specific currency you pick. You can also choose the exchange rate based on a date you enter in the formula.

Use these sample formulas to convert the values of the total amount (transaction currency) field. The output is FLOAT:

Note: To convert values from other fields, replace foreigntotal with a different field ID.

```
1 | TO_NUMBER({foreigntotal#converted})
```

- Displays the currency amounts in your account's base currency. The conversion uses the exchange rate that's active when you run the query or workbook visualization.

```
1 | TO_NUMBER({foreigntotal#converted[1]})
```

- Displays the currency amounts in the currency ID specified in the brackets [1]. The conversion uses the exchange rate active on the date when you run the query or workbook visualization.

```
1 | TO_NUMBER({foreigntotal#converted[1;2017-01-30]})
```

- Displays the currency amounts in the currency specified in the brackets [1], using the exchange rate set for a specific date. Enter the date in YYYY-MM-DD format.

Currency Conversion from the User Interface

In a workbook visualization, you can convert currency values right from the UI for amount fields from transaction and transaction line records. These options aren't available in a dataset.

To convert currencies in a workbook visualization, click the Field Menu icon  next to a transaction or transaction line field in the Layout Panel or column header, then select **Currency**. The following options are presented:

- **Display Original:** displays each amount in the transaction's original currency. In table views and pivot tables, this is the default for transaction and transaction line fields.
- **Apply Conversion:** changes each amount using the currency and date you pick in the Convert To window. By default, it uses the current date and your account's or subsidiary's base currency. In charts, this is the default for transaction and transaction line fields.
- **Conversion Options...:** opens the Convert To window, where you pick the currency and date to use for conversion. The rates are drawn from your account's currency exchange rates table. For more information, see the help topic [Currency Exchange Rates List Page](#).

Note: If you check the **Anchor Relative to Today** box, the current date is used for conversion every time the workbook is opened.

If you're creating a calculated measure with fields that have values in multiple currencies, you can apply conversion in the measure definition too. For more information, see [Calculated Measures](#).

Formula Fields and Calculated Measures

Unlike formula fields, which are created at the dataset level and evaluate results row by row, calculated measures are made at the workbook level using aggregated results. For example, if you create a formula field for sales amounts, each row in the dataset will show one value based on your formula. If you then use that formula field as a measure in a pivot table or chart, the results are simply an aggregate of the formula field values. In contrast, calculated measures use the aggregated results of base measures and arithmetic operations to show things like year-over-year variances and profit percentages. You can't get these kinds of results using formula fields alone.

The following examples show why you get a different result in each case:

- [Creating a formula field and using it as a measure](#)
- [Creating a calculated measure](#)

Creating a formula field and using it as a measure

In your dataset, you can use this definition to create a formula field that shows profit percentage:

% Profit Formula = {estgrossprofit}/{amount}

The formula field column gives a percentage value for each entry. For example, the % Profit (formula) column shows a value for each of the three entries for Sales Order 100537: 18%, 100%, and 100%.

Dataset: Sales (Ordered)_GrossProfit

Criteria summary: Show details

Date

Date on or after same day last fiscal year

Drop fields here to add criteria. Add New Group of Criteria

Reset Criteria

DATE	EST. GROSS PROFIT	AMOUNT	% PROFIT (FORMULA)	TRANSACTION	TRANSACTION ...	QUANTITY
13.5.2020	648.40	\$3,602.00	18.0011 %	Sales Order #SORD100537	Sales Order	1
13.5.2020	4,500.00	\$4,500.00	100.0000 %	Sales Order #SORD100537	Sales Order	450
13.5.2020	294.00	\$294.00	100.0000 %	Sales Order #SORD100537	Sales Order	7
9.6.2020	179.00	\$895.00	20.0000 %	Sales Order #SORD100538	Sales Order	1
9.6.2020	2.34	\$11.69	20.0171 %	Sales Order #SORD100538	Sales Order	1
21.10.2020	4.13	\$18.75	22.0267 %	Sales Order #SORD100539	Sales Order	1
21.10.2020	4.13	\$18.75	22.0267 %	Sales Order #SORD100539	Sales Order	1
21.10.2020	4.13	\$18.75	22.0267 %	Sales Order #SORD100539	Sales Order	1
21.10.2020	22.92	\$81.40	28.1572 %	Sales Order #SORD100540	Sales Order	1
21.10.2020	330.00	\$330.00	100.0000 %	Sales Order #SORD100540	Sales Order	30
21.10.2020	51.52	\$110.00	46.8364 %	Sales Order #SORD100540	Sales Order	1
11.1.2021	296.51	\$500.00	59.3020 %	Sales Order #SORD100541	Sales Order	5
19.1.2021	5.00	\$10.20	49.0196 %	Sales Order #SORD100542	Sales Order	1
19.1.2021	205.35	\$1,026.75	20.0000 %	Sales Order #SORD100542	Sales Order	1

If you use this formula field as a measure in a pivot table, the returned value is 72.67%, which is not entirely accurate.

$$(18\% + 100\% + 100\%) / 3 = 72.67\%$$

The screenshot shows the SuiteAnalytics Workbook interface. On the left, the 'DATASET' pane lists fields: % Profit (formula), Amount, Amount (Net), Class, Customer, Date, Est. Gross Profit, Item, Location, Quantity, and Sales Rep. The 'LAYOUT' pane shows 'Transaction' in the Rows section and '% Profit (formula) (Average)' in the Measures section. The main pivot table displays data for various sales orders, with the first row highlighted in red.

Transaction	Est. Gross Profit (Sum)	Amount (Sum)	% Profit (formula) (Avg)
Sales Order #SORD100537	5,442.40	\$8,396.00	72.67 %
Sales Order #SORD100538	181.34	\$906.69	20.01 %
Sales Order #SORD100539	12.39	\$56.25	22.03 %
Sales Order #SORD100540	404.44	\$521.40	58.33 %
Sales Order #SORD100541	296.51	\$500.00	59.30 %
Sales Order #SORD100542	210.35	\$1,036.95	34.51 %

Creating a calculated measure

In your pivot table, you can use this definition to create a calculated measure:

$$\% \text{ Profit (calculated measure)} = \text{Est. Gross Profit (Sum)} / \text{Amount (Sum)}$$

The calculated measure column gives a percentage value based on the aggregated values from Est. Gross Profit (Sum) and the Amount (Sum).

$$(648.40 + 4,500 + 294) / (3,602 + 4,500 + 294) = 64.82$$

The % Profit (calculated measure) in the pivot table gives 64.82%, while the % Profit (formula) (Avg) column gives 72.67%.

This screenshot shows the SuiteAnalytics Workbook interface with an additional calculated measure. The 'Measures' section in the 'LAYOUT' pane now includes '% Profit (calculated me...'. The pivot table has been updated to include this new measure, with the first row highlighted in red.

Transaction	Est. Gross Profit (Sum)	Amount (Sum)	% Profit (formula) (Avg)	% Profit (calculated measure)
Sales Order #SORD100537	5,442.40	\$8,396.00	72.67 %	64.82 %
Sales Order #SORD100538	181.34	\$906.69	20.01 %	20.00 %
Sales Order #SORD100539	12.39	\$56.25	22.03 %	22.03 %
Sales Order #SORD100540	404.44	\$521.40	58.33 %	77.57 %
Sales Order #SORD100541	296.51	\$500.00	59.30 %	59.30 %
Sales Order #SORD100542	210.35	\$1,036.95	34.51 %	20.29 %

In this example, the formula field measure only shows an average of the percentages from the dataset $((18\% + 100\% + 100\%) / 3 = 72.67\%)$. With the calculated measure, the amount is found by first aggregating Est. Gross Profit and Amount fields first and then performing the division (that is $648.40 + 4500 + 294 = 5442.4$, $3602 + 4500 + 294 = 8396$, so $5442.4 / 8396 = 64.82\%$). Therefore, the calculated measure returns the expected value.

For more information about formula fields, see [Formula Fields](#).

For more information about calculated measures, see [Calculated Measures](#).

Workbook and Dataset Templates

SuiteAnalytics Workbook currently offers over 20 predefined workbook and dataset templates.

A dataset template comes with all the required fields, record types, and filters you need, so you can create a workbook without setting up a complex dataset first. Workbook templates also include predefined workbook visualizations like table views, pivot tables, and charts, so you can easily analyze and visualize data from the related dataset templates. In a workbook template, you can add more visualizations based on the connected dataset template or any other datasets you have access to.

By default, only users with the Analytics Administrator permission can access workbook and dataset templates. If you don't have the Analytics Administrator permission, you can only use these templates if a someone with that permission shares them with you. You also need access to the underlying dataset template to open a workbook template. Additionally, some templates are feature dependent. If you don't see a listed templates in your account, reach out to your account administrator.

If you have the Analytics Administrator permission, you can get to the templates from the Analytics Home page under either the Workbooks or Datasets subtabs.

For more information about the Analytics Administrator permission, see [The Analytics Administrator Permission](#).

To see the dataset templates currently available in SuiteAnalytics Workbook, see [Dataset Templates](#).

To see the workbook templates currently available in SuiteAnalytics Workbook, see [Workbook Templates](#).

The sales (ordered), sales (invoiced), and manufacturing workbook and dataset templates are based on custom analytical record types made for SuiteAnalytics Workbook. These record types combine data from transaction, transaction line, and transaction accounting line record types.

To see the analytical record types currently available in SuiteAnalytics Workbook, see [Analytical Record Types](#)

Dataset Templates



Important: By default, only users with the Analytics Administrator permission can access the workbook and dataset templates in your account. If you don't have the Analytics Administrator permission, ask someone who has the permission to share the templates with you.

SuiteAnalytics Workbook offers over 20 dataset templates. Each comes with preselected fields, joined record types, and filters so you can quickly create a custom workbook or your own version of the dataset. Additionally, each dataset template has an accompanying, predefined workbook template. For information about workbook templates, see [Workbook Templates](#).

Here are the dataset templates you can use in SuiteAnalytics Workbook. If you have the Analytics Administrator permission, access Templates on the Datasets subtab of the Analytics Home page:



Important: Some templates are feature dependent. If you don't see one of the listed templates in your account, contact your account administrator.

- [Actual Time Dataset](#)
- [Customers by Item Dataset](#)
- [Employee Bonus History and Current Compensation Dataset](#)
- [Employee Tax Jurisdictions Dataset](#)

- Estimated Revenue Dataset
- Intelligent Item Recommendations Dataset
- Items by Customer Dataset
- Manufacturing Dataset
- Merchandise Hierarchy Dataset Template
- Recurring Revenue Dataset
- My Transactions Dataset
- Performance Management — Goals Dataset
- Performance Management — Reviews Dataset
- Pick Task Inventory Balance Dataset
- Performance Management – Reviews Scheduler Dataset Template
- Personal Information Access Logs Dataset and Workbook
- Pick Task Inventory Balance Dataset
- Purchases (Billed) Dataset
- Purchases (Ordered) Dataset
- Reviews Scheduler by Relative Date Dataset Template
- Sales (Ordered) Dataset
- Sales (Invoiced) Dataset
- SuiteCommerce Top Searches Dataset Template
- SuiteCommerce Top Searches with No Results Dataset
- Supply Chain Dataset
- System Notes v2 Dataset Template
- Time-Off Analysis Dataset
- Transaction Detail Dataset
- Warehouse Inventory: Counts & Adjustments Dataset
- Warehouse Inventory: Inbound Dataset
- Warehouse Inventory: Outbound Dataset
- Web Store Orders Dataset
- Workplace Tax Jurisdictions Dataset
- Customer 360 Datasets

Workbook Templates



Important: By default, only users with the Analytics Administrator permission can access the workbook and dataset templates in your account. If you don't have the Analytics Administrator permission, ask someone who has the permission to share the templates with you.

SuiteAnalytics Workbook offers over 20 workbook templates, all based on available dataset templates in your account. Each template comes with different visualizations to help you analyze your data, including a Table View with all the fields from the underlying dataset. Use these templates to quickly analyze and visualize different parts of your company data. To access the underlying dataset template, click the dataset name in the Dataset Panel from any of the workbook visualizations and select **Open Dataset**.

Some templates might also be based on a set of linked datasets. For more information, see [Dataset Linking in SuiteAnalytics Workbook](#).

The following workbook templates are currently available in SuiteAnalytics Workbook. If you have the Analytics Administrator permission, access Templates on the Workbooks subtab of the Analytics Home page:



Important: Some templates are feature dependent. If you don't see one of the listed templates in your account, contact your account administrator.

- [Actual Time Workbook](#)
- [Customers by Item Workbook](#)
- [Employee Bonus History and Current Compensation Workbook](#)
- [Employee Tax Jurisdictions Workbook](#)
- [Intelligent Item Recommendations Workbook](#)
- [Items by Customer Workbook](#)
- [Manufacturing Workbook](#)
- [Merchandise Hierarchy Workbook Template](#)
- [Recurring Revenue Workbook](#)
- [My Transactions Workbook](#)
- [Performance Management — Goals and Reviews Workbook](#)
- [Personal Information Access Logs Dataset and Workbook](#)
- [Purchases \(Billed\) Workbook](#)
- [Purchases \(Ordered\) Workbook](#)
- [Sales \(Ordered\) Workbook](#)
- [Sales \(Invoiced\) Workbook](#)
- [SuiteCommerce Top Searches Workbook](#)
- [SuiteCommerce Top Searches with No Results Workbook Template](#)
- [Supply Chain Workbook](#)
- [System Notes v2 Workbook Template](#)
- [Time-Off Analysis Workbook](#)
- [Transaction Detail Workbook](#)
- [Warehouse Inventory: Counts & Adjustments Workbook](#)
- [Warehouse Inventory: Inbound Workbook](#)
- [Warehouse Inventory: Outbound Workbook](#)
- [Web Store Orders Workbook](#)
- [Workplace Tax Jurisdictions Workbook](#)
- [Customer 360 Workbooks](#)

Analytical Record Types

Analytical record types are custom types made specifically for SuiteAnalytics Workbook and only exist in the [The Analytics Data Source and SuiteAnalytics Workbook](#). These record types are based on standard record types but have their own set of fields and predefined criteria. They combine data from the

transaction, transaction line, and transaction accounting line record types. The sales (ordered), sales (invoiced), and manufacturing templates use analytical record types.

The following analytical record types are currently available in SuiteAnalytics Workbook:

- [Manufacturing Component Analytical Record Type](#)
- [Manufacturing Transaction Analytical Record Type](#)
- [Sales \(Invoiced\) Analytical Record Type](#)
- [Sales \(Ordered\) Analytical Record Type](#)
- [Customer 360 Analytical Record Types](#)

Workbook-based Portlets

With the Analytics portlet, SuiteAnalytics Workbook lets you view your workbook data on your dashboards. The Analytics portlet lets you add charts, pivot tables, and table views from workbooks listed under My Workbooks and Shared with Me on the Workbook Listing page. You can add up to 10 Analytics portlets to your dashboards, each with its own workbook visualization. SuiteAnalytics Workbook also lets you translate Analytics portlet names through Manage Translation.

For more information about each of the portlet types, see the following topics:

- [Chart-based Portlets](#)
- [Pivot-based Portlets](#)
- [Table View Portlets](#)

For more information about how to add and set up an Analytics portlet, see the help topics [Adding an Analytics Portlet](#) and [Setting Up the Analytics Portlet](#).

For more information about translatable Analytics portlet names, see the help topic [Adding Translatable Analytics Portlet Names](#).

SuiteAnalytics Workbook Tutorial

Use this tutorial to walk through the authoring process for SuiteAnalytics Workbook.

In this tutorial, you'll create a sample workbook with visualizations based on two different datasets. One uses the sales (ordered) record type, and the other uses the item record type. By combining these datasets in a workbook, you'll quickly see if you have enough inventory to fulfill all the orders in your account.

The table below lists the steps to create the datasets and workbook visualizations. Each step builds on the last, so you must complete them in order:

Step	Description
Step One: Select a Root Record Type	Create your first dataset by choosing a root record type.
Step Two: Add Fields and Join Record Types	Choose the fields and record types that you want to include in your dataset.
Step Three: Filter Your Dataset	Apply criteria filters to refine your dataset query results.
Step Four: Create a Workbook Based on Your Dataset	Continue your analysis by creating a workbook based on your dataset.
Step Five: Pivot Your Dataset Query Results	Create a pivot table using your dataset query results.
Step Six: Connect a Second Dataset to Your Workbook	Create and join an additional dataset to your workbook.
Step Seven: Set up a Table View	Create a table view using the query results from your second dataset.

Note: This tutorial assumes you've got the right permissions and features turned on to view sales order and item data. If you're not sure which permissions you need to view sales order records or fields, download the worksheet: [NetSuitePermissionsUsage.xls](#). If you don't have a needed permission, contact your NetSuite administrator.

Select a Root Record Type

The following steps show you how to select a root record type for your initial dataset. The record types you see depend on the features turned on in your account and the permissions for your NetSuite role. If you don't see the sales (ordered) record type, contact your administrator.

To select a root record type:

1. Click the **Analytics** tab in the NetSuite navigation menu.
2. On the Analytics Home page, click the Datasets subtab and select **New Dataset**.
3. Select a record type from the list.

The Dataset Builder opens with preselected fields on the Data Grid based on the record type you choose.

For the purposes of this tutorial, select the sales (ordered) record type.

Continue to Step Two: [Add Fields and Join Record Types](#).

Add Fields and Join Record Types

If you haven't already, complete Step One: [Select a Root Record Type](#).

The steps below show you how to add fields and join record types to your dataset. Only fields in the Data Grid can be used to build tables, pivot tables, and charts.

By default, the Data Grid shows preselected fields based on the root record type you chose. Fields you add will show up highlighted at the top of the Fields list.

To add fields and join record types:

1. Add fields from the root record type to the Data Grid. You have three options:
 - Drag the fields from the Fields list to the Data Grid.
 - Double-click the fields in the Fields list.
 - Type the name of the fields in the search bar at the top of the Fields list, then drag or double-click them to add them to the Data Grid.

For the purposes of this tutorial, you don't need to add any additional fields from the sales (ordered) record type.

2. Join related record types to the dataset by adding their fields to the Data Grid.
 - a. By default, the Records list on the Dataset Builder shows all related record types that you have access to in your account.
Click the arrow next to any record type in the Records list to view additional related record types.
 - b. Click a record type name to update the Fields list.
 - c. Double-click or drag fields to the Data Grid.



Note: Depending on how the record types are related in your dataset, you might see data duplication. For more information, see [Guidelines for Joining Record Types in SuiteAnalytics Workbook](#).

For the purposes of this tutorial, add the status field from the transaction record type.

3. Create and add formula fields to the Data Grid.
 - a. Click the **Formulas** link above the Records list.
 - b. Click **New Formula** in the Fields list.
 - c. In the Formula Field window, enter the formula field name.
 - d. Select an output type for the formula field values.



Note: Each output type only works with certain fields and formula functions. If you pick an incompatible output type, your formula won't work. For more information, see [Formula Fields](#).

- e. In the Formula field, enter the field IDs and SQL formula functions for your formula. Or double-click field IDs or formula functions from the Functions and Fields subtabs to add them to the expression.

Note: The Fields subtab only includes fields from the root record type and any related types you've accessed in the Dataset Builder. If you want to use fields from another record type, you'll need to access it in the Dataset Builder or type in the field ID by hand.

- f. To validate the formula, click **Validate**.
- g. If there are no errors, click **Apply** to add the formula field to the dataset.
For the purposes of this tutorial, no formula fields are required.
4. Remove any fields you do not want to include in the dataset.
 - a. Click the Field Menu icon  in the column you want to remove from the Data Grid.
 - b. Select **Remove Column** from the dropdown list.
 For the purposes of this tutorial, remove the memo and partner fields.

Warning: Any changes you make to a dataset are automatically carried over to workbook visualizations based on it. If you remove a field from the Data Grid, it can cause failures in connected visualizations. Exercise caution when removing a field from the grid.

Continue to Step Three: [Filter Your Dataset](#).

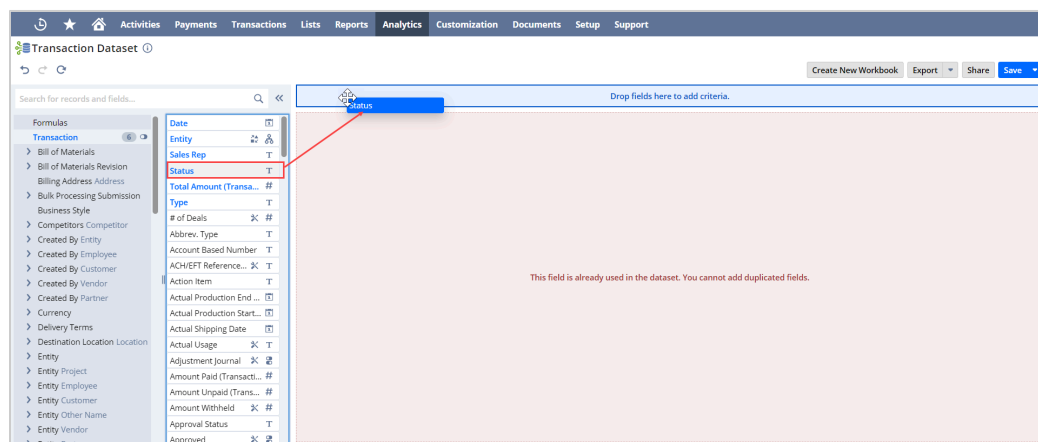
Filter Your Dataset

If you haven't already, complete Step Two: [Add Fields and Join Record Types](#).

The following steps show you how to filter your dataset query results. Any criteria filters you add are automatically applied to both new and existing workbook visualizations based on this dataset. For example, if you set a filter to exclude sales orders created in the last week, data from those orders won't show in any workbook visualizations, old or new, that use this dataset.

To filter your dataset:

1. On the Dataset Builder, drag a record field or formula field from the Fields list to the Criteria Builder above the Data Grid.
(Optional) If the field you want to use is already in the Data Grid, drag the field header from the grid to the Criteria Builder.



The Filter window appears.

2. In the Filter window, pick the filter conditions you want to apply. You can get up to four filtering options, depending on the field type:
 - Values: Existing values or dates from the source data, or custom values
 - Ranges/Date Ranges: Range of values or dates available in the source data
 - Relative Conditions/Relative Dates: Conditions relative to the existing values in the source data
 - Conditions/Specific Dates: Specific value or date and an expression

For the purposes of this tutorial, no filter conditions are required.

Note: If you want to group filter conditions, click **New Group**. Grouped filters count as one item and are checked in the order they appear in the Criteria Builder. For more information, see [Grouping Filters](#).

3. Repeat steps 1–2 for each filter you want to define.

By default, filters are added using an AND operator. To change the relationship between filters and filter groups, click **AND** and select **OR** from the popup window.

4. After you set each filter, the Data Grid updates to show only data that matches your criteria, and a summary appears above the grid.

Important: If you want to convert field values in multiple currencies, do it within a workbook visualization or by using a custom formula field with the TO_NUMBER function. For more information, see [Currency in Datasets and Workbooks](#).

5. Click **Save**.
The Save Dataset As window appears.
6. Enter a name and description for the dataset in the fields provided.
7. Click **Save**.

Continue to Step Four: [Create a Workbook Based on Your Dataset](#).

Create a Workbook Based on Your Dataset

If you haven't already, complete Step Three: [Filter Your Dataset](#).

The following steps show you how to create a workbook based on your completed dataset.

Note: If you want to make more workbooks from the dataset you saved in Step Three, go to the Analytics Home page, click New Workbook in the Workbooks subtab, then pick the dataset to open a new workbook.

If you want to edit the dataset from your workbook, click its name in the Dataset Panel in any workbook visualization.

To create a workbook based on your dataset:

1. On the Dataset Builder, click **Create New Workbook**.
A workbook opens. By default, the workbook is named after the root record type on the dataset.
2. Select a workbook visualization you want to create: either a Table View, Pivot table, or Chart.
For this procedure, select **Pivot**.

Continue to Step Five: [Pivot Your Dataset Query Results](#).

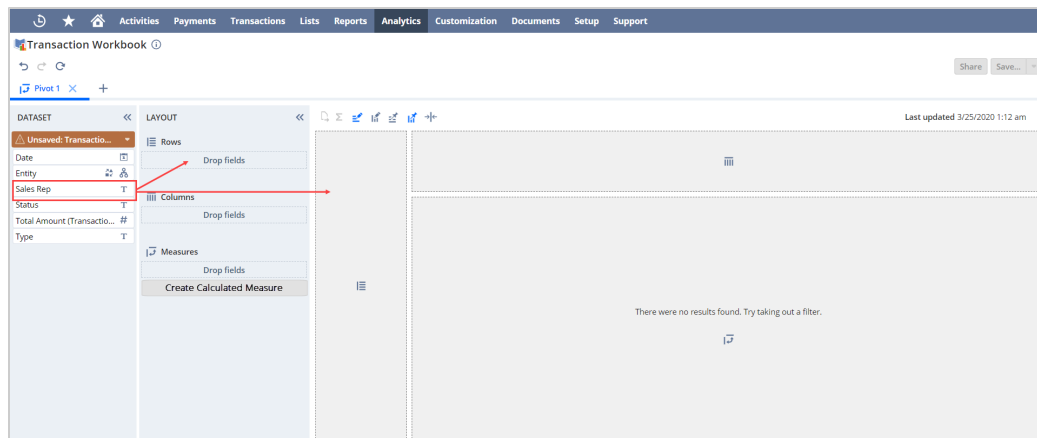
Pivot Your Dataset Query Results

If you haven't already, complete Step Four: [Create a Workbook Based on Your Dataset](#).

The following steps show you how to pivot your dataset query results to create a pivot table.

To pivot your source data:

1. On the Pivot tab, drag fields from the Fields list to the Rows, Columns, or Measures tabs in the Layout panel. Or, drag them straight into the Pivot Table Viewer.

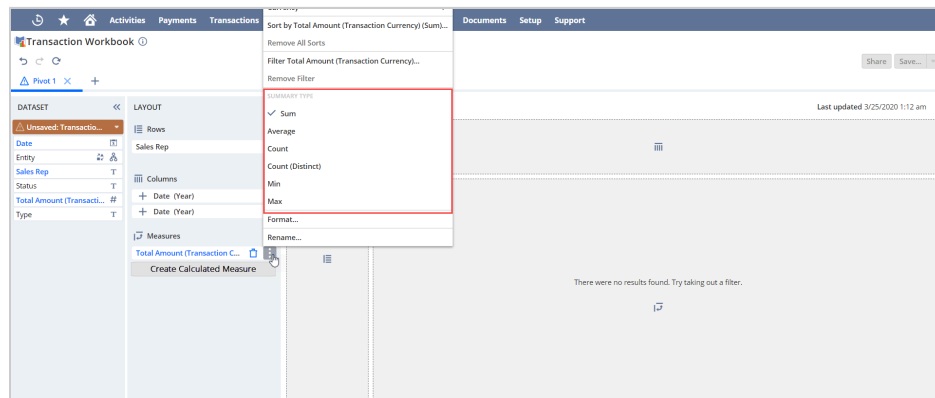


For this tutorial, add the item and status fields to Rows, and add the amount (net) and quantity fields in Measures.

Note: If you add hierarchical fields, you'll be prompted to pick a display type for them. Based on where you add the field and the display type, you can add subtotals at each level. For more information, see [Hierarchical Fields](#).

2. (Optional) To create a calculated measure, click **Create Calculated Measure** in the Layout Panel. You can also click the Field Menu icon  next to a measure in the Pivot Table Viewer and select **Create Calculated Measure**. Calculated measures are displayed with a calculator icon . For more information, see [Calculated Measures](#).
3. Select the summary type and format options for any date or numerical fields you add to the pivot table.

1. Click the Field Menu icon  next to the field you want to format in the Layout panel.
2. Select a summary type from the popup window.



3. (Optional) Select **Currency...** to view the currency consolidation or conversion options for fields with values in multiple currencies.

For more information, see [Currency in Datasets and Workbooks](#).



Important: If your fields have values in multiple currencies, you'll need to convert them to a single currency before doing arithmetic or any numeric manipulation.

4. (Optional) Click **Format...** to customize the numeric values for a field.

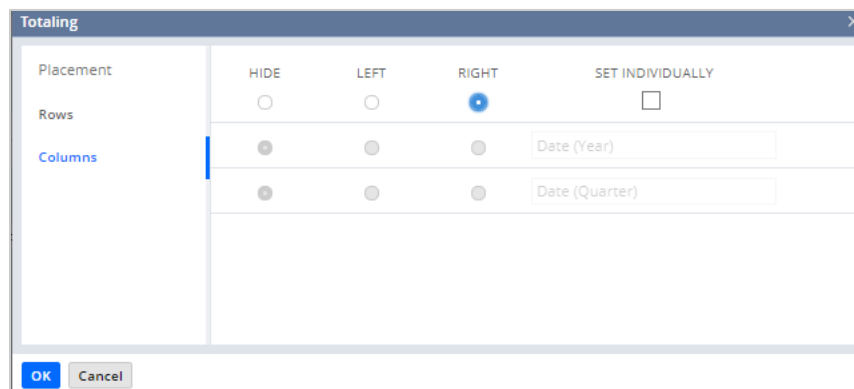
For more information about numeric formatting options, see [Customizing Numeric Values](#).

For the purposes of this tutorial, select the following summary types. No numeric formatting is required:

- Amount (Net) (Sum)
- Quantity (Sum)

4. Add totals and grand totals to the pivot table.

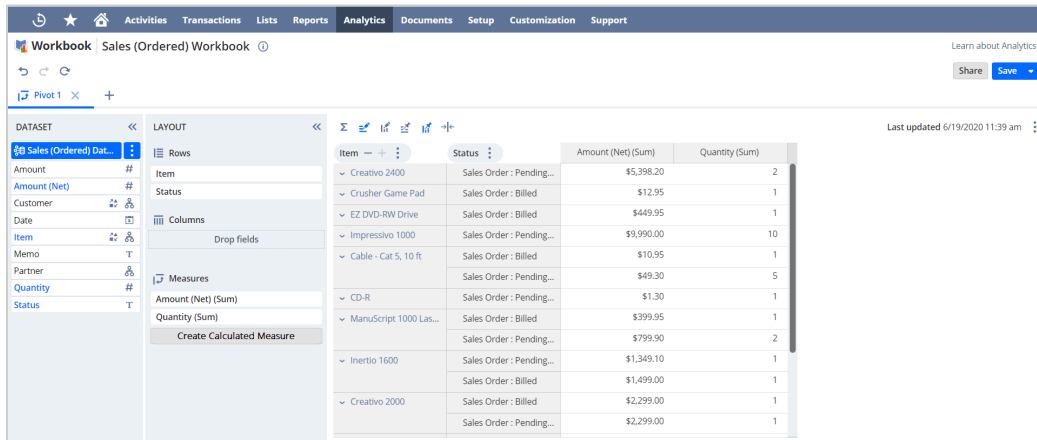
1. Click the Totaling icon .
2. In the Totaling window, select where you want the totals or grand totals for each field to appear. If there are multiple fields that can be totaled in the rows or columns, check the **Set Individually** box to choose where the totals for each field appear on the pivot table.



3. Click **OK**.

For the purposes of this tutorial, no totals are required.

5. Click the Refresh icon  to generate the pivot table.



Item	Status	Amount (Net) (Sum)	Quantity (Sum)
Creative 2400	Sales Order : Pending...	\$5,398.20	2
Crusher Game Pad	Sales Order : Billed	\$12.95	1
EZ DVD-RW Drive	Sales Order : Billed	\$449.95	1
Impressivo 1000	Sales Order : Pending...	\$9,990.00	10
Cable - Cat 5, 10 ft	Sales Order : Billed	\$10.95	1
CD-R	Sales Order : Pending...	\$49.30	5
Manuscript 1000 Lasi...	Sales Order : Billed	\$1.30	1
	Sales Order : Pending...	\$399.95	1
	Sales Order : Pending...	\$799.90	2
Inertio 1600	Sales Order : Pending...	\$1,349.10	1
	Sales Order : Billed	\$1,499.00	1
Creative 2000	Sales Order : Billed	\$2,299.00	1
	Sales Order : Pending...	\$2,299.00	1

6. (Optional) Filter the data displayed in the pivot table.

Note: Filter conditions created on the Pivot tab only affect the data you see in the pivot table. They don't change the underlying dataset or other workbook visualizations.

- Click the Field Menu icon  next to the field you want to filter. Depending on whether it's set as a column, row, or measure, or if you click the Field menu icon from the Fields List or the Layout Panel, the following options are available:
 - Top 10: display only the top 10 rows or columns based on the measures defined for the table.
 - Bottom 10: display only the bottom 10 rows or columns based on the measures defined for the table.
 - Filter [Field Name] by...: enables you to define a custom measure-based filter for the selected row or column.
 - Filter [Field Name]: enables you to define a custom value-based filter based on specific values within the table results.
 - Add as Filter...: enables you to define a custom value-based filter based on specific values within the table results.

- The results in the table are updated automatically.

For more information, see [Workbook Visualization Filters](#).

For the purposes of this tutorial, no pivot table filters are required.

7. (Optional) Apply conditional formatting to the pivot table measures to highlight your results.

Important: If your measure field has values in multiple currencies, you need to convert or consolidate them before applying conditional formatting. For more information, see [Currency in Datasets and Workbooks](#).

- Click the Field Menu icon next to the measure field you want to highlight in the Layout Panel and point to **Conditional Formatting**.
- Click **Manage Conditional Formatting**, then select the operators, values, and colors or icons for the rule. You can click the  add icon to create multiple rules and apply different colors or icons to the same measure or column. For more information, see [Conditional Formatting](#).

 **Note:** To apply conditional formatting to percentage values, use decimal format when defining the values for the rule. For example, rather than **greater or equal to 20%**, the rule should be defined as **greater or equal to 0.2**.

- c. Check **Apply to subtotals and grand totals** if you want to apply your rules to the measure subtotals and grand totals.
- d. Click **Apply**.
- e. Repeat steps A-D for each measure you want to highlight.

For the purposes of this tutorial, no conditional formatting is required.

- 8. (Optional) Click the Export icon to save a CSV file of your pivot table.

Continue to Step Six: [Connect a Second Dataset to Your Workbook](#).

Connect a Second Dataset to Your Workbook

If you haven't already, complete Step Five: [Pivot Your Dataset Query Results](#).

The following steps show you how to create and add an additional dataset from your workbook.

To connect a second dataset to your workbook:

1. Click the add **+** icon to open a new workbook visualization tab.
2. Click **Connect Dataset**.
3. On the Select a dataset page, click **New Dataset**.
4. Select a record type for the dataset.
For the purposes of this tutorial, select the item record type.
5. In the Dataset Builder, add the quantity on hand and quantity back ordered fields from the item record type to the Data Grid.
6. Click **Save**.
7. In the Save Dataset As window, enter a name and description for the dataset, then click **Save**.
8. Click **Apply to workbook**.

Continue to Step Seven: [Set up a Table View](#).

Set up a Table View

If you haven't already, complete Step Six: [Connect a Second Dataset to Your Workbook](#).

The following steps show how to create a table view using the query results from your second dataset.

To create a table view:

1. Select **Table**.
2. Double click the names of the fields in the order that you want them to appear in the table.
For this tutorial, add the fields in the following order: item name, quantity on hand, quantity ordered.

3. Click **Save**.
4. In the Save Workbook As window, enter a name and description for the workbook in the fields provided and click **Save**.

All selections made in each workbook visualization are saved.



Important: You must save any datasets created as part of the workbook authoring process before you can save the workbook. For more information, see [Saving and Sharing Workbooks and Datasets](#)

5. (Optional) Share the workbook with other users in your account.
 - a. Click **Share**.
 - b. In the Share Workbook window, select the users or roles you want to share the workbook with.
 - c. Click **Share**.

For more information about sharing workbooks, see [Accessing and Sharing Workbooks and Datasets](#).
6. (Optional) If you want to edit any datasets linked to the workbook, click the dataset name in the Dataset Panel in any workbook visualization.

For more information, see [Editing a Dataset](#).

Navigating SuiteAnalytics Workbook

The SuiteAnalytics Workbook user interface includes an entry point, a Dataset Builder, and three workbook visualization tabs. Click the following links to learn more about each area of the interface:

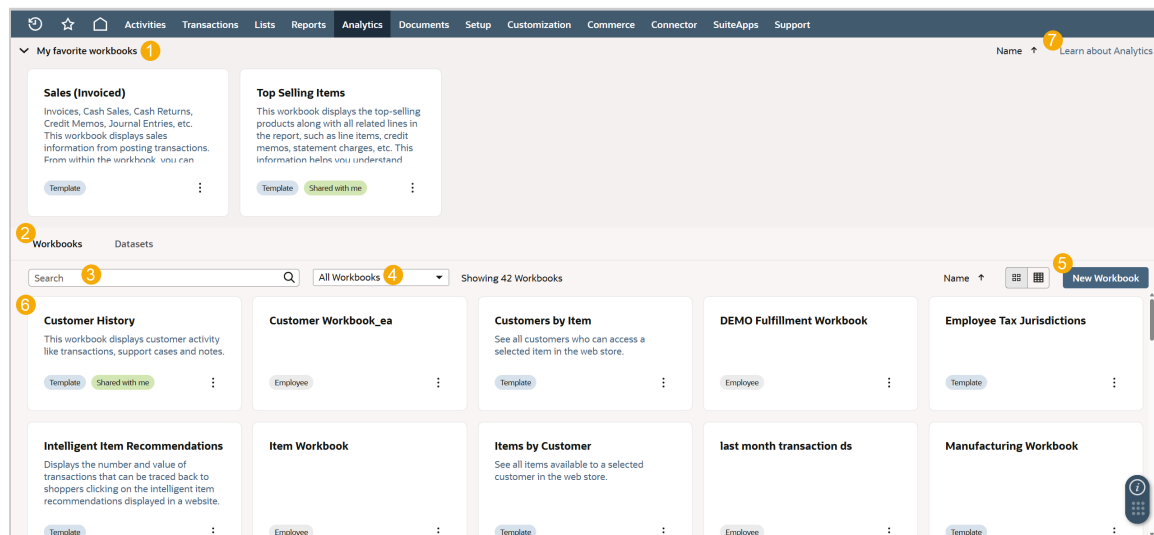
- [Analytics Home Page](#)
- [Dataset Builder](#)
- [Table Tab](#)
- [Pivot Tab](#)
- [Chart Tab](#)

Analytics Home Page

The Analytics Home page appears when you click the Analytics tab in the NetSuite navigation menu. The page has the Workbook and Dataset subtabs, so you can open and view any workbooks and datasets you have access to. If you have the Analytics Administrator permission, you can also work with predefined workbook and dataset templates and any workbooks or datasets other users in your account have created. For more information about the Analytics Administrator permission, see [The Analytics Administrator Permission](#).

The image below shows the parts of the Analytics Home page:

 **Note:** The options available on this page vary depending on whether you are in the Workbooks or Datasets subtab.



1. **Favorite Workbooks** — Includes any workbooks you have marked as a favorite. Click the icon in the upper right corner to sort the workbooks by name, in ascending or descending order.
2. **Workbooks/Datasets subtabs** — Switch between workbooks and datasets in your account.
3. **Search** — Search for saved workbooks or datasets using a field name, user name, criteria filter, or any other workbook or dataset element. Searches are only executed across the selected list (see 4).

For example, to search workbooks that you own, make sure My Workbooks is selected in the list of the Workbooks subtab.

4. **Workbook/Dataset lists** — Specify the workbooks or datasets you want to view. You can only view templates and objects from other users if you have the Analytics Administrator permission:
 - **All Workbooks/ Datasets**- Every workbook or dataset that you have access to in your account.
 - **My Workbooks/ Datasets**- Workbooks or datasets that you have created.
 - **Shared with me**- Workbooks and datasets that other users have created and shared with you.
 - **Employee Workbooks/ Datasets**- Workbooks and datasets created by other users in your account.
 - **Templates**- Predefined workbook and dataset templates. For more information, see [Workbook and Dataset Templates](#).
5. **New Workbook/Dataset** — On the Workbook subtab, click New Workbook create a new workbook using an existing dataset or a new one you create. On the Dataset subtab, click New Dataset to select a record type and create a new dataset. For more information, see [Custom Workbooks and Datasets](#).
6. **Workbook/Dataset Viewer** — All workbooks and datasets that you have access to are presented in the viewer. On the Workbook subtab, click the icons next to the **New Workbook** button to switch between a list view and tile view of the workbooks in your account.
 - **List view** – Shows the workbook name, details, description, owner, and when it was last opened or modified.
 - **Tile view** – Shows the workbook name, description, and type.

In the Workbook/Dataset Viewer, you can do the following:

 - To see more details about a dataset, such as a summary of its fields, filters, or the joined record types, click **Details**. In the Details Panel, you can also create a new workbook based on the dataset.
 - To see more details about a workbook, such as a list of each visualization and their datasets, click **Details**. In the Details Panel, you can also open each dataset the visualizations use.
 - To delete or share a workbook or dataset, click the menu icon and select **Share** or **Delete**. For datasets, you can also select **Create Workbook**.
 - To mark a workbook as favorite, do this:
 - In the tile view, click the menu icon and select **Mark as Favorite**.
 - In the list view, click the Star icon next to the workbook name.
7. **Learn about Analytics** — Click **Learn about Analytics** to open a panel with all SuiteAnalytics Workbook videos and their descriptions. To watch a video, click the thumbnail. It will open in a new window.

Dataset Builder

The Dataset Builder appears after you select a root record type for a new dataset, or when you click the dataset name in the Dataset Panel of a workbook. On this tab, you combine record types, fields, and criteria filters to create queries. The results of these queries are the basis for all workbook visualizations in your account.

In the Dataset Builder, your query results are displayed in a table in the Data Grid on the right. Record types, fields, custom formula fields, and configuration options appear in the Records and Fields lists on the left. Only fields in the Data Grid can be used to build workbook visualizations such as pivot tables and charts.

Above the Data Grid, the Criteria Builder lets you create criteria filters to narrow down the results in the grid. Criteria filters affect both your dataset query results and the data shown in workbook visualizations based on the dataset.

To build your dataset, drag fields from the Fields list to the Data Grid. To create a filter condition, drag fields from the Fields list or the Data Grid to the Criteria Builder. In the grid, you can sort and filter the values. After you create your query, you can apply or preview the results in a workbook in several ways.

- For information about how to use the Dataset Builder to create a query, see [Defining a Dataset](#).
- For information about creating criteria filters to refine your query results, see [Dataset Criteria Filters](#).
- For information about creating a workbook based on your dataset query results, see [Creating a Workbook](#).

The image below shows the parts of the Dataset Builder:

Note: The appearance of the Dataset Builder varies slightly if you access it from within a workbook or on its own. See the following table for more information.

The screenshot shows the Oracle NetSuite Dataset Builder interface. The top navigation bar includes tabs for Activities, Transactions, Lists, Reports, Analytics, Documents, Setup, Customization, Commerce, Connector, SuiteApps, and Support. The main interface is divided into several sections:

- Dataset Information (1):** Located at the top left, it shows the dataset name 'Transaction Dataset' and its internal ID.
- Undo, Redo, and Refresh (2):** Buttons for managing the dataset's state.
- Search (3):** A search bar to find record types and fields.
- Records list (4):** A list of record types and fields, including 'Transaction Line', '[M-H] Dept.', '[M-H] Product', '[M-H] Sport', '[M-H] Store', 'Activity Code', 'Approval Status', 'Assignee', 'Associated Sales Transaction...', 'Associated Transaction Fulfillment...', 'Associated Transaction Payment...', and 'Associated Transaction Store...'. Each item has a number of fields in parentheses.
- Formulas (5):** A section for creating custom formula fields.
- Criteria Builder (6):** A section for creating filter conditions.
- Data Grid (7):** A table displaying the dataset's results. The columns are DATE, MEMO, ENTITY, TRANSACTION, TYPE, and TOTAL. The rows show various transactions, such as 'ACME Shipping Co.', 'Elena's Investments', 'Great Wall', 'Acme Medical Supply', 'Hertz Car Rental', 'ACME Shipping Co.', 'Dental Office', 'TR Fictious Company 1', 'Acme Medical Supply', 'Prefix Numbering Test', and 'ACME Shipping Co.'.
- Actions (8-11):** Buttons for 'Create New Workbook', 'Export', 'Share', and 'Save'.
- Learn about Analytics (12):** A link to learn more about analytics.
- Help (13):** A question mark icon for help.

- 1. Dataset Information** ⓘ — Click to view or edit the dataset name, description, or internal ID.
- 2. Undo, Redo, and Refresh** — Undo or redo your last changes, and refresh the data in the Data Grid.
- 3. Search** — Search for record types and fields using keywords, field labels, or internal field IDs. The search looks at the root record type and record types up to three joins away. To search other record types, click **Show more results**.
- 4. Records list** — Lists all the related record types you can access, based on the root record type selected for the dataset. If a record type has a number next to it, that shows how many fields from it are in the Data Grid. You can perform the following actions from the Records list:

Note: By default, record types and fields are listed alphabetically, except in a transaction dataset. There, the transaction line and transaction accounting line record types show up at the top.

- To join a record type to the dataset, add one of its fields to the Data Grid or use it in a filter. For more information, see [Guidelines for Joining Record Types in SuiteAnalytics Workbook](#).
- To create or view custom formula fields, click **Formulas**. For more information about formula fields, see [Formula Fields](#).

- To see more information about a record type, point to its name and click the Information icon. ⓘ.
- To see the fields of a record type and update the Fields list, click the record type name.
- To see more related record types, click the Arrow icon ► next to any record type.
- To see only record types that have fields in the Data Grid, click the toggle icon.

5. **Fields list** — The Fields list shows all the fields you can access for the record type you've selected in the Records list. If you click **Formulas** in the Records list, the Fields list shows any custom formula fields created for the dataset. At the top of the Fields list, any fields from the selected record type that are in the Data Grid are highlighted and listed alphabetically. Fields used in a filter are highlighted but not shown at the top of the list. Next to each field, these icons give you more information:

- : This icon means it's a custom field. For more information, see the help topic [Custom Fields](#).
- : This icon means it's a field that contains values that exist on multiple record types. For example, the customer field on the sales (ordered) record type, that has the same data as the Name/ID field in the Customer record.
- : This icon means it's a hierarchical field. These fields have a parent-child relationship in NetSuite, offer different display options, and can be used to create filters based on parent or child values. For more information, see [Hierarchical Fields](#).
-    : These icons show the type of values in the field— text, number, date, or Boolean.

With the Fields list, you can do the following:

- To collapse the Records and Fields lists and expand the Data Grid, click the double arrows ⇐⇐.
- To see more information about a field, point to it and click the ⓘ icon.
- To add a field to your dataset, double-click it or drag it from the Fields list to the Data Grid.
- To create a filter, drag a field from the Fields list to the Criteria Builder above the Data Grid.
- (Optional) If the field you want is already in the Data Grid, drag its header from the grid to the Criteria Builder.



Important: When you add fields from related record types to the Data Grid or use them in filters, that record type is joined to the dataset. Depending on how the related record type connects to the root record type, some joins can cause unexpected results such as data duplication. If joining record types is new to you, see [Guidelines for Joining Record Types in SuiteAnalytics Workbook](#) for more information.

6. **Criteria Builder** — The Criteria Builder shows the filters you've created for the dataset in the order that they are applied to the Data Grid, as well as the operators that connect them. For example, the filters "Cleared is true AND Status any of Open, Paid In Full" mean the cleared field is checked first, then the status field.

Criteria filters narrow down your dataset query results and are automatically applied to any workbook visualizations based on the dataset. For example, if you filter out all invoices from your dataset, invoice data is also removed from any connected workbook visualizations.

You can add filters to the Criteria Builder by dragging fields from the Fields list to the builder, then setting the filter conditions in the Filter window. You can also drag a column header from the Data Grid to the Criteria Builder. For more information, see [Dataset Criteria Filters](#). In the Criteria Builder, you can do the following:

- To edit a filter, click its name.
- To delete a filter, point to it and click Delete .
- To change the order of the filters, point to the one you want to move and click the arrows.

- To group filters, click **New Group**. For more information, see [Grouping Filters](#).
 - To change the operator between filters, click **AND** or **OR** and select a different operator from the popup window.
 - To find a field in the Fields list, point to its filter, click the Field Menu icon  and select **Show Location in Field List**.
 - To reset your filters, click **Reset Criteria**.
 - To hide the Criteria Builder and expand the Data Grid, click the arrow next to the criteria summary.
7. **Data Grid** — Your dataset query results are displayed in the Data Grid. By default, the grid shows preselected fields based on the root record type you selected for the dataset. You can add fields to the grid by double-clicking them on the Fields list or dragging them to the grid. The number of rows per page in the grid depends on the **Number of Rows in List Segments** setting in your NetSuite user preferences, up to 500 rows per page. Only fields in the grid can be used in workbook visualizations based on the dataset. In the grid, you can do the following:
- To sort values in the grid, click the Field Menu icon  in the column header and select **Sort Ascending**, **Sort Descending**, or **Edit Sort...** for advanced options.
- For more information, see [Advanced Sorting Options](#).
-  **Note:** If you sort your results based on a field that contains CLOB values, Workbook currently only evaluates the first 250 characters for each result. Results with more than 250 characters may not be sorted properly.
- To filter the values in the grid, click the Field Menu icon  next to the field you want, then select **Filter...** from the list. The filter window appears with up to four filter types, depending on the values of the field. For more information, see [Filter Types](#).
-  **Note:** Filter conditions and sorting options set up in the Data Grid only affect what is shown in the grid and don't change values in any connected workbook visualizations. If you want to change your dataset query results and any related visualizations, set up criteria filters using the Criteria Builder above the Data Grid. For more information, see [Dataset Criteria Filters](#).
- To move a column, drag its header where you want it. Alternatively, click the Field Menu icon  next to the column and select a move option.
 - To remove a column, click the Field Menu icon  next to it and select **Remove Column**.
 - To rename a column, click the Field Menu icon  next to it and select **Rename....**
 - To change a column's width, drag the right edge of its header until it's as wide as you want.
 - To see the records linked in the grid, click the link. The NetSuite record opens in a new browser tab.
 - To see other pages, click the arrows at the bottom of the grid.
 - To see a summary of the values for any column, click its header. The summary is shown at the bottom of the grid.
 - To find a field in the Records and Fields lists, click the Field Menu icon  and select **Show Location in Field List**.
8. **Create New Workbook/ Apply to Workbook** — Click **Create New Workbook** to create a workbook based on the current dataset. If you opened the dataset from an existing workbook, click **Apply to Workbook** to apply your changes before saving them. This lets you preview how your changes will affect the connected workbook.
9. **Export** — Click **Export** and choose one of these options:

- Export data to CSV



Note: To export data from the Data Grid to a CSV file, you need the Export Lists permission.

- Export dataset to SDF XML
 - Export dataset to SuiteQL TXT
10. **Share** — Share your dataset with users or roles. For more information, see [Accessing and Sharing Workbooks and Datasets](#)
 11. **Save** — Save your dataset, including any joined record types, custom formula fields, or filters. If you don't own the dataset you're working in, you need to save a copy using Save As.
 12. **Learn about Analytics** — Click **Learn about Analytics** to open a panel with all SuiteAnalytics Workbook videos and their descriptions. To watch a video, click the thumbnail and it opens in a new window.
 13. **Real-Time / Cached Response** — If you're experiencing long load times for your datasets or workbooks, you can turn on Cached Data in Datasets. For more information, see [Optimized Data Refresh](#).

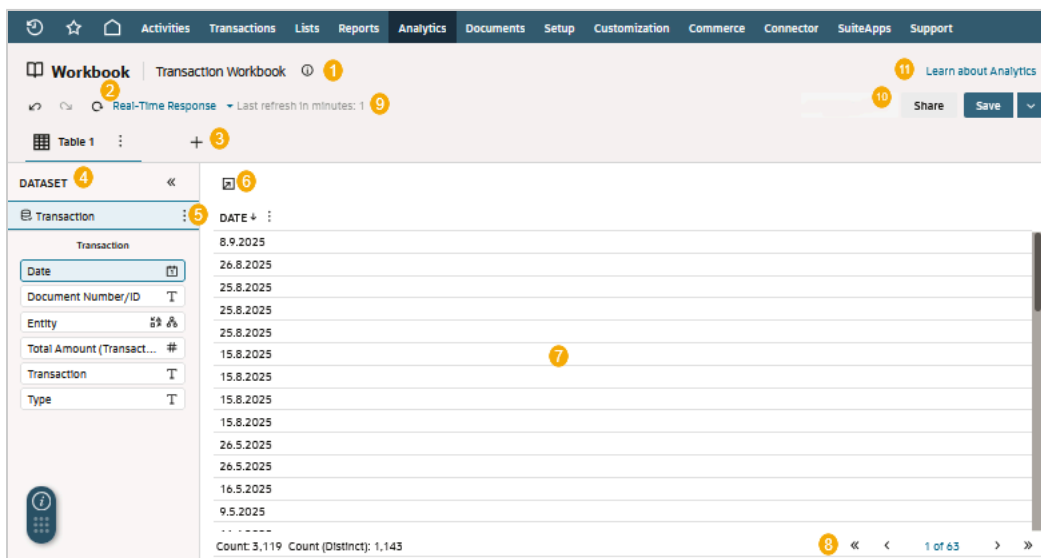
Table Tab

The Table tab lets you explore your dataset query results without changing the source data in any workbook visualizations or setting a layout. All fields in the dataset appear in the Dataset Panel on the left. To build your table, drag fields from the Dataset Panel to the Viewer.

You can create multiple table views in a single workbook using the dataset query results.

For information about how to explore your data in a table view, see [Workbook Table Views](#).

The image below shows the parts of the Table View tab:



1. **Workbook Information** — Click to view or edit the workbook name, description, internal ID, or add a translatable portlet name to your Analytics portlet. This information is available from anywhere in the workbook.

For more information about translatable Analytics portlet names, see the help topic [Adding Translatable Analytics Portlet Names](#).

2. **Undo, Redo, and Refresh** — Undo or redo your last changes, and refresh the data in the current workbook visualization. The undo and redo options work for any action in the workbook, not only the current tab. For more information about data refreshing, see [Data Refresh in SuiteAnalytics Workbook](#).
3. **Add** — Click the add icon to add a new workbook visualization.
4. **Dataset Panel** — The Dataset Panel is on the far left area and has all the fields from the selected dataset. To build a visualization, drag fields from the Dataset Panel to the Layout Panel or the Table Viewer. You can also edit the dataset or switch to a different one using the Dataset Selector at the top.
 - To view or edit a dataset, click its name in the Dataset Selector.
 - To see more information about a field, point to it and click the  icon.
 - To collapse the Dataset Panel and expand the Viewer, click the double arrows .
5. **Dataset Selector** — This list lets you access and edit the selected dataset, choose a different one, or connect a dataset that's not already used in another visualization.
 - To view or edit a dataset, click its name in the Dataset Selector. Alternatively, click the menu icon and select **Open Dataset**.
 - To connect or create a new dataset, click the arrow and select **Connect Dataset**.
 - To save or discard changes to the selected dataset, click the menu icon and select **Save Changes** or **Discard Changes**. You need to save or discard changes to all connected datasets before you can save the workbook.
6. **Export** — Click this icon to export the table as a CSV file.



Important: To help prevent CSV injection, exported tables may have additional formatting applied. For more information, see [Known Limitations in SuiteAnalytics Workbook](#).

7. **Table Viewer** — The Table Viewer shows your table as soon as you add fields. In the Table Viewer, you can customize how columns look, and sort or filter your data.
 - To customize how your table looks, you can do the following:
 - To rename a field, click the Field Menu icon  and select **Rename...**



Note: This change applies only to the column name in the current table view.

- To change column order, drag them where you want or click the Field Menu icon  and select a move option.
- To remove a column, click the Field Menu icon  in its header and select **Remove Column**.
- To filter or sort your data, click the Field Menu icon  next to the field you want and do the following:



Note: If you sort your results based on a field that contains CLOB values, Workbook currently only evaluates the first 250 characters for each result. Results with more than 250 characters may not be sorted properly.

- To sort the table, select **Ascending** or **Descending**.
- For advanced sorting, select **Add Sort...** For more information, see [Advanced Sorting Options](#).

- To filter your data, select **Filter...** and set the filter conditions. For more information, see [Value-based Filters](#).
 - To add conditional formatting, click the Field Menu icon  in the column header, point to **Conditional Formatting**, and select **Manage Conditional Formatting**. For more information, see [Conditional Formatting](#)
- 8. **Navigation Panel** — Shows how many rows are in your table view and lets you move through pages. To see other pages, click the arrows at the bottom of the viewer.
- 9. The data refresh details you see depend on whether the Cached Data in Datasets feature is enabled or not:
 - **Cached Response / Real-time Response Data from [time]** — Displayed when the Cached Data in Datasets feature is enabled. It tells you if the visualization is showing real-time or cached data, and when it was last updated.
 - **Last Updated** — Displayed when the Cached Data in Datasets feature is disabled. It shows when the data in the current visualization was last updated. To get the latest results, click the **Refresh** icon.

For more information, see [Data Refresh in SuiteAnalytics Workbook](#).
- 10. **Workbook Menu** — You can open this menu from any workbook visualization.
 - Click **Share** to share the workbook and any connected datasets with other users in your account.

For more information, see [Accessing and Sharing Workbooks and Datasets](#)

- Click **Save** to save your workbook, including all your selections in each visualization. You can only save your workbook if the connected datasets have been saved too. If you don't own the connected datasets, save copies with different names.
- 11. **Learn about Analytics** — Click **Learn about Analytics** to open a panel with all SuiteAnalytics Workbook videos and their descriptions. To watch a video, click the thumbnail and it opens in a new window.
- 12. **Tab Menu** — Hover over the Tab menu icon  and select **Rename Tab** or **Delete Tab**. You can also rename your table view by double-clicking the tab name.

Pivot Tab

The Pivot tab lets you pivot your dataset query results. All fields in the dataset appear in the Dataset Panel on the left. To build your pivot table, drag fields from the Dataset Panel to the Layout panel, then click the Refresh icon . The pivot table is shown in the Pivot Table Viewer on the right.

You can create multiple pivot tables using the same dataset query results.

For information about how to use the Pivot tab to pivot your dataset query results, see [Creating a Workbook](#).

The image below shows the parts of the Pivot tab:

The screenshot displays the SuiteAnalytics Workbook interface for a 'Transaction Workbook'. The top navigation bar includes tabs for Activities, Transactions, Lists, Reports, Analytics, Documents, Setup, Customization, Commerce, Connector, SuiteApps, and Support. The main workspace is divided into several panels: a left sidebar for the Dataset Panel and Layout Panel, a central pivot table visualization, and a right sidebar for filters and measures. The pivot table shows data for 'Date (Year)' with columns for 2022, 2023, 2024, and 2025. The rows list various transactions, including 'Return Authorization ...', 'Purchase Order #5819', 'Commission #18', 'Purchase Order #5820', 'Purchase Order #5823', 'Commission #20', 'Purchase Order #5817', 'Purchase Order #5818', 'Purchase Order #5816', and 'Commission #19'. The pivot table cells show values for 'Total Amount (Transa...)'.

1. **Workbook Information** — Click to view or edit the workbook name, description, internal ID, or add a translatable portlet name to your Analytics portlet. This information is available from anywhere in the workbook.
- For more information about translatable Analytics portlet names, see the help topic [Adding Translatable Analytics Portlet Names](#).
2. **Undo, Redo, and Refresh** — Undo or redo your last changes, and refresh the data in the current workbook visualization. The undo and redo options work for any action in the workbook, not only the current tab. For more information about data refreshing, see [Data Refresh in SuiteAnalytics Workbook](#).
 3. **Add** — Click the add icon to add a new workbook visualization.
 4. **Dataset Panel** — The Dataset Panel is on the far left area and has all the fields from the selected dataset. To build a visualization, drag fields from the Dataset Panel to the Layout Panel or the Table Viewer. You can also edit the dataset or switch to a different one using the Dataset Selector at the top.
 - To view or edit a dataset, click its name in the Dataset Selector.
 - To see more information about a field, point to it and click the icon.
 - To collapse the Dataset Panel and expand the Viewer, click the double arrows .
 5. **Dataset Selector** — This list lets you access and edit the datasets connected to the visualization, select a different one, or connect a dataset that's not already used in another visualization. In visualizations with linked datasets, you can also edit the dataset link.
 - To view or edit a dataset, click its name in the Dataset Selector. Alternatively, click the menu icon and select **Open Dataset**.
 - To connect or create a new dataset, click the arrow and select **Connect Dataset**.
 - To edit the dataset link, click the dataset menu icon and select **Edit Dataset Link**.
 - To save or discard changes to the selected dataset, click the menu icon and select **Save Changes** or **Discard Changes**. You need to save or discard changes to all connected datasets before you can save the workbook.
 6. **Layout panel** — The Layout panel shows your pivot table layout. For date and numeric fields, the summary type is shown next to the field. Every time you make changes to your layout, click the Refresh icon to see the updates.

- To add fields to the layout, drag them from the Dataset Panel to the section you want. You can add multiple fields to the same table component. For more information, see [Grouping Pivot Table Fields](#).
 - To add a calculated measure, click **Create Calculated Measure** and set up your calculations. Calculated measures have a calculator  icon. For more information, see [Calculated Measures](#).
 - If you're in a NetSuite account with multiple currencies or subsidiaries and your workbook has fields with values in different currencies, click the Field Menu icon  next to the field and select **Currency...** to convert or consolidate. You need to do this to run calculations on those fields. For more information, see [Currency in Datasets and Workbooks](#).
 - To create a value-based filter, click the Field Menu icon  and select **Filter [Field name]...** You also create measure-based filters by selecting the **Top 10 [Field name]**, **Bottom 10 [Field name]**, or **Filter [Field name] by** options. For more information, see [Workbook Visualization Filters](#).
 - To rename a field, click the Field Menu icon  and select **Rename...**
 - To format numbers in the table, click the Field Menu icon  and select **Format....** For more information, see [Customizing Numeric Values](#).
 - To change the summary type for dates or numbers, click the Field Menu icon  next to the field and select the summary type you want.
 - To apply conditional formatting to your pivot table measures, click the Field Menu icon  next to the field you want to highlight, then point to **Conditional Formatting** and select **Manage Conditional Formatting**. For more information, see [Conditional Formatting](#)
 - To collapse the Layout panel and expand the Pivot Table Viewer, click the double arrows .
 - To add a grand total for a field, click the Field Menu icon  and select **Show Grand Total**.
7. **Pivot Table menu** — This menu has additional formatting options for the pivot table. You can do the following:
- Export icon  — Export the current pivot table as an XLSX or CSV file.



Important: To help prevent CSV injection, exported tables may have additional formatting applied. For more information, see [Known Limitations in SuiteAnalytics Workbook](#).



Note: When you export a pivot table to XLSX, formatting and conditional formatting customizations aren't kept. If your pivot table is in compact mode, you can export it to XLSX only.

- Totaling icon  — Add totals and grand totals to your pivot table. You can add column totals to the right or left, and row totals to the top or bottom.
 - Highlight Rows icon  — When enabled, rows are highlighted when you point to them.
 - Highlight Columns icon  — When enabled, columns are highlighted when you point to them.
 - Row Lock icon  — When enabled, the top row stays visible while you scroll.
 - Column Lock icon  — When enabled, the first column stays visible while you scroll.
 - Compact Mode icon  — If you define multiple fields as rows or add a multi-column hierarchy, turn on compact mode to combine them into one column.
8. **Pivot Table filter summary** — Shows a summary of the filter conditions applied to the pivot table. Filter conditions created on the Pivot tab only affect the pivot table, not the dataset.

9. **Pivot Table Viewer** — Shows your pivot table after you update the layout and click Refresh . You can also add fields directly to the Pivot Table Viewer instead of dragging them to the Layout panel.

In the Pivot Table Viewer, you can do the following:

- To resize a column or row, drag its edge until it's the size you want.
- To change the width of a column based on its values, double-click the column header.
- To add a total for a column or row, click the Field Menu icon  and select **Show Grand Total**.
- To create a measure-based filter, click the Field Menu icon  and select **Top 10 [Field name]**, **Bottom 10 [Field name]**, or **Filter [Field name]**.
- To create a value-based filter for the table, click the Field Menu icon  and select **Filter [Field Name]...**

For more information, see [Workbook Visualization Filters](#).

- To rename a field, click the Field Menu icon  and select **Rename...**
- To sort the table, click the Field Menu icon  and select **Sort A to Z**, **Sort Z to A**, or **Sort [Field name...]**.



Note: If you sort your results based on a field that contains CLOB values, Workbook currently only evaluates the first 250 characters for each result. Results with more than 250 characters may not be sorted properly.

- To create a calculated measure, click the Field Menu icon  next to a measure and select **Create Calculated Measure**.

10. The data refresh details you see depend on whether the Cached Data in Datasets feature is enabled or not:

- **Cached Response / Real-time Response Data from [time]** — Displayed when the Cached Data in Datasets feature is enabled. It tells you if the visualization is showing real-time or cached data, and when it was last updated.
- **Last Updated** — Displayed when the Cached Data in Datasets feature is disabled. It shows when the data in the current visualization was last updated. To get the latest results, click the **Refresh** icon.

For more information, see [Data Refresh in SuiteAnalytics Workbook](#).

11. **Workbook Menu** — You can open this menu from any workbook visualization.

- Click **Share** to share the workbook and any connected datasets with other users in your account.

For more information, see [Accessing and Sharing Workbooks and Datasets](#)

- Click **Save** to save your workbook, including all your selections in each visualization. You can only save your workbook if the connected datasets have been saved too. If you don't own the connected datasets, save copies with different names.

12. **Learn about Analytics** — Click **Learn about Analytics** to open a panel with all SuiteAnalytics Workbook videos and their descriptions. To watch a video, click the thumbnail and it opens in a new window.

13. **Tab Menu** — Hover over the Tab menu icon  and select **Rename Tab** or **Delete Tab**. You can also rename your table view by double-clicking the tab name.

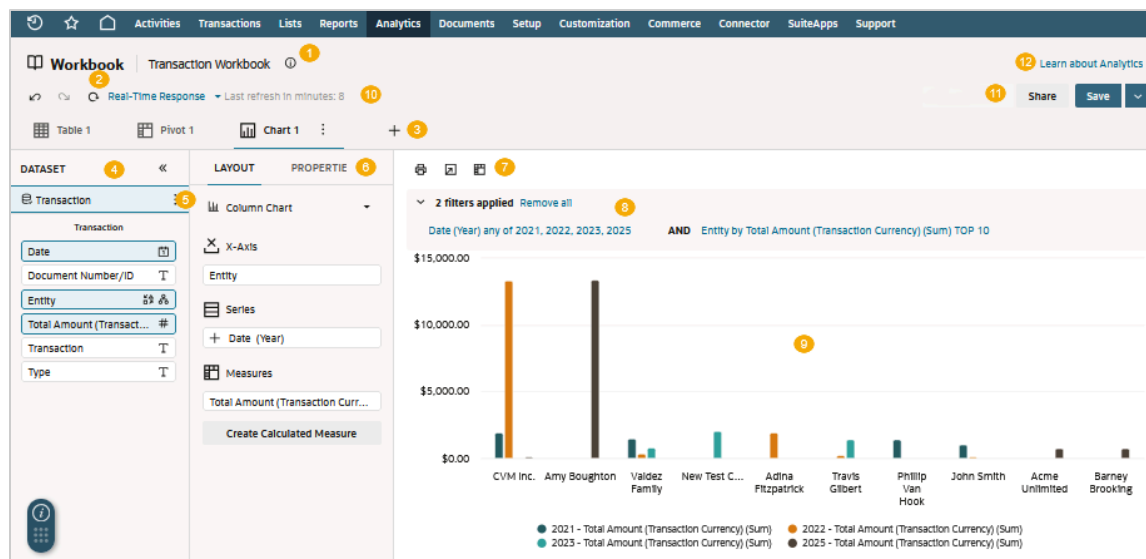
Chart Tab

The Chart tab lets you create visualizations of your dataset query results. All fields in the dataset appear in the Dataset Panel on the left. To build your chart, drag fields from the Dataset Panel to the Layout panel, then click the Refresh icon . The chart is displayed in the viewer on the right.

You can create multiple charts using the same dataset query results.

For information about how to use the Chart tab to create a chart with your dataset query results, see [Workbook Charts](#).

The image below shows the parts of the Chart tab:



1. **Workbook Information** — Click to view or edit the workbook name, description, internal ID, or add a translatable portlet name to your Analytics portlets. This information is available from anywhere within the workbook.
For more information about translatable Analytics portlet names, see the help topic [Adding Translatable Analytics Portlet Names](#).
2. **Undo, Redo, and Refresh** — Undo or redo your latest edits, and refresh the data in the workbook visualization you are currently in. The undo and redo options apply to any action within the workbook and not only the current tab you are in. For more information about data refreshing, see [Data Refresh in SuiteAnalytics Workbook](#).
3. **Add** — Click the add icon to add a new workbook visualization.
4. **Dataset Panel** — The Dataset Panel is on the far left area and has all the fields from the selected dataset. To build a visualization, drag fields from the Dataset Panel to the Layout Panel or the Table Viewer. You can also edit the dataset or switch to a different one using the Dataset Selector at the top.
 - To view or edit a dataset, click its name in the Dataset Selector.
 - To see more information about a field, point to it and click the  icon.
 - To collapse the Dataset Panel and expand the Viewer, click the double arrows .
5. **Dataset Selector** — This list lets you access and edit the datasets connected to the visualization, select a different one, or connect a dataset that's not already used in another visualization. In visualizations with linked datasets, you can also edit the dataset link.
 - To view or edit a dataset, click its name in the Dataset Selector. Alternatively, click the menu icon and select **Open Dataset**.

- To connect or create a new dataset, click the arrow and select **Connect Dataset**.
 - To edit the dataset link, click the dataset menu icon and select **Edit Dataset Link**.
 - To save or discard changes to the selected dataset, click the menu icon and select **Save Changes** or **Discard Changes**. You need to save or discard changes to all connected datasets before you can save the workbook.
6. **Layout panel** — The Layout panel shows your chart layout. For date and numeric fields, the summary type is shown next to the field. Every time you change your layout, click the Refresh icon  to see the updates.
- To select a chart type, click the chart type link and select an option. The following options are available:
 - Column Chart
 - Bar Chart
 - Area Chart
 - Line Chart
 - Stacked Column Chart
 - Stacked Bar Chart
 - Stacked Area Chart

For more information, see [Chart Types](#).
 - To show measures as a percentage of the totals for each record, click **Show percentage**. You can do this for stacked charts only.
 - To add fields to the layout, drag them from the Dataset Panel to the section you want. As with pivot tables, you can group multiple fields in the same chart component to change the level of detail. For more information, see [Grouping Pivot Table Fields](#).
 - To add a calculated measure, click **Create Calculated Measure** and set up your calculations. Calculated measures are displayed with a calculator  icon. For more information, see [Calculated Measures](#).
 - If you're in a NetSuite account with multiple currencies or subsidiaries and your workbook has fields with values in different currencies, click the Field Menu icon  next to the field and select **Currency...** to convert or consolidate. You need to do this to run calculations on those fields. For more information, see [Currency in Datasets and Workbooks](#).
 - To create a value-based filter for the chart, click the Field Menu icon  and select **Filter [Field name]...** You can also create measure-based filters if you select the **Top 10 [Field name]**, **Bottom 10 [Field name]**, or **Filter [Field name]** by options. For more information, see [Workbook Visualization Filters](#).
 - To rename a field, click the Field Menu icon  and select **Rename...**
 - To format numbers in the chart, click the Field Menu icon  and select **Format...** For more information, see [Customizing Numeric Values](#).
 - To change the summary type for dates and numbers, click the Field Menu icon  next to the field and select the summary type you want.
 - To change a field's the summary type, click the Field Menu icon  and select the summary type you want.
 - To add a title, subtitle, or axis labels, click the Properties subtab and fill in the fields.
 - To collapse the Layout panel and expand the Chart Viewer, click the double arrows .
7. **Chart menu** — This menu has additional options for the chart in the viewer. The following options are available:

- Print  — Print the current chart.
 - Export  — Export the current chart as an SVG file.
8. **Chart filter summary** — Shows a summary of the filters applied to the chart. Filter conditions created on the Chart tab only affect the chart, not the dataset.
 9. **Chart Viewer** — Shows your chart after you update the layout and click Refresh .

In the Chart Viewer, you can do the following:

 - To see the exact amount for a measure, point to the bar or data point you want in the chart.
 - To hide specific data points, click the corresponding color in the legend below the chart.
 10. The data refresh details you see depend on whether the Cached Data in Datasets feature is enabled or not:
 - **Cached Response / Real-time Response Data from [time]** — Displayed when the Cached Data in Datasets feature is enabled. It tells you if the visualization is showing real-time or cached data, and when it was last updated.
 - **Last Updated** — Displayed when the Cached Data in Datasets feature is disabled. It shows when the data in the current visualization was last updated. To get the latest results, click the **Refresh** icon.

For more information, see [Data Refresh in SuiteAnalytics Workbook](#).
 11. **Workbook Menu** — You can open this menu from any workbook visualization.
 - Click **Share** to share the workbook and any connected datasets with other users in your account.

For more information, see [Accessing and Sharing Workbooks and Datasets](#)
 - Click **Save** to save your workbook, including all your selections in each visualization. You can only save your workbook if the connected datasets have been saved too. If you don't own the connected datasets, save copies with different names.
 12. **Learn about Analytics** — Click **Learn about Analytics** to open a panel with all SuiteAnalytics Workbook videos and their descriptions. To watch a video, click the thumbnail and it opens in a new window.
 13. **Tab Menu** — Hover over the Tab menu icon  and select **Rename Tab** or **Delete Tab**. You can also rename your table view by double-clicking the tab name.

SuiteAnalytics Workbook Glossary

This glossary contains descriptions of the elements used for workbook and dataset authoring.

Analytical Record Type: Record types created specifically for Workbook that are made up of fields from various record types and predefined criteria. For more information, see [Analytical Record Types](#).

Analytics data source: The data source used by SuiteAnalytics Workbook. Record types and fields in this data source might have different locations or labels than the data sources used for saved searches and reports. For more information, see [Analytics Data Source Overview](#).

Calculated measure: A measure that you can create in your pivot table or chart using arithmetic operations with the measures available in the connected dataset. For more information, see [Calculated Measures](#).

Charts: A workbook visualization that enables you to visualize your dataset query results using predefined chart and graph types, such as line graphs and bar charts. For more information, see [Workbook Charts](#).

Common keys: Common keys are fields that contain the same data in two different datasets. You setup common keys when you want to link two datasets in a workbook. When you select the **Create Dataset Link** option within a workbook, NetSuite automatically assigns fields with matching names and data types as common keys. For more information, see <Common Keys>.

Conditional formatting: Enables you to highlight results for fields containing NUMBER and STRING values in your workbook table views and pivot tables. For more information, see [Conditional Formatting](#)

Criteria Builder: The area above the Data Grid in the Dataset Builder. You use this area to create criteria filters, which change the dataset query results and the values presented in any workbooks that are based on the dataset.

Criteria filters: Filter conditions that are applied to a dataset using the Criteria Builder. Criteria filters are automatically propagated to any workbooks and workbook visualizations that are based on the dataset, when the dataset is saved. For more information, see [Dataset Criteria Filters](#).

Data Grid: The Data Grid is where you add fields to your dataset. You can only build workbook visualizations such as table views, pivot tables, and charts using fields that have been added to the underlying dataset of a workbook, and therefore the Data Grid.

Dataset: Datasets are the basis for all workbooks and workbook visualizations in your account. In a dataset, you combine record type fields and criteria filters to create a query. The results of this query act as the source data for the workbook visualizations you create. A single dataset can be used in multiple workbook visualizations, so to prevent data discrepancies, they can only be edited or deleted by dataset owners or users with the Analytics Administrator permission. Also, you can use multiple different datasets in the same workbook, enabling you to analyze different metrics. For more information, see [Defining a Dataset](#).

Dataset Builder: The area of the user interface where you build your datasets by creating a query. The Dataset Builder contains the Records and Fields lists, the Data Grid, and the Criteria Builder. For more information, see [Dataset Builder](#).

Dataset Panel: The leftmost area of any workbook visualization, containing all of the fields of the selected dataset. You can access and edit the underlying dataset, switch to a different dataset, or create a new dataset from this panel if the selected one doesn't contain the fields you need. Use the fields in this panel to build your table views, pivot tables, and charts.

Filters: Filter conditions applied only to a specific workbook visualization, such as a table view, pivot table, or chart. These filters do not affect the underlying dataset or any other visualizations within the workbook. For example, if you apply a filter to a pivot table, it doesn't change the data presented in the workbook's charts. To filter the values presented throughout the workbook or in the underlying dataset, you need to create a criteria filter.

Join: The action of combining fields from multiple record types to define a dataset. To complete a join, select a record type from the Records list in the Dataset Builder, then add one of the record type fields to the Data Grid or use it to build a criteria filter. There are certain behaviors you should be aware of when you join record types in a dataset, such as unwanted data duplication. For more information, see [Guidelines for Joining Record Types in SuiteAnalytics Workbook](#).

Join path: Some record types are only accessible through a specific join path. For example, to join the transaction accounting line record type in a dataset, you must first access the transaction line record type through the transaction record type. In this scenario, the join path is transaction> transaction line> transaction accounting line.

Linked datasets: After you connect a dataset to a workbook, you can use it as the basis for any visualizations you create. You can also link two connected datasets, so that you can compare data from each within the same visualization. For more information, see [Dataset Linking in SuiteAnalytics Workbook](#).

Pivot Table: A workbook visualization that enables you to pivot your dataset query results by defining measures and dimensions, so that you can analyze different subsets of data. For more information, see [Workbook Pivot Tables](#).

Records and Fields lists: The leftmost area of the Dataset Builder. These lists display any of the record types and fields that you can join in a dataset, based on the selected root record type on the dataset, your NetSuite permissions, and the features enabled in your account.

Records Catalog: A cross-channel, contextual tool that includes all records available for use in NetSuite. Within the Records Catalog, you can find all available records, their fields, sublists, IDs, APIs, joins, and search filter fields for those records.

Table View: A workbook visualization that enables you to view your dataset query results in a simple table. For more information, see [Workbook Table Views](#).

Workbook: Workbooks are where you analyze the results of your dataset queries using different visualizations, such as table views, pivot tables, and charts. All workbook visualizations are based on a dataset, and you can use different datasets for each visualization in a single workbook. This enables you to analyze different metrics within a single workbook. You can access the underlying dataset from the Dataset Panel in any workbook visualization. When you author a workbook visualization you can either define a new custom dataset, or use any of the existing datasets that you have access to in your account. For more information, see [Creating a Workbook](#).

SuiteAnalytics Workbook FAQs

See the questions and answers below for information about SuiteAnalytics Workbook.

How can I use condition-based filters in currency fields?

Condition-based filters are available for percentage, integer and float fields that do not return currency values. If you want to use this option with currency fields, customize your field using the TO_NUMBER formula and change the output type.

For more information about how to customize a currency field, see [Advanced Sample Formula Fields](#).

How can I use relative dates in Workbook formulas?

The SuiteAnalytics Workbook Formula Builder lets you create custom formula fields using relative dates.

For more on Workbook formulas and examples using relative dates, see [Calculating Amounts for Relative Date Ranges](#) and [Advanced Sample Formula Fields](#).

How do I remove a dataset from a workbook after it's been connected?

You can't manually disconnect a dataset after you connect it to a workbook. When you save and close a workbook, any datasets you didn't use in a visualization are removed automatically.